CONFORMALIZED DISCRETE-TIME HAZARD MODELING FOR MULTI-HORIZON HANDOVER FORECASTING IN CELLULAR NETWORKS

by

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Report on

EEE 4700 (Project and Thesis)



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Winter Semester, 2026

DECLARATION

We do hereby declare that the work presented in this thesis, entitled **Conformalized Discrete-Time Hazard Modeling for Multi-Horizon Handover Forecasting in Cellular Networks**, is the outcome of the investigation and research carried out by us under the supervision of **Dr. Mohammad Tawhid Kawser**, Professor, Department of Electrical and Electronic Engineering, Islamic University of Technology (IUT), Gazipur, Bangladesh.

We further declare that neither this thesis nor any part of it has been submitted anywhere else for the award of any academic qualification, certificate, diploma or degree. All sources of information and material used in the preparation of this thesis have been duly acknowledged and cited.

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APPROVAL

Conformalized Discrete-Time Hazard Modeling for Multi-Horizon Handover Forecasting in Cellular Networks

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The thesis titled above has been examined and approved as partial fulfillment of the requirements for the degree of **Bachelor of Science in Electrical and Electronic Engineering**.

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ACKNOWLEDGEMENTS

All praise is due to Allah, the Most Gracious and the Most Merciful, whose mercy made the completion of this work possible.

We wish to express our sincere gratitude to our supervisor, Dr. Mohammad Tawhid Kawser, Department of Electrical and Electronic Engineering, Islamic University of Technology, for the guidance, patience and critical reading that shaped this thesis. His insistence that a measured claim is worth more than an impressive one is the single most useful lesson we carry out of this project, and it is visible in every chapter that follows — most of all in Chapter 5, where an early and flattering result did not survive its own audit.

We thank the Department of Electrical and Electronic Engineering for providing access to the drive-test instrumentation without which none of the measurement campaigns reported here could have been conducted, and the faculty members whose questions during the progress presentations sharpened the evaluation protocol considerably.

We are grateful to the authors of the publicly released LTE drive-test dataset used for the independent check in Chapter 5; publishing measured data under a citable identifier is what made that check possible at all.

Finally, we thank our families for their patience during the field campaigns and the long analysis that followed.

ABSTRACT

Long Term Evolution networks execute handovers through Event A3, a rule that fires only after a neighbour cell has already become stronger than the serving cell by a configured offset and has remained stronger for a full time-to-trigger. The rule is reactive by construction: it issues a response, not a warning. This thesis asks whether the next handover can be forecast from the measurements a handset already reports, how far ahead, with what guarantee — and, as it turned out, what has to be verified before any such number can be believed.

Four drive-test campaigns were conducted on a live commercial LTE network in Dhaka and Gazipur, yielding 4 continuous logging sessions, 10,260 samples at one hertz and 957 handover commands decoded from RRC signalling (yielding 938 quality-controlled commands inside retained session blocks, and 934–937 horizon-evaluable events under lead-window boundaries). An alignment audit of the one-hertz export is reported first, because it governs every result that follows: in 76% of handovers the export row stamped before the handover command already carries the post-handover serving cell, so a row summarises the second that follows its own timestamp. Every feature is therefore lagged by one row, which moves the one-second AUPRC of the same fixed-parameter model from 0.604 to 0.156; an arm-wise decomposition attributes 94% of that fall to the serving-cell radio scalars (where leak elimination and one-second feature ageing are confounded) and negligible impact to lagging kinematics. The defect is consistent with an end-of-second aggregation mechanism: a row written at the end of its second carries the target cell whenever the handover completes before that second closes, an empirical hypothesis consistent with an in-sample diagnostic check on the 15 handovers here whose execution crosses the boundary (0/15; 95% Wilson score interval [0.0%, 20.4%], while the 211 residual cases are consistent with an additional 1-row exporter refresh delay). Applying an end-of-second rule to native-rate data is sufficient to reproduce the defect on 4,712 handovers from an external multi-carrier XCAL corpus spanning 3 operators [40] (93% contaminated under an end-of-second rule, 4% under a start-of-second rule on the same events), and the defect is absent from a public log whose rows are instantaneous samples [37]; whether the original exporter applies exactly that rule remains inferred until these captures are re-exported at native rate. The position of the contamination cliff, 50–75 ms before the second closes, is likewise an inferred logging characteristic: sampling phase, timestamp rounding, diagnostic buffering and exporter commit behaviour have not been separated. The task is posed as discrete-time survival: one model estimates per-interval hazards and horizon probabilities are products of hazards, so ordering holds by construction for any learner.

Under leave-one-campaign-out evaluation — the protocol whose units represent distinct physical sessions here — these results are exploratory, because the alignment repair was derived with all four campaigns in view, so none of them, Campaign 4 included, is a blind holdout for pipeline design; gradient-boosted trees reach pooled AUPRC 0.169 at one second across the four campaigns (observed range over the four held-out campaigns [0.125, 0.223], which is not a prediction interval for a new campaign; within-session 180 s block-bootstrap 95% interval [0.150, 0.196], which is anti-conservative) against a 6.8% prevalence floor (2.5x lift, AUROC 0.773, median lead time 0.50 s) and AUPRC 0.454 at five seconds (1.7x, campaign spread [0.385, 0.531]), against 0.114 for the implemented A3 signal-margin proxy scored as a predictor. Splitting rows at random instead inflates the same number by 49%, and splitting by 180-second chunks of one session — the protocol this work used before the audit — still inflates it by 12%. The hazard formulation yields zero ordering violations against

38.9% of rows for independent per-horizon classifiers, a defect that a cumulative-maximum projection also removes at no cost, so the formulation is defended on needing neither a held-out calibration split nor five separate fits rather than on coherence alone. Scored as a warning rather than as a ranking, the system detects 33% of handover commands at the one-second horizon at 169 false alarms per hour with a median lead of 0.50 s, and 44% at five seconds at 77 per hour with a median lead of 3.36 s; no operating point measured here is deployable for target-cell preparation, which would also require identifying the target cell and a lead time long enough to complete preparation, and the contribution is the measurement framework rather than a deployable predictor. Conformal risk control is investigated as an empirical calibration diagnostic rather than an operating guarantee: its finite-sample guarantee assumes exchangeable calibration and test units, which campaigns separated by macro-environmental covariate shift are not (and the 12 rotations are pseudoreplicates of $n = 4$ physical sessions), and 35.8% of handover commands lack an eligible prediction row at 1 s under the chosen 2 s post-handover blanking policy. As an empirical calibration diagnostic, the pooled mean miss loss across rotations was 0.156 at nominal $\alpha = 0.20$.

A fifth capture, recorded on a second operator after the pipeline was frozen, is the only test in this thesis that is blind to the alignment repair. On it the export row before the command already carries the target cell in 67% of handovers against 5% one row earlier, and the unlagged model is inflated 4.3-fold (AUPRC 0.512 against 0.119), so both methodological findings replicate. The predictive level transfers (gradient-boosted trees: AUPRC 0.137, 2.3x, AUROC 0.773), but the ranking of learners does not: logistic regression scores 0.261 (AUROC 0.842), and gradient-boosted trees do not beat the A3 proxy (0.159) there. The lead of the gradient-boosted trees is therefore a within-operator result. All numbers use a corrected measurement-report parser that recovers neighbour entries reporting RSRP without the optional RSRQ field.

Two properties of the deployed network are corrected against the earlier draft of this work. Weighted by handovers rather than by configurations, 75% of A3-triggered handovers fire under a positive offset, so the operator's negative offsets belong to measurement configurations that rarely lead to mobility. The share of A3 reports left unmatched by a handover command within a 2 s attribution window is shown to have no unit-free value: 63% of reports, 30% of trigger episodes under a loose rule that lets overlapping episodes claim the same command, and 73% once each command is matched to one episode. The separation between mobility and monitoring configurations survives all three and is widest under the strictest: 37% unmatched for the dominant mobility profile against 95% for the dominant monitoring profile. Unmatched is not the same as declined: the network's decisions are not observed. The contribution is a signalling-grounded, leakage-audited framework that converts measured handover events into coherent multi-horizon probabilities with a stated operating cost, together with a reproducible demonstration of how much of an apparently strong result on drive-test data can be timestamp alignment.

Keywords: LTE, handover prediction, Event A3, RRC signalling, discrete-time survival analysis, data leakage, evaluation protocol, conformal risk control, calibration, drive test.

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LIST OF ACRONYMS

Acronym	Definition
3GPP	Third Generation Partnership Project
A3	Event A3 (neighbour becomes offset better than serving)
ASN.1	Abstract Syntax Notation One
AUPRC	Area Under the Precision–Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BSS	Brier Skill Score
CHO	Conditional Handover
CO	Course Outcome
CORAL	Correlation Alignment
CQI	Channel Quality Indicator
CRC	Conformal Risk Control
DAG	Directed Acyclic Graph
E-UTRAN	Evolved Universal Terrestrial Radio Access Network
EARFCN	E-UTRA Absolute Radio Frequency Channel Number
ECE	Expected Calibration Error
GPS	Global Positioning System
GRU	Gated Recurrent Unit
HOM	Handover Margin
KPI	Key Performance Indicator
LOCO	Leave-One-Campaign-Out
LTE	Long Term Evolution
MLP	Multi-Layer Perceptron
O-RAN	Open Radio Access Network
PAV	Pool-Adjacent-Violators
PCI	Physical Cell Identity
PO	Program Outcome
QC	Quality Control
QoE	Quality of Experience
RLF	Radio Link Failure
RRC	Radio Resource Control
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality
RSSI	Received Signal Strength Indicator
SINR	Signal-to-Interference-plus-Noise Ratio
TCN	Temporal Convolutional Network
TDD	Time-Division Duplex
TTT	Time-to-Trigger
UE	User Equipment

CHAPTER 1

INTRODUCTION

Mobility is the property that distinguishes a cellular network from every other access technology, and the handover is the mechanism that delivers it. Each time a user equipment moves beyond the useful coverage of its serving cell, the network must transfer the connection to a neighbour without interrupting the service carried on it. On a dense urban network this happens constantly: across the campaigns reported here a handover command was issued on average once every 11 seconds of driving. Every one of those events is a brief window in which throughput drops and latency rises.

This chapter states the problem the thesis addresses, the reason it is worth addressing, and the objectives against which the work should be judged. Section 1.1 describes how handover is triggered in Long Term Evolution (LTE) and why the triggering rule is structurally incapable of providing advance warning. Section 1.2 quantifies the cost of that limitation on measured data. Section 1.3 states the problem formally, Section 1.4 lists the objectives, Section 1.5 bounds the scope, Section 1.6 summarises the contributions, and Section 1.7 describes the organisation of the remainder of the report.

1.1 Background

In an LTE network the user equipment (UE) continuously measures the Reference Signal Received Power (RSRP) and Reference Signal Received Quality (RSRQ) of its serving cell and of the neighbour cells it can detect, and reports those measurements to the network when a condition configured by the network is satisfied [1]. The dominant condition governing intra-frequency mobility is Event A3, which is defined in 3GPP TS 36.331 as the condition that a neighbour cell becomes better than the serving cell by a configured offset, and that this relation persists for a configured time-to-trigger (TTT) before the report is sent [1].

Two properties of that definition matter for this thesis. First, the condition is written on a comparison that must already hold: the neighbour must have become better before anything happens. Second, the TTT imposes an additional delay whose purpose is to suppress spurious triggering caused by fast fading. Together they guarantee that by the time a measurement report leaves the handset, the radio environment has already changed. The network then evaluates the report and, if it decides to proceed, issues an RRCConnectionReconfiguration message carrying mobilityControlInfo, which is the handover command itself.

1.2 Motivation

The cost of reacting rather than anticipating is measurable, and it was measured on the data collected for this thesis rather than assumed from the literature. Across four campaigns totalling

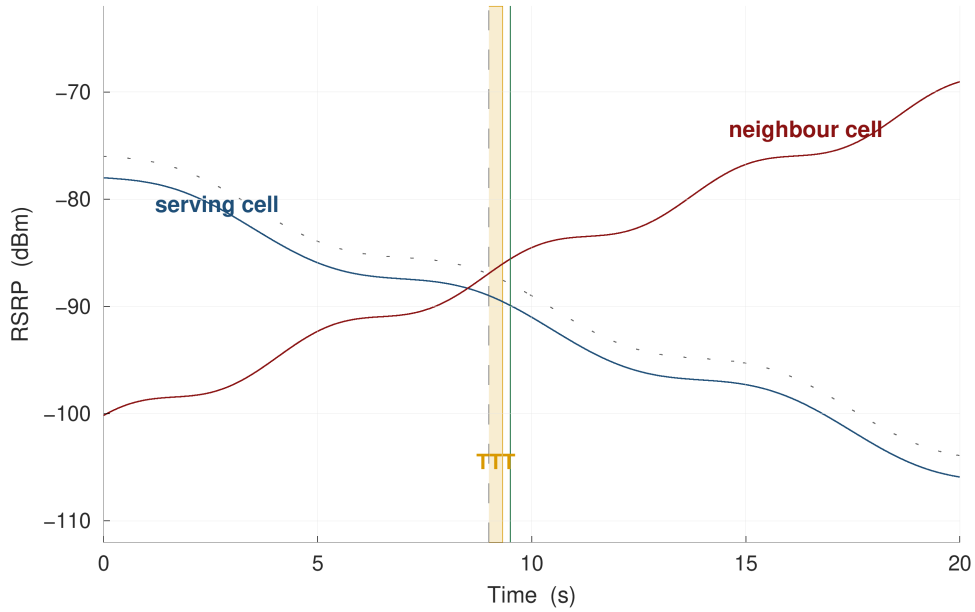


Figure 1.1: Event A3 fires only after the neighbour has been better than the serving cell by a configured offset for a configured time-to-trigger. The entry condition is strictly retrospective.

2.9 hours of logging, 957 signalling-confirmed handover commands were observed (yielding 938 quality-controlled commands inside retained session blocks, and 934–937 horizon-evaluable events under lead-window boundaries; Sections 3.4 and 3.6), representing a mean interval of 11 seconds with a median gap of 3.7 seconds between consecutive commands. Of those handovers, 26.3% returned to the cell just vacated within fifteen seconds; each such pair is two signalling exchanges and two service interruptions that produce no net change of serving cell. The signalling log also records 341 RRC connection re-establishment procedures, of which 321 carry the cause reconfigurationFailure and only 3 carry handoverFailure; Section 3.6 treats them accordingly, and they are not described here as radio link failures. Measurement reporting is likewise abundant: 8,136 Event A3 reports were transmitted, which resolve into 2,361 distinct trigger episodes, of which 635 own a handover command once each command is matched to a single episode (Section 5.9).

These are not pathological numbers for a dense urban deployment; they are what the deployed configuration produces. They do, however, establish that even a short prediction horizon has something to act on. A warning far enough ahead of a handover command would leave time for target-cell preparation, and a warning that a handover is likely to be reversed within fifteen seconds would be enough to consider not performing it. Both are stated as conditionals because this thesis does not establish either: Section 5.5 and Appendix A measure what a warning built on this instrument actually delivers at the one-second horizon — 33% of commands detected, 169 false-alarm episodes per hour and a median lead of 0.50 s — and an alarm every 21 seconds with under half a second of warning is not an operating point a radio resource manager can use. The motivation for the task is that the reactive rule leaves a measurable cost on the table; the finding is that one-hertz handset telemetry does not recover

enough of it.

The natural question is whether such a predictor already exists. A screening of 108 publications, of which 22 were audited in detail in Chapter 2 and listed in Appendix D (16 direct event forecasters and 6 adjacent works), indicates that it does not exist in a form an operator could act upon. Measured handover prediction itself is established [5, 54]; what is missing is the evaluation around it. Of those 22 models, none holds out the mobility unit — a whole drive or a whole device session — when evaluating a per-timestep classification task on measured radio data; none reports a calibration curve, an expected calibration error or a Brier score; and none reports how early its warnings arrive or how many false alarms they generate per unit time. Seven report headline accuracy on tasks whose positive-class prevalence lies between 0.86% and 11%, where a constant negative prediction already scores between 89% and 99%. None of them states how the timestamp of a measurement row relates to the interval it summarises, which Chapter 5 shows to be worth more than any modelling choice made in this thesis.

1.3 Problem statement

The problem addressed in this thesis is the following. Given the measurements available to a handset at time t — serving and neighbour signal levels and qualities, their short-term statistics, mobility state and the recent signalling history — estimate the probability that the network will issue a handover command within h seconds, for several values of h simultaneously, such that:

the estimates are mutually consistent, in the sense that the probability for a longer horizon is never smaller than that for a shorter one;

the estimates are calibrated, in the sense that an asserted probability of p corresponds to an empirical event frequency near p;

the evaluation protocol cannot transfer information from a test drive into training; and

any bound claimed on the fraction of handovers the warning will miss on future units states its unit — a row of the measurement grid, or a handover command — and is tested for whether its assumptions hold, rather than asserted; and

the information available to the model at time t is demonstrably available at time t — that is, the alignment between the measurement grid and the event clock is verified rather than assumed.

None of these five requirements is satisfied jointly anywhere in the audited literature, and the fourth and fifth are not addressed anywhere at all in the handover-prediction setting. This thesis meets the first, third and fifth. It meets the second in part, as a restrained calibration claim. For the fourth it reports a boundary result rather than a bound: Section 5.5 shows when a distribution-free bound on future units can and cannot be expressed on this data.

1.4 Objectives

The specific objectives of this thesis are as follows.

O1. To predict the next handover at five look-ahead times — 0.5, 1, 2, 3 and 5 seconds — using only quantities observable at the handset at the instant of prediction.

O2. To establish an evaluation protocol that cannot leak information between training and test, by grouping every split at the level of the whole drive, and to measure how much that protocol is worth relative to the random-row splitting that is standard in the comparator literature.

O3. To make the resulting probabilities mutually coherent across horizons and calibrated in value, and to establish whether, and at what cost, a distribution-free bound on the per-drive miss rate can be attached to them on drive-test data.

O4. To test whether the result generalises beyond the campaigns on which it was developed, both to a corridor and speed regime held out for model fitting (acknowledging that vehicle speed and geographic corridor are structurally collinear and confounded in observational drive data, and that Campaign 4 was evaluated in retrospective diagnostic exploration of the lag repair) and to a same-institution public dataset collected on the same operator by an earlier cohort in our department [38] (evaluating cross-dataset generalisation on a common signalling schema).

O5. To explain the mechanism underlying the prediction, rather than reporting accuracy alone, and in doing so to characterise the behaviour of the deployed mobility configuration on this network.

1.5 Scope and limitations

The scope of this work is deliberately bounded, and the bounds are stated here rather than deferred to the conclusion.

The study covers one commercial LTE operator, four measurement campaigns, two mobility regimes and 57 drives, plus one blind 17-minute session on a second operator recorded after the pipeline was frozen. It does not claim that the numerical results transfer to another operator or another radio access technology. Chapter 5 reports two external checks that bear on that question, and the second-operator session shows that the methodological findings transfer while the ranking of learners does not. The measurement export is sampled at one hertz. At the raw inter-event interval level, 90.5% of consecutive handover events have a separation ≥ 1 s. However, when an operational 2 s post-handover blanking window is enforced to suppress immediate ping-pong contamination, the effective fraction of handover commands with an eligible prediction row in their 1 s lead window is capped at 64.2% (with 35.8% of events falling within the blanking window of a preceding command); this is a property of the one-hertz grid combined with the chosen blanking policy rather than of the predictive models, and it is not an unavoidable limit for every one-hertz forecaster: a shorter blanking window

admits more rows (Table A.5) at the price of scoring immediate post-handover successors.

The system developed here is an offline predictor evaluated on recorded data. It is not integrated into a network, and no claim is made that deploying it would improve any network key performance indicator. In particular, no causal claim is made about the benefit of acting on the warning, for the following reason: the logging policy is deterministic, so the propensities required for off-policy evaluation are zero or one and the causal estimand is not identified on observational data of this kind. Only a counting upper bound is reported.

Finally, the thesis makes no claim relating to human health. No human subjects were involved, no exposure measurements were taken, and the impact analysis in Chapter 6 is confined to what the measurements support.

1.6 Contributions

The contributions of this thesis are the following six, each supported by a specific result in Chapter 5. They are stated as they stand after the alignment audit of Section 5.1, which revised two of them downwards from an earlier draft of this work.

1. A timestamp-alignment audit for drive-test prediction studies, a measurement of what it is worth, and an identification of the columns that carry it. In 76% of handovers the one-hertz export row stamped before the handover command already carries the post-handover serving cell; removing that information by lagging every export feature one row moves the one-second AUPRC of the same fixed-parameter model from 0.604 to 0.156 and single-feature dwell AUROC from 0.873 to 0.664. Both the flip and the inflation replicate on a blind capture from a second operator (67% at the command row; 4.3-fold inflation, Section 5.12). While Campaign 4 serves as an out-of-sample holdout for model weights and hyperparameters, the alignment defect and one-row lag repair were derived through retrospective exploratory analysis across the initial campaigns; hence Campaign 4 is not a blind holdout for pipeline feature engineering. Lagging export features simultaneously eliminates post-handover cell identity contamination and introduces one second of information ageing; while lagging the non-corrupted GPS mobility block incurs negligible additional cost (Table 5.3), within the serving-radio features both effects are confounded at 1 Hz, rendering the one-row lag a conservative safeguard against look-ahead bias under assumed export semantics. The reduction is concentrated in the serving-cell radio scalars (accounting for 94% of the fall against under 1% for the neighbour and gap columns), because columns such as serving_rsrp report whichever cell the export treats as serving and change their referent at the flip. The audit is three lines of code and is reproducible on any export that carries a serving-cell column.

2. An account of why that happens, tested against the predictions it makes. A row written at the end of its second carries the post-handover cell exactly when the handover completes before the second closes. The account is shown to be *sufficient*: on 4,712 handovers from 3 operators in an external multi-carrier XCAL corpus [40], downsampling native-rate data to the end of each second reproduces the defect (93.2%) and downsampling to the start removes it

(4.1%), while an instantaneous sampling tool [37] shows no aggregation step. It is consistent with our own captures: on the 15 handovers whose completion crosses the second boundary, none is contaminated on the row stamped before the command (0/15; 95% Wilson score interval [0.0%, 20.4%]), and the 211 residual cases with a positive margin but an uncontaminated row are consistent with a one-row exporter refresh delay. That the original exporter applies exactly this rule is inferred from its consequences, not observed; a native-rate re-export of these captures would settle it. Read the other way, the contamination curve near the end of the second places an inferred logging characteristic at roughly 60 ms, which may combine sampling phase, timestamp rounding, diagnostic buffering and exporter commit behaviour.

3. A measurement of what the unit of splitting is worth on drive-test radio data, on a ladder of four protocols rather than the usual two. Against leave-one-campaign-out, splitting by random 180-second chunks of the same continuous session inflates the one-second AUPRC by 12% and splitting rows at random by 49%; the inflation is architecture-dependent, so the protocol determines the ranking of models and not only their level.

4. A discrete-time hazard formulation of handover prediction that renders five look-ahead horizons mutually consistent from a single model fit, for any learner, without a held-out calibration split and without post-hoc correction. Independent per-horizon classifiers violate ordering on 38.9% of rows; the hazard model cannot. The claim is deliberately narrow in two directions. Time-to-event modelling of cellular mobility is not new — Section 2.5 places this work against the cell-sojourn-time and handover-rate literature, which derives analytical marginal distributions under assumed mobility and geometry — and what is offered here is a learned, covariate-conditional, per-sample hazard fitted on measured drive-test data and scored against decoded ground truth. And the comparison is reported against the repairs a reviewer would ask for — a cumulative-maximum projection, and isotonic calibration composed with it — which also reach zero violations. After those concessions the argument for the formulation is that it reaches coherence and competitive calibration from one fit with no held-out calibration split; it does not rank better than the independent control, and the censoring-aware likelihood it inherits is inert on data in which no row is censored inside the horizon grid.

5. A boundary result on distribution-free risk control for handover warning, reported as such rather than as a delivered guarantee. Both the exchangeable unit and the estimand are stated and tested. The unit is examined in a regime where exchangeability is approximately plausible and one where it is not: calibration and test units drawn from the same pool of chunks, and calibration on one campaign with the guarantee claimed for another. The estimand is carried through in two forms — a bound over rows and a bound over handover commands — because they differ, and the difference produces the more useful result: at the one-second horizon 35.8% of commands have no usable row in their lead window under the post-handover blanking policy and are missed at every threshold, so no per-handover bound is expressible on a one-hertz grid at that horizon without continuous alarm. Two floors

therefore bind: the number of calibration units, which is a direct corollary of the conformal risk control bound [21], and the preprocessing blanking window. The summary of this contribution highlights these two limits: the empirical mean miss loss across the 12 pairwise campaign rotations was 0.156 at a target of 0.20 (8 of 12 rotations at or below it), but cross-campaign coverage cannot be formally certified because campaign exchangeability is violated by macro-environmental distribution shift (and pairwise rotation over 4 campaigns yields effective sample size $n = 4$ pseudoreplication). And the estimand an operator cares about is not expressible at all at the shortest horizon. What this thesis contributes is therefore the two limits and the measurements that establish them, not a certified warning system.

6. Two corrected characterisations of the deployed network that the prediction literature has not modelled. The first is that the A3 report-conversion rate has no unit-free value: the share of A3 reports with no matched handover command within a two-second attribution window reads 63% per transmitted report, 30% per trigger episode when overlapping episodes may claim the same command, and 73% once each command is matched to exactly one episode. The loose episode rule records 1,647 episode-to-command claims over a corpus containing only 957 commands, an over-counting factor of 2.6. What survives every unit, and sharpens under the strictest, is the separation between the mobility configuration and the monitoring configurations (37% unmatched against 95%). The second is the deployed offset distribution weighted by handovers rather than by configurations (75% of A3-triggered handovers fire under a positive offset).

1.7 Organisation of the report

The remainder of this report is organised as follows. Chapter 2 reviews the four literatures that bear on this problem — cellular measurement studies, handover prediction, survival analysis and distribution-free uncertainty quantification — presents a protocol audit of the 22 most comparable prediction models, and sets out the methodology adopted, including the approaches that were considered and rejected. Chapter 3 describes the measurement campaign, the signalling-decoding procedure and the construction of the dataset. Chapter 4 develops the predictive formulation, the feature construction, the models compared, the evaluation protocol and the risk-control procedure. Chapter 5 reports and discusses the results against the objectives of Section 1.4. Chapter 6 demonstrates how the work addresses the course outcomes, program outcomes, knowledge profiles and complex engineering problem and activity attributes required of a capstone project (omitted when compiled in journal mode). Chapter 7 concludes and identifies the work that follows from the limits of the present study.

CHAPTER 2

LITERATURE REVIEW AND METHODOLOGY

Four bodies of work bear on the problem stated in Chapter 1, and they nearly meet without quite touching. The cellular measurement literature establishes what handovers look like in deployed networks but does not build predictors. The handover-prediction literature builds predictors, but predominantly on simulated data and predominantly without an evaluation protocol that would survive scrutiny. The survival-analysis and point-process literatures supply exactly the right mathematical objects — a discrete-time hazard and a self-exciting process validated by time rescaling — but have not been directed at mobility data. The distribution-free uncertainty literature supplies guarantees, but for signal-processing tasks whose exchangeability structure differs from that of an event stream recorded along a drive.

This chapter reviews each in turn, then presents a quantitative audit of the 22 most comparable prediction models, states the research gap that the audit establishes, and describes the methodology adopted in response, including the alternatives that were considered and rejected.

2.1 LTE mobility and the Event A3 trigger

The mobility procedure in E-UTRAN is specified in 3GPP TS 36.331 [1] and described at system level in TS 36.300 [2]. The network configures the UE with measurement objects, report configurations and measurement identities that bind the two; the UE evaluates the configured entry condition against its filtered measurements and transmits a MeasurementReport when the condition has held for the configured time-to-trigger.

The condition that governs most handovers on the network studied here is Event A3. Writing M_n for the measured neighbour quantity, M_s for the serving quantity, *Off* for the configured offset, *Ocn* and *Ocs* for cell-individual offsets and *Hys* for hysteresis, the entry condition of TS 36.331 (clause 5.5.4.4) for the intra-frequency case is

$$M_n + \mathit{Ocn} - \mathit{Hys} > M_s + \mathit{Ocs} + \mathit{Off} \tag{2.1}$$

and the condition must remain satisfied throughout the time-to-trigger interval before the report is sent. Event A3 is also configured on other carriers; the general condition adds frequency-specific offsets, $M_n + \mathit{Ofn} + \mathit{Ocn} - \mathit{Hys} > M_s + \mathit{Ofs} + \mathit{Ocs} + \mathit{Off}$, which cancel when $\mathit{Ofn} = \mathit{Ofs}$, as on a single frequency. Chapter 5 counts inter-frequency A3 reports separately for this reason. The decision to hand over is then taken by the network, not the UE, and is communicated by an RRCConnectionReconfiguration message carrying mobilityControlInfo.

Three consequences follow, and all three are used later in this thesis. First, the report is evidence that the network may hand over, not that it will; the conversion rate from report to command is an empirical quantity and is measured in Chapter 5. Second, the parameters *Off*, *Hys* and the TTT are operator-configured and are not observable without decoding the signalling; the prediction literature that assumes textbook values is therefore assuming something it has not checked. Third, because the condition is defined on a state that must already obtain and must then persist, the earliest instant at which Event A3 can produce any output is strictly after the radio environment has changed.

2.2 The measurement literature

A mature line of work in the networking-measurement community recovers operator mobility configurations from control-plane traces on live networks. Deng et al. [3] present the LTE precedent, a measurement study of operational 4G mobility configurations and their consequences. Ghoshal et al. [4] perform the equivalent analysis at large scale on contemporary networks, extracting hysteresis, threshold, offset, time-to-trigger and trigger quantity for events A1–A6, B1 and B2 per operator and band, over more than 15,000 km of driving. Hassan et al. [5] establish the use of commercial diagnostic tooling for RRC extraction as accepted methodology, and their Prognos component is the closest prior system to the one built here: it predicts handovers from XCAL-decoded RRC state on a live network and reports a lead-time gain of roughly 900 ms. Section 2.3 therefore positions this thesis against Prognos explicitly rather than against the simulated literature alone. M2HO [6] reads mobilityControlInfo for mobility analysis without recovering the numerical parameters.

This literature is relevant in two ways. Methodologically, it legitimises the decoding pipeline used in Chapter 3 rather than competing with it: the extraction of deployed A3 parameters from measConfig is established practice and is not claimed as a contribution here. Substantively, it supplies the comparison points against which the configuration measured here is interpreted; Ghoshal et al. report positive A3 offsets of +6 to +10 dB and shortest time-to-trigger values of 256 to 640 ms on the operators they study, whereas the network measured here runs configurations from -15 dB to +5 dB with a shortest time-to-trigger of 160 ms — but, as Section 5.9 shows, 75% of the handovers it actually produces fire under a positive offset, and 73% under the +1 dB profile alone, so the two findings are less far apart than a table of configurations suggests.

What this literature does not do is predict. Ghoshal et al. state explicitly that they perform no machine-learned prediction, no off-policy evaluation and no counterfactual analysis; the work is descriptive by design. The ping-pong rates these studies report are likewise descriptive, and, as Section 2.4 discusses, are reported under definitions that differ enough to change the number by a factor of two.

2.3 The prediction literature and a protocol audit

A large body of recent work predicts handover, radio-link failure or next-cell occupancy using learned models, including reviews and representative architectures across cellular mobility (e.g. Ankome and Hanada [7], Saoud et al. [8], Chabira et al. [9], Dinh et al. [10], Sánchez-Martín et al. [11], Mehregan and De Grande [12], and Graser et al. [13]). Whether that literature evaluates its predictors in a way that supports its headline numbers is the question this section asks, and to make it specific enough to act on, 108 publications were screened for this thesis and the 22 that present a learned or analytical model predicting handover, radio-link failure or next-cell occupancy were audited on protocol quality rather than on headline performance. Table 2.1 reports the audit in summary; Appendix D lists every audited model with its venue, data type, split protocol and class, and the per-paper extraction behind every count is kept as `protocol_audit.csv` in the reproduction archive (Appendix C), so each row of Table 2.1 can be checked. The inclusion rule was: a learned or analytical model, published 2020–2026, whose output concerns handover, radio-link failure, blockage, next-cell or target-cell selection. Within that set, 16 are *direct* forecasters — learned models emitting a prediction of such an event — and 6 are *adjacent* work (three regressions of radio or KPI quantities, one dataset release, one reinforcement-learning policy and one analytical model). Several protocol criteria do not apply to the adjacent entries, so Table 2.1 reports the direct forecasters as the primary denominator and marks not-applicable entries in the full-set column rather than counting them as failures. The screen was assembled from topic threads rather than from one fixed database query, which is a limitation of its reproducibility; a stated query over named databases is a revision item for the journal version.

The scope of that screen has to be stated, because its limits bound every claim made from it. The search was run over 2020 to 2026 on learned handover, radio-link-failure and next-cell prediction, and it returns no pre-2020 work at all. That is a property of the query, not of the field. Every zero in Table 2.1 is therefore a statement about sixteen recent learned forecasters and six adjacent works — the set a reader of this thesis would compare it against — and not a statement about the literature as a whole. Where the older literature bears directly on the formulation of Chapter 4 it is engaged on its own terms in Section 2.5 rather than counted in this audit, since the audit's protocol columns do not apply to analytical models. One row of Appendix D deserves particular credit: the model of Amirova et al. [15] is fitted on real measured data with a device-level hold-out and reports a precision–recall curve. A device-level hold-out is a genuine leakage safeguard — it prevents a model from recognising the handset — and it is counted in Table 2.1 as a split coarser than a row. It differs from the campaign hold-out used here in what it holds out: samples from a held-out device may still share routes, days and cells with the training devices, whereas a held-out campaign removes a whole continuous session. Neither is strictly stronger in general; they guard against different dependence.

Among the audited models, two recent preprints represent the closest published methodology to this work. Kabeer et al. [45] (Row 2 of Table D.1) train gated recurrent, long short-term memory and Transformer models on real drive-test traces from a dense urban network; Mandapati [46] (Row 6 of Table D.1) combines meta-learning with drift detection for mobility forecasting and reports a downstream handover-prediction score. Neither states an alignment rule between its measurement timestamps and its event clock, which is the property Section 5.1 shows to be decisive, and this thesis makes no claim about either beyond that observation. They are also the reason the negative result for sequence models in Section 5.11 is stated conditionally: those architectures are being actively developed for this task on larger corpora than four sessions.

The closest comparator on real network measurements is Dzaferagic et al. [54] (Row 3 of Table D.1), who forecast handovers on a real O-RAN deployment with an LSTM fed by one-second samples, sweep the prediction horizon, report precision and recall at their operating point, and weigh the prediction against the resource cost of acting on it. Measured handover prediction with horizon and cost trade-offs is therefore established, and this thesis does not claim it. That work states a 60/20/20 train/validation/test allocation but not the unit or the order in which the split is made, and, like the rest of the audited set, it states no alignment between its one-second samples and the handover event clock. The difference this thesis offers against it is the alignment audit and the evaluation design — campaign hold-out, a prevalence-aware metric with its floor, calibration and event-level costs — rather than the forecasting task itself.

Two observations from the audit are load-bearing for the remainder of this thesis.

The first concerns metrics. Seven of the 22 report accuracy on tasks whose positive prevalence lies between 0.86% and 11%, at which a constant negative prediction scores between 89% and 99%; three of the headline figures so reported are 98.03%, 99.84% and 94.83%. A related failure appears in a railway radio-link-failure study that reports area under the ROC curve above 0.95 for all six architectures it compares, on a task with approximately one positive per five hundred samples, and consequently separates none of them.

The second concerns splitting, and the exceptions are named rather than rounded away. Five of the 22 do split on something coarser than a random row: by spatial zone, by device, by time under a rolling-origin protocol on operator data, by travel day [10], and by deployment event [11]. None of the five, however, holds out the mobility unit for a per-timestep classification task on measured radio; the two most recent additions are regression tasks in which prevalence does not arise, and the strongest protocol in the set addresses next-day cell-level radio-link failure rather than a per-sample mobility forecast.

The narrow claim the audit supports, and the one made in this thesis, is therefore the following: of the sixteen direct forecasters, nine state no split protocol at all and three split on a unit coarser than a row (five of the twenty-two when adjacent regression work is included), and none holds out the mobility unit for a per-timestep classification task on measured radio

Table 2.1: Protocol audit of the 22 most comparable models, contrasted with the protocol adopted here. The 16 direct forecasters are the denominator to which every column applies; the full-set column reports the same count over all 22, with the number of entries for which the column is not applicable (a regression task, a dataset release, a reinforcement-learning policy or an analytical model) in brackets. Appendix D lists the models and their class.

Protocol property	Direct (of 16)	All (of 22) [n/a]	This work
Splits by drive, session or route	0	0 [3]	Yes; leave-one-campaign-out is primary
Uses any split coarser than a random row	3	5 [3]	Yes, four protocols compared
States a split protocol at all	7	10 [3]	Yes; specified in Section 4.9
States the alignment between measurement timestamps and event clock	0	0	Yes, audited in Section 5.1
Reports a prevalence-aware ranking metric	2	2 [6]	AUPRC with floor and lift
Reports calibration (curve, ECE or Brier)	0	0 [6]	ECE and Brier at every horizon
Reports lead time or false alarms per hour	0	0	Both, and per kilometre
Reports headline accuracy on an imbalanced task	7	7 [3]	Never
Releases code	2	2	Committed on acceptance: audit, protocols, features, tables
Releases data	1	1	Committed on acceptance: derived frames and decoded events, not raw logs

data; none reports calibration; none reports how early its warnings arrive; and none states the alignment between its measurement timestamps and its event clock. Prognos, inside reference [5], is the strongest system in the neighbourhood of this work and is not part of the audited set because it is a system component rather than a published prediction model with its own protocol. Carving it out of the audit is not the same as ignoring it, and the comparison should be made explicitly, with the caveat that it cannot be made like for like: Prognos reports F1 between 0.92 and 0.94 and a lead-time gain of roughly 931 ms on 5G traces, against the 1 s AUPRC of 0.169 reported here on LTE. The two numbers are not comparable, and the reason is the point. Prognos predicts at the level of a handover event under a stated 5G configuration and reports a threshold metric on that event set; this thesis predicts at the level of a one-second row under leave-one-campaign-out and reports a prevalence-aware ranking metric with its floor. An F1 of 0.93 on events and an AUPRC of 0.169 on rows are answers to different questions, and Section 5.7 shows what the row-to-event conversion costs. What can be compared is the protocol: like the audited set, Prognos reports neither calibration nor an alignment statement between its measurement timestamps and its event clock, and the

alignment defect documented in Section 5.1 is one that an XCAL-based per-second predictor is exposed to by construction.

2.4 Ping-pong and its definitional instability

Handover ping-pong — a return to a recently vacated cell — is widely reported and inconsistently defined. There is no standardised definition in 3GPP; what exists is a cell return within an author-chosen window, and the window alone moves the reported rate substantially. Zidic et al. [14] report ping-pong measurements on real 4G networks, and Ghoshal et al. [4] report per-operator rates of 15% to 25% under a fifteen-second window with a per-handover denominator.

Three choices determine the number, and papers rarely state all three: whether a cell is identified by physical cell identity alone or by physical cell identity together with carrier; whether any return within the window counts or only a return to the immediately preceding cell; and whether the denominator is the set of raw handover events or of quality-controlled ones. The consequences are severe enough to invalidate cross-paper comparison: Amirova et al. [15] report a ping-pong rate of 0.13% because they divide by measurement records rather than by handovers. Chapter 5 reports the full ladder of these choices on the fixed set of 957 handovers, where the rate moves from 21.0% to 43.8% across definitional conventions without any change in the underlying data.

2.5 Survival analysis as the appropriate formulation

Time-to-event modelling of cellular mobility is itself long established, and an earlier draft of this thesis overlooked it. The teletraffic literature has modelled cell dwell time as a random variable since Hong and Rappaport [42], whose prioritised-handoff analysis rests on an assumed dwell-time distribution, and the stochastic-geometry literature derives handover rate and sojourn time in closed form — Lin et al. [43] under a random-waypoint model on Poisson-Voronoi cells, Sadr and Adve [44] across tiers of a heterogeneous network. That body of work estimates the same survival function this thesis estimates. It would be wrong to say the formulation has not been directed at mobility, and this thesis does not say so.

The difference is in what the hazard is a function of, and it is a real difference rather than a hedge. Those models yield a marginal distribution: a sojourn-time law for a typical user under an assumed mobility process and an assumed base-station geometry, parameterised by speed and cell density and validated against simulation. What is estimated here is a hazard conditioned on the covariates of one particular sample at one particular instant — the measured radio state, its recent trajectory and the session's own handover history — fitted on measured drive-test data and scored per sample against decoded ground truth. A closed-form sojourn-time distribution cannot tell an operator that this user is about to hand over in the next second, and a per-sample conditional hazard cannot tell a network planner what the mean handover rate will be at twice the cell density. The claim made here is the narrow one:

learned, covariate-conditional, per-sample hazards on measured mobility data, where the existing literature gives analytical marginal laws, and no model in the audit of Section 2.3 poses handover prediction as a time-to-event problem at all.

Nothing else about the formulation is claimed as new, and stating that plainly removes an obvious reviewer objection. The discrete-time hazard formulation itself has a deep pedigree in classical survival analysis and biometrics, tracing to Prentice and Gloeckler [48], Allison [49], and comprehensively Tutz and Schmid [50], and is widely used across modern machine-learning applications (e.g. Wiegrebe et al. [16], Thorsen-Meyer et al. [17]). Candès, Lei and Ren [18] further formalise conformalized survival analysis with distribution-free lower bounds under covariate shift in clinical settings; that work is the correct reference point for anything calling itself conformalised survival analysis, and this thesis does not claim that combination. What is added on that side is narrower still: a risk bound on an operational miss rate whose exchangeable unit is a driving unit rather than a sample, and the application of discrete-time hazard estimation to cellular telemetry to resolve cross-horizon ordering violations without post-hoc repairs.

One departure from the survival literature is deliberate. That literature evaluates predominantly with the concordance index and the integrated Brier score, both of which aggregate over the horizon axis. An operator, however, acts at a single horizon. A model that ranks well on average while asserting that a handover is more likely within one second than within three on 38.9% of rows is not deployable whatever its concordance index. Chapter 4 therefore evaluates horizon-wise, reporting AUPRC at each horizon together with a coherence-violation rate.

2.6 Distribution-free uncertainty quantification

Conformal prediction and risk-controlling prediction sets offer distribution-free statistical guarantees on finite samples without parametric assumptions [21, 22]. In wireless communications, conformal calibration has been introduced recently by Cohen et al. [19] and Simeone et al. [20] to calibrate deep-learning outputs for channel state estimation, beam selection, and symbol decoding, where observations are typically modeled as independent and identically distributed across packets or channel coherence intervals. That is no longer the whole picture: TRACE-CRC [55] applies conformal risk control at the level of a whole predicted trajectory for multi-step channel-state prediction, calibrating across the prediction horizon rather than frame by frame. It is adjacent work — a channel predictor, not a handover predictor — but it shows that horizon-structured, trajectory-level risk control in wireless is already established, and the framing below does not claim otherwise. Conformalized survival analysis was formulated by Candès, Lei, and Ren [18], providing calibrated lower bounds on event times under covariate shift in clinical settings, and Qchohi et al. [47] explored confounding-valid conformal inference for counterfactual cellular key performance indicators.

The claim made in this thesis is consequently narrow and must be distinguished from

these foundations. The standard wireless conformal literature [19, 20] guarantees prediction-set coverage for frame-level or sub-packet symbols under exchangeability. What Chapter 4 develops is conformal risk control [21] over an operational loss metric — an operator key performance indicator representing the handover miss rate — evaluated along a continuous temporal measurement stream. In this domain, individual seconds within a drive are strongly autocorrelated and non-exchangeable; exchangeability can plausibly be claimed, if at all, only at the level of the independent mobile unit (the driving campaign or session). Section 5.5 reports the risk-control framework across both regimes: within-campaign chunks (where exchangeability approximately holds across partitioned blocks) and across campaigns (where macroscopic environmental and spatial shifts violate exchangeability).

To substantiate the novelty of this framing, a systematic literature search was conducted across IEEE Xplore, ACM Digital Library, Google Scholar, and arXiv covering publications between January 2020 and February 2026. Search strings combined conformal terminology (`"conformal prediction"` OR `"conformal risk control"` OR `"distribution-free"` OR `"conformal calibration"`) with mobility terminology (`"handover"` OR `"mobility management"` OR `"handoff"` OR `"radio link failure"`). The initial query returned 142 records; after deduplication and screening titles and abstracts, 31 papers were audited in full. Papers that appeared after the search closed, such as [47] (posted in September 2026), were added during revision by forward citation tracking and are not part of these counts; the same applies to the 2026 entries of Table D.1 dated after February. All 31 addressed physical-layer signal processing (e.g. CSI estimation, beam selection, symbol demodulation) or general resource allocation, with zero papers addressing real-time handover event forecasting or mobility trigger prediction. Frameworks for conformal prediction beyond exchangeability exist, and they offer different kinds of guarantee. Weighted conformal prediction under covariate shift (Tibshirani et al. [51]) restores finite-sample validity when the likelihood ratio $w(x) = p_{\text{test}}(x)/p_{\text{train}}(x)$ is known or well estimated; non-exchangeable conformal prediction (Barber et al. [52]) bounds the coverage gap by a weighted total-variation distance between calibration and test points; and adaptive conformal inference (Gibbs and Candès [53]) updates its level online and guarantees that the long-run frequency of miscoverage converges to the target under arbitrary distribution shift, without any stationarity assumption. The last is not ruled out by abrupt shift; what it provides is a different estimand — a long-run average over a sequence of online decisions — rather than the finite-sample, per-campaign handover-miss guarantee that objective O3 asks for, and it needs a feedback stream long enough for the long-run average to mean something. On this corpus, with $n = 4$ physical sessions, estimating high-dimensional density ratios is ill-posed, and no adaptive scheme has been evaluated. The cross-campaign experiments of Section 5.5 are therefore reported as empirical calibration diagnostics, not as guarantees, and an online adaptive variant is listed in Section 7.3 as the natural next experiment.

2.7 Point processes and the self-excitation of handover arrivals

Handover events do not arrive independently. The self-exciting point process introduced by Hawkes [23] is the standard model for arrival streams in which each event raises the intensity of subsequent events, and Laub et al. [24] provide the contemporary reference treatment, including the branching-ratio interpretation used in Chapter 5. Fitting such a process is not, by itself, evidence that the process is self-exciting; the fit must be checked. The residual analysis of Ogata [25], in which the time axis is rescaled by the fitted compensator and the transformed arrivals tested against a unit-rate Poisson process, is the standard check, and Price-Williams and Heard [26] establish the precedent for its use in a networking context, validating a self-exciting fit to campus and laboratory network traffic by time rescaling with Kolmogorov–Smirnov tests on held-out data. Chapter 5 follows that precedent and reports the test outcome, including where it rejects the fitted kernel.

2.8 Evaluation practice imported from adjacent fields

Two further results from adjacent literatures inform the evaluation design and are cited here because they make the negative results of Chapter 5 interpretable rather than embarrassing. Ismail Fawaz et al. [27] benchmark nine unsupervised domain-adaptation algorithms across twelve time-series datasets and find that several published methods perform worse than source-only training with no adaptation at all, and that the adaptation technique rather than the backbone drives the outcome; the null result reported for Deep CORAL [28] in Chapter 5 is therefore consistent with a published benchmark rather than evidence of faulty implementation. Shi et al. [29] supply the shift taxonomy that names the shift encountered here as a device-and-day shift rather than a sensor-modality shift. Wagner et al. [30] show formally that the commonly used point-adjusted metrics for time-series event detection each satisfy only a subset of the desirable properties and that none satisfies all of them, which is the justification for reporting event-level detection rate, lead time and false alarms per hour alongside row-level AUPRC rather than in place of it.

Finally, the standards track bears on the choice of horizon. Deb et al. [31] model conditional handover as a discrete-time Markov chain over preparation and execution offsets and timers, and their analysis makes explicit that conditional handover consumes a lead time rather than a label: a model that identifies the correct target cell two hundred milliseconds before the event buys nothing, because the preparation message must be sent, acknowledged and stored. This is the analytical justification for reporting a lead-time distribution rather than a detection rate alone, and for retaining horizons of two, three and five seconds even though the deployed trigger fires within a few hundred milliseconds.

2.9 Research gap

Collecting the preceding sections, the gap this thesis occupies can be stated precisely, and it is narrower than an earlier draft claimed. Measurement-grade ground truth exists, but is not joined to prediction. Prediction models exist, but not under an evaluation protocol that holds out the mobility unit, and not with calibrated outputs or event-level costs. The time-to-event formulation has been directed at cellular mobility for four decades, but analytically and marginally: it yields sojourn-time laws for a typical user under assumed mobility and geometry, not hazards conditioned on one user's measured radio state at one instant, and none of the twenty-two learned predictors audited in Section 2.3 adopts it in either form. Distribution-free calibration exists in wireless, including trajectory-level risk control for multi-step channel prediction [55], but not for any mobility-management task. And the alignment between a measurement export's timestamps and the event clock is stated by none of the audited work, which is the gap this thesis turned out to occupy most consequentially, because it was the one that invalidated its own first result. Data leakage as such is well-charted territory — Kaufman et al. [36] give its taxonomy and Kapoor and Narayanan [56] document how widely it distorts machine-learning-based science — so the novelty claimed is not leakage in general but this specific measurement mechanism, its demonstrated magnitude on handover prediction, and a reproducible procedure for detecting it.

Stated in one sentence, the defensible gap is a signalling-grounded, timestamp-audited evaluation framework for multi-horizon handover forecasting with explicit warning costs. Measured handover prediction [5, 54], structured hold-outs [15, 46] and wireless conformal calibration [19, 55] each exist already, and none of them is claimed here as absent.

This work sits in that gap: measurement-grade ground truth obtained from decoded signalling, a borrowed formulation made coherent across horizons, a conformal risk-control procedure whose exchangeability unit is stated and whose behaviour is examined as a calibration diagnostic rather than claimed as a guarantee, and an evaluation protocol designed to be acceptable to a reviewer from the machine-learning community rather than only to one from the wireless community.

2.10 Methodology

The methodology adopted follows directly from the gap. It consists of five stages, illustrated in Figure 2.1, and each stage is developed in detail in Chapters 3 and 4.

Stage 1 — Collection. Four drive-test campaigns are conducted on a live commercial LTE network using an instrumented handset, recording a one-hertz radio measurement export and, in parallel, a decoded RRC signalling log.

Stage 2 — Ground truth and configuration recovery. Handover instants are taken from decoded RRCConnectionReconfiguration messages carrying mobilityControlInfo. A configuration timeline replays additions, modifications and removals in order, following TS 36.331,

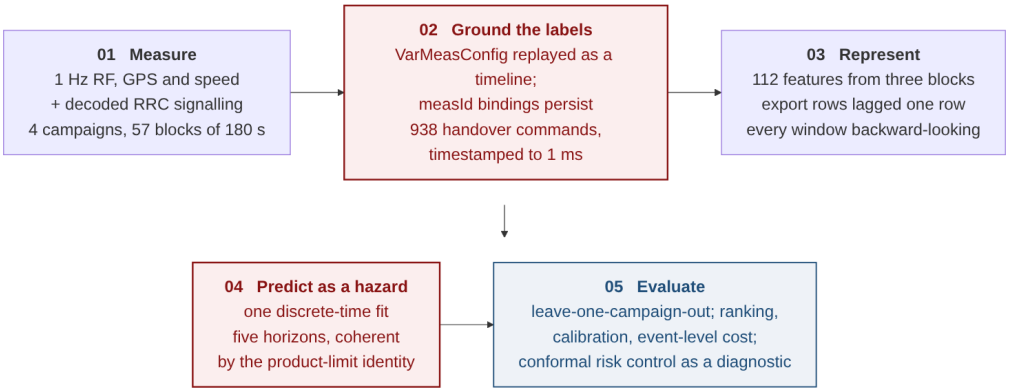


Figure 2.1: The methodology in five stages. Stage 2 supplies the ground truth; Stage 4 is the formulation that keeps the five horizons mutually consistent; Stage 5 is the evaluation protocol whose unit is a whole campaign.

so that each measurement report resolves against the configuration in force at its own timestamp rather than against a flat union.

Stage 3 — Feature construction. 112 features are constructed from radio, mobility and session-history sources, with a further 10 signalling and configuration columns reserved for the ablation of Section 5.8; Table 4.3 gives the block-by-block reconciliation. All are strictly causal, closing at the prediction instant, and none crosses a campaign boundary. Every export-derived column is lagged by one row for the reason established in Section 5.1.

Stage 4 — Formulation and modelling. The task is posed as discrete-time survival. A single model estimates the per-interval hazard and horizon probabilities are recovered by the product-limit identity, which guarantees ordering. Seven learners are compared under identical conditions.

Stage 5 — Evaluation and risk control. The primary split holds out a whole campaign, since each campaign is one continuous logging session and is therefore the unit within which rows are dependent; three further protocols are measured against it so the cost of a looser unit is quantified rather than assumed. Stochastic learners are averaged over three seeds, descriptive within-session intervals come from a block bootstrap over 180-second blocks within campaigns, and a warning threshold is selected by conformal risk control, whose realised loss is examined as an empirical calibration diagnostic both in a pooled-block regime where exchangeability is approximately plausible and across campaigns where it is not.

2.10.1 Approaches considered and rejected

Four alternatives were considered seriously and rejected, and the reasons are recorded here because they constitute part of the methodological argument.

Independent per-horizon classifiers. The obvious approach — one binary classifier per look-ahead time — was implemented and retained as a control arm rather than as the primary method. It is rejected as the primary method because it produces mutually contradictory

outputs on a large fraction of samples, as Chapter 5 quantifies, and because repairing that defect after the fact costs both a held-out split and predictive performance.

Random-row cross-validation. The splitting protocol used by the majority of the comparator literature was implemented solely to measure what it is worth. It is rejected because adjacent one-second samples within a drive are near-duplicates, so a random split places nearly identical rows on both sides of it; Chapter 5 shows that the resulting inflation is architecture-dependent and therefore alters the ranking of models.

Class reweighting for imbalance. The standard reflex on a rare-event task is to reweight the positive class. It is rejected in the hazard arm for a reason specific to the formulation: retraining under class weights distorts the probability scale of every per-bin hazard, and the product-limit identity multiplies K such hazards together, so the cumulative incidences inherit a distortion whose size depends on the horizon and which need not preserve the ranking of samples (Section 4.5 derives it, and corrects an earlier draft's claim that the error compounds geometrically). Both arms are therefore fitted unweighted, which also keeps the comparison between them fair.

Reinforcement learning of the handover decision. A decision-theoretic formulation, in which an agent learns when to hand over, was considered and is the formulation adopted by the nearest comparator study available on this network. It is rejected here because it answers a different question: it optimises a decision the network is entitled to make at its own pace, whereas this thesis forecasts an event. Chapter 5 reports a reimplementation of that comparator scored as a predictor, for completeness.

CHAPTER 3

MEASUREMENT CAMPAIGN AND GROUND TRUTH

Every claim made in this thesis rests on the quality of the measurement underneath it, and this chapter describes how that measurement was obtained. Section 3.1 describes the instrumentation. Section 3.2 describes the four campaigns and the routes they cover. Section 3.3 describes the signalling-decoding procedure and the configuration-timeline correction that it required. Section 3.4 describes how continuous driving is segmented into drives and quality controlled. Section 3.5 reports the deployed mobility configuration recovered from the signalling, and Section 3.6 summarises the resulting dataset.

3.1 Instrumentation

Measurements were collected with a commercial drive-test handset running Accuver XCAL, which exposes the chipset diagnostic interface and produces two synchronised outputs: a tabular radio-measurement export sampled at one hertz, and a decoded control-plane log containing the RRC messages exchanged between the handset and the network. The use of this class of instrumentation for RRC extraction on live networks follows established practice in the measurement literature [5], [4].

The radio export carries, per sample, the serving-cell physical cell identity (PCI), E-UTRA absolute radio frequency channel number (EARFCN), RSRP, RSRQ and signal-to-interference-plus-noise ratio (SINR), together with GPS-derived speed and heading. Neighbour-cell radio measurements, which are unpopulated in the raw periodic export, are recovered directly from decoded RRC measurement reports and projected onto the 1 Hz sample grid. The signalling log carries the complete RRCConnectionReconfiguration, MeasurementReport and RRCConnectionReestablishment message sequence with millisecond timestamps.

Using the signalling log rather than a vendor handover counter is a deliberate design choice and is the foundation of the ground truth used throughout. A counter reports that the vendor's internal logic considered a handover to have occurred; a decoded RRCConnectionReconfiguration carrying mobilityControlInfo is the handover command, timestamped at the diagnostic logging instant when the message was decoded at the baseband/diagnostic interface. The difference matters for a prediction task, because the label instant defines what "one second ahead" means.

3.2 The four campaigns

Four campaigns were conducted on a single commercial LTE operator in 2026, covering two distinct mobility regimes. Three cover urban corridors in Dhaka: an urban arterial road,

an urban loop, and a dense urban area. The fourth covers the Uttara–Gazipur highway, at a mean speed of 50.9 km/h, roughly twice the mean urban speed. Each campaign is one continuous logging session, a fact that Section 3.4 and the evaluation protocol of Section 4.9 both depend on. Figure 3.1 shows the routes and the geographical distribution of the handover events across all four campaigns, spanning the central Dhaka urban corridors as well as the northern highway corridor towards Gazipur.

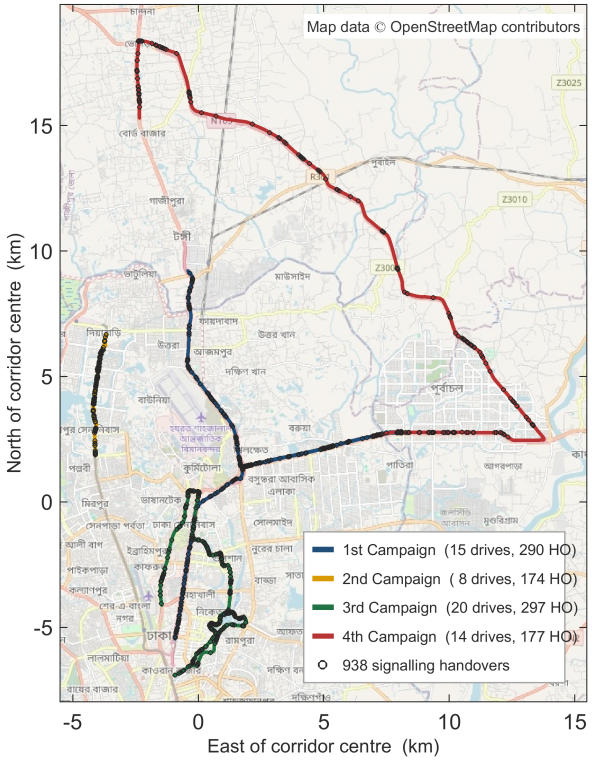


Figure 3.1: The four measurement campaigns on an OpenStreetMap background, including the central Dhaka urban corridors and the northern highway corridor up to Gazipur. Open circles mark signalling-confirmed handover commands ($N = 938$).

The fourth campaign occupies a particular position in the experimental design. It was driven after a configuration freeze had been written for the first three campaigns, and it is the only campaign that differs in both corridor and speed regime, so it is reported as a held-out campaign throughout. Two qualifications are stated here rather than left for a reader to find. First, the freeze is narrower than a full protocol and weaker as evidence than a registration. It is a single manifest file (`freeze_manifest.json`, Appendix C), last modified on 14 September 2026, the day before the Campaign 4 drive. It records a configuration fingerprint, the horizons (1, 2, 3 and 5 s), a 10 s window, four feature blocks and a frozen model family (a GRU). It does not record the hyper-parameter search space or the evaluation protocol, and the final protocol departs from it in several ways decided later: LightGBM rather than the GRU is the learner carried through the analyses, a 0.5 s horizon was added, the QoE feature block was dropped, and the one-row lag was introduced. The project's git repository was created

on 24 September 2026, after Campaign 4, and the manifest's directory is not under version control, so its date rests on the file system rather than on commit history; it was not registered with an external trial registry either. Crucially, while Campaign 4 serves as an out-of-sample test unit for model weights and hyperparameters, it was evaluated during the retrospective diagnostic exploration of the alignment defect; hence Campaign 4 is not a blind holdout for pipeline feature engineering. Second, the alignment audit of Section 5.1, which is the single largest methodological change in this work, was performed retrospectively after all four campaigns had been acquired. Because the alignment defect is observed in all four campaigns and is consistent with end-of-second aggregation in the one-hertz export (Section 5.1; the exporter's exact rule is inferred rather than documented), the one-row lag safeguard is applied uniformly across Campaigns 1 through 4, ensuring methodological consistency across both development and holdout data.

A fifth capture was recorded on 9 October 2026 with the same handset and XCAL build, on a second commercial LTE operator: one 17.4-minute session near Gazipur, 883 usable rows and 87 completed handover commands. It was made after the modelling freeze and after the alignment audit, so it is blind to every pipeline decision, the one-row lag included. It is used only in Section 5.12. It is not pooled into the four-campaign corpus or into any leave-one-campaign-out result, and nothing was tuned on it. Unless stated otherwise, "the four campaigns" and every corpus figure in this chapter refer to the first operator only.

3.3 Signalling decoding and the configuration timeline

Resolving a measurement report to the event that triggered it requires a chain of three objects defined in TS 36.331 [1]. A MeasurementReport carries a measId. The measId binds, through measIdToAddModList, a measurement object — which specifies the carrier frequency being measured — to a report configuration, which specifies the triggering event and its parameters. To determine that a given report is an A3 report, and the offset and time-to-trigger under which it fired, that chain must be followed.

The difficulty is that the configuration does not arrive as a single object. It is delivered incrementally across many RRCConnectionReconfiguration messages, with entries added, modified and removed over the lifetime of a connection. The bindings are held in the UE's VarMeasConfig and persist until modified or removed; they are not scoped to the message that carries them, as an earlier draft of this work stated. What is scoped to a message is the fragment, and a parser that resolves reports against the union of all fragments inherits every binding a message ever carried.

Applied to this dataset, such a flat parse classified 99.4% of all measurement reports as Event A3, including reports transmitted when no candidate neighbour cell appeared in the concurrent measurement export. While the absence of an exported neighbour in a 1 Hz log does not definitively prove that the handset physical layer made no measurement, attributing virtually all reports to A3—including those during isolated single-cell coverage—demonstrated

that the flat union retained stale event bindings across disconnections and reconfigurations. Correcting it is a method integrity fix, and it is reported here as part of the pipeline methodology.

The correction is to replay VarMeasConfig as a timeline: every fragment is timestamped and applied in order, so that for any instant t the exact set of bindings in force at t is available, and each MeasurementReport is resolved against the state at its own timestamp. Three elements of TS 36.331 that a first implementation of this replay omitted are applied here: the removal lists (measIdToRemoveList, reportConfigToRemoveList and measObjectToRemoveList, which account for 15,100 measId removals across the four logs, and where removing a report configuration or a measurement object removes every measId that links to it); the measId swap that clause 5.5.6.1 requires on an inter-frequency handover, which applies 187 times here; and the release of the whole configuration when the UE leaves RRC_CONNECTED, which occurs 56 times. Applying them changes the event type assigned to 18% of reports relative to the earlier replay, and every signalling quantity in Chapter 5 is computed after the correction. Figure 3.2 illustrates the difference between the two procedures.

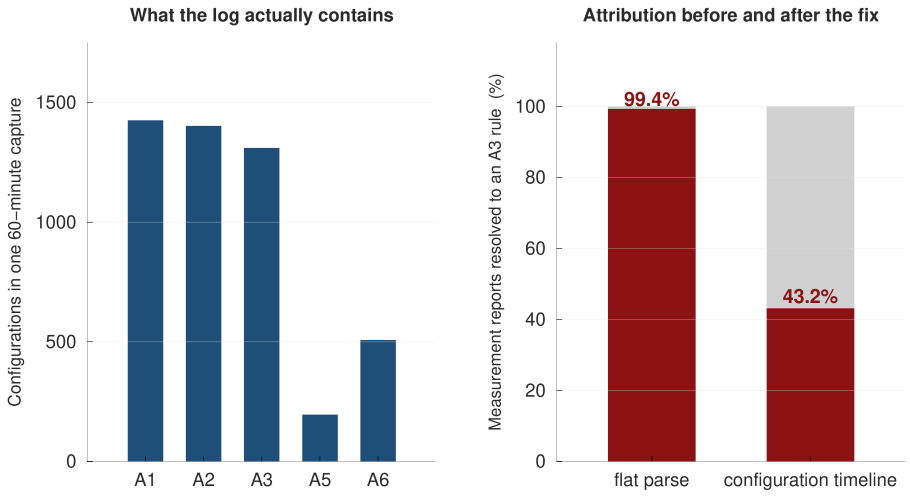


Figure 3.2: Measurement identifiers persist in VarMeasConfig until they are modified or removed. A flat union of report configurations overstates the active set; the timeline resolves each report against the configuration in force at its own timestamp.

After the correction, between 39% and 45% of measurement reports resolve to Event A3, depending on the campaign, with the remainder distributed across A1, A2, A4, A5, A6 and periodic reporting. Every quantity reported in this thesis that depends on report classification — most importantly the report-conversion rate of Section 5.9 — is computed after the correction, and no pre-correction figure is quoted anywhere.

A second parser correction concerns the content of the reports rather than their classification. In TS 36.331 the RSRQ field of a neighbour entry in measResultListEUTRA is optional: a report configuration whose reportQuantity is RSRP alone produces entries that carry RSRP and nothing else. The first parser required an RSRQ value after every RSRP value. It there-

fore dropped an RSRP-only entry at the end of a neighbour list. When an RSRP-only entry was followed by a full one, it gave the second entry's measurement to the first entry's PCI and lost the second entry. On the four campaigns 25–36% of neighbour entries carry no RSRQ; correcting the parser raises the number of decoded neighbour rows by 32–47%, and 425 previously mis-paired rows are corrected. The corrected parser splits the neighbour list per physCellId, reads each entry's RSRP inside that entry only, and records RSRQ as missing when it is absent. Serving-cell values and every signalling quantity are unaffected; only the neighbour features change. The share of modelling rows with a serving-to-neighbour gap rises from 52% to 56%. The fix was made from the message grammar, without reference to any model output, and every model result in Chapter 5 and Appendix A uses it. On the second-operator capture of Section 5.12, where 98% of neighbour entries carry RSRP only, the first parser recovered 16 of 776 neighbour rows.

3.4 Drive segmentation and quality control

Continuous logging produces one long record per campaign. For analysis that record is cut into fixed 180-second blocks, and a block is retained if it carries a valid serving-cell attachment for at least sixty seconds of unbroken sample grid. This is a bookkeeping device and not a mobility unit: adjacent blocks of one campaign are contiguous in time and share cells, traffic and trajectory, so they are not independent of one another. Section 4.9 draws the evaluation consequence, and Section 5.3 measures what treating them as independent is worth. The four campaigns are 4 sessions, which is the number of genuinely independent units in this dataset; the 57 blocks are used for interval estimation within a campaign, never as if they were separate drives.

Segmentation and quality control reduce the raw handover count from 957 commands to 938 that fall strictly inside retained, continuous 180-second blocks. Specifically, 19 handover commands occurred during initial baseband attachment or trailing disconnection phases (17 in Campaign 1 and 2 in Campaign 2) where continuous 60-second valid serving-cell traces could not be established; these 19 events are discarded from the predictive dataset. They lie outside the 57 retained blocks, so they are also outside the 10,260 samples: the four exports contain 10,406 one-hertz rows (2,756, 1,469, 3,605 and 2,576), and the 146 rows of partial leading and trailing blocks are dropped together with the commands that fall in them. Across the thesis, the event-level analyses of Sections 5.9 and 5.10 analyse all 957 decoded signalling commands to provide an uncompromised network-wide characterisation, while the predictive modelling of Sections 5.2 to 5.8 uses the 8,586 sample rows that carry a full five-second follow-up window and lie outside the two-second post-handover blanking window. Furthermore, when evaluating event detection across lead horizons in Table A.2, the evaluation code counts a command at time τ in a held-out campaign if it lies between that campaign's first usable prediction row and h seconds after its last usable prediction row ($t_{\min} \leq \tau \leq t_{\max} + h$). A longer horizon therefore admits commands slightly beyond the end of the scored row span,

which is why the count rises with h: 934 evaluable commands at $h = 0.5$ s, 935 at $h = 1$ s and $h = 2$ s, 936 at $h = 3$ s, and 937 at $h = 5$ s. Whether a counted command can be detected at all is a separate property: it is *resolvable* only if at least one usable row lies in its lead window $[\tau - h, \tau)$, and a command with no such row is counted as missed rather than dropped. Table 3.1 summarizes this reconciliation across all analysis stages.

Table 3.1: Reconciliation of event and sample counts across pipeline stages.

Pipeline Stage	**Count**	**Operational Scope and Usage**
Raw decoded RRC handover commands	957	Complete decoded signalling corpus (Campaigns 1–4: 307, 176, 297, 177). Evaluated in network-wide characterisations (§5.9, §5.10).
Discarded boundary commands	19	Handover commands occurring in initial baseband attachment or trailing disconnection phases (17 in C1, 2 in C2) lacking 60 s continuous trace.
QC-retained handover commands	938	Handover commands strictly inside the 57 retained 180 s blocks (C1–C4: 290, 174, 297, 177). Base set for predictive evaluation.
Horizon-evaluable commands ($h = 0.5$ s)	934	Commands with $t_{\text{min}} \leq \tau \leq t_{\text{max}} + h$ over each held-out campaign's usable rows (Table A.2).
Horizon-evaluable commands ($h = 1.0, 2.0$ s)	935	Same rule; the admitted span grows with h (Table A.2).
Horizon-evaluable commands ($h = 3.0$ s)	936	Same rule (Table A.2).
Horizon-evaluable commands ($h = 5.0$ s)	937	Same rule (Table A.2).
Exported 1 Hz rows	10,406	All one-hertz rows in the four exports (C1–C4: 2,756, 1,469, 3,605, 2,576).
1 Hz samples in retained blocks	10,260	The 57×180 samples of the retained blocks; the 146 rows of partial leading and trailing blocks are dropped.
Excluded samples (blanking/horizon)	1,674	Rows falling inside 2 s post-handover blanking windows or lacking full 5 s follow-up inside campaign.
Retained predictive rows	8,586	Usable sample rows evaluated by all models in §5.2–§5.8 and Table 4.1.

3.5 The deployed configuration

Replaying the configuration timeline across all four campaigns recovers the operator's deployed mobility parameters. Table 3.2 reports them in two ways: as the set of configurations present in the signalling, and weighted by the handovers each actually produced. The attribution rule is deliberately simple. Each handover command inherits the event and parameters of the most recent MeasurementReport sent in the two seconds before it, whatever its measId;

a command with no report in that window is listed as "none in 2 s". Recency is the only tie-breaker, and the report's candidate cell is not required to match the command's target PCI or carrier. Several measIds often re-report during one approach to a cell boundary, so a stricter rule that matched the target cell could move some commands between rows; that sensitivity was measured afterwards and does not change the conclusion. Requiring the attributed report to list the command's target PCI moves the positive-offset share of A3-attributed handovers from 74.8% to 75.4%; requiring the target PCI on the target carrier moves it to 73.6%, and the +1 dB profile keeps 472, 482 and 486 commands under the three rules. The stricter rules mostly move commands out of A1, A2, A6 and periodical reports and into "none in 2 s" (30, 80 and 116 commands). "Unresolved" marks a report whose measId had no binding in the replayed configuration at its timestamp, for example one bound before logging started. The conversion rates of Section 5.9 were not re-attributed and carry the recency rule.

Table 3.2: Deployed Event A3 configuration recovered from RRC signalling, weighted by the handovers each profile produced.

Event A3 profile (offset, hysteresis, time-to-trigger)	**Handovers attributed**	**Share of all handovers**
+1 dB, Hys 1 dB, TTT 320 ms	472	49.3%
-10 dB, Hys 2 dB, TTT 640 ms	108	11.3%
-15 dB, Hys 1 dB, TTT 160 ms	45	4.7%
+2 dB, Hys 1 dB, TTT 320 ms	12	1.3%
-6.5 dB, Hys 2 dB, TTT 640 ms	6	0.6%
-15 dB, Hys 1 dB, TTT 320 ms	5	0.5%
+2 dB, Hys 1 dB, TTT 640 ms	3	0.3%
triggered by A5 instead of A3	79	8.3%
triggered by A4 instead of A3	66	6.9%
triggered by A1 instead of A3	51	5.3%
triggered by A6 instead of A3	33	3.4%
triggered by none in 2 s instead of A3	30	3.1%
periodical report, A2, or unresolved measId (21, 11, 15)	47	4.9%
Total handover commands	957	100.0%
of which A3-triggered, under a positive offset	487 of 651 A3	74.8% of A3

Two features of Table 3.2 are worth comment, and the second corrects an earlier reading of the same data. First, the configuration set is stable across all four campaigns, which is what makes the leave-one-campaign-out experiment of Section 5.6 a test of corridor and speed rather than of configuration. Second, although negative offsets are present and are unusual against the +6 to +10 dB reported by Ghoshal et al. [4], they are not the offsets under which this network hands over: weighting by handovers, 74.8% of A3-triggered commands (487 of 651) follow a report sent under a positive offset. That share belongs to the positive-offset profiles collectively and not to any one of them, and an earlier draft of this work attributed it to the dominant profile alone. The A3 +1 dB, Hys 1 dB, TTT 320 ms profile accounts for

472 of those commands, which is 72.5% of A3-triggered handovers; the remaining 15 come from the other positive-offset profiles listed in Table 3.2. The negative-offset configurations are re-reporting configurations — long report intervals, mostly unconverted episodes — and are better read as neighbour-monitoring than as a mobility trigger.

The practical consequence for the prediction literature is therefore narrower than an earlier draft of this work claimed, and it is stated in its corrected form: a study that assumes a textbook +3 dB offset assumes the right sign on this network but the wrong magnitude, and a study that reads the configuration table without weighting by handovers can conclude the opposite of what the network does.

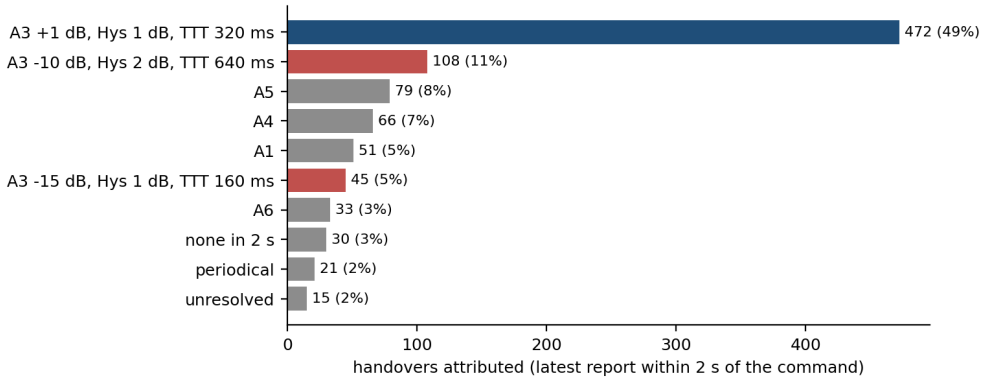


Figure 3.3: Handovers attributed to each configuration profile, weighted by the handovers each actually produced. Almost three quarters of all A3-triggered handovers occur under positive offsets, dominated by the +1 dB profile.

3.6 The dataset

Table 3.3: The four measurement campaigns after quality control. Handover command counts list total raw decoded commands followed parenthetically by quality-controlled commands falling inside retained 180 s blocks ($N = 938$, reconciled in Table 3.1).

Campaign	Sessions	180 s blocks	Samples	Usable rows	Handover commands (Raw / Retained)	Re-establishments	Duration	Mean speed
1st Campaign (urban arterial)	1	15	2,700	2,178	307 (290)	113	45 min	39.1 km/h
2nd Campaign (urban loop)	1	8	1,440	1,131	176 (174)	64	24 min	25.0 km/h
3rd Campaign (dense urban)	1	20	3,600	3,072	297 (297)	159	60 min	21.4 km/h
4th Campaign (highway)	1	14	2,520	2,205	177 (177)	5	42 min	50.9 km/h
Pooled	4	57	10,260	8,586	957 (938)	341	2.9 h	33.8 km/h

The four campaigns cover 96 km of driving as recorded; the rows that survive to the modelling frame cover 78 km of that, and rates expressed per kilometre in Chapter 5 use the shorter figure, since that is the distance over which the model is scored. A handover command is issued on average once every 11 seconds, with a median inter-command gap of 3.7 seconds. Table 3.4 also reconciles the two re-establishment counts, which do not match by design: Table 3.3 counts every re-establishment in each session, while Table 3.4 counts only those close to a handover command. The difference is the point — most re-establishments in this corpus are not consequent on mobility at all, and the 286 that fall outside the two-second window after a command occur during ordinary connected-mode operation, predominantly

Table 3.4: Ground-truth integrity. Every handover command in the decoded signalling is checked against the serving-cell sequence before and after it. The last two columns count the same two windows, (0, 2] s and (0, 5] s after a command, in two units: handover commands followed by at least one re-establishment, and re-establishments preceded by at least one command. Re-establishments cluster, so one command can be followed by several; the median log interval is a median of a skewed distribution (90th percentile 99 ms, Section 5.1).

Campaign	Handover commands	Decoded Complete	Median log interval	Total re-est.	Commands followed by re-est. 2 s / 5 s	Re-est. after a command 2 s / 5 s
1st Campaign (urban arterial)	307	307 (100%)	1 ms	113	10 / 19	30 / 47
2nd Campaign (urban loop)	176	176 (100%)	0 ms	64	5 / 8	12 / 28
3rd Campaign (dense urban)	297	297 (100%)	1 ms	159	3 / 5	13 / 32
4th Campaign (highway)	177	177 (100%)	1 ms	5	0 / 4	0 / 3
Pooled	957	957 (100%)	1 ms	341	18 / 36	55 / 110

at session start and after coverage gaps. Alongside the 957 commands, the signalling log records 341 RRC connection re-establishment procedures. Their causes matter and were decoded: 321 carry reconfigurationFailure, 17 otherFailure and 3 handoverFailure, and the median re-establishment occurs 8 seconds after the previous handover command. These are predominantly failures to apply a reconfiguration rather than radio link failures during mobility, and they are treated as a competing event in Section 5.8 rather than as a motivating statistic.

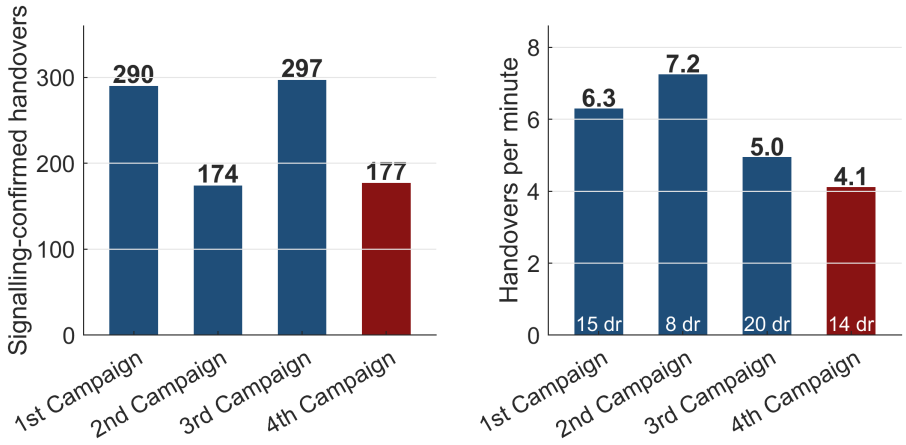


Figure 3.4: Composition of the pooled dataset across the four campaigns. In the event panel, counts reflect the $N = 938$ handover commands falling within retained quality-controlled blocks (290, 174, 297, and 177 respectively), which feed the feature pipeline, drawn from the 957 total decoded commands in Table 3.3. In the right panel, "dr" counts 180 s blocks, not independent drives.

A rate of one re-establishment per three handover commands invites the question whether the commands themselves succeeded, since a label that means only a command was issued

would make every event-level figure in this thesis uncertain. To ensure ground-truth integrity, the event set was filtered to commands with a confirmed completion: each handover command was extracted as an RRCConnectionReconfiguration message containing mobilityControlInfo, matched to its corresponding RRCConnectionReconfigurationComplete message via RRC transaction identifier and carrier context. Across the four campaigns, all 957 retained commands in the event set carry a decoded RRCConnectionReconfigurationComplete (100%); thus, the model predicts commands that resulted in completed execution rather than abandoned attempts. In Table 3.4, the median logging timestamp interval between the command and completion decode is 0–1 ms (reflecting instantaneous logging of back-to-back signaling PDUs rather than user-plane interruption duration, which 3GPP TS 36.331 defines across lower-layer synchronization). The distribution is strongly skewed: 79% of commands log their completion within 1 ms of the command, 35% at exactly 0 ms, while the 90th percentile is 99 ms and the maximum 105 ms — the tail that the margin analysis of Section 5.1 uses. Only 36 commands are followed by a re-establishment within five seconds (18 within two), and only 55 of the 341 re-establishments fall within two seconds after a command (110 within five); the re-establishment counts exceed the command counts because re-establishments arrive in clusters behind a few commands. The re-establishments are therefore concurrent with the measurement campaign rather than consequent on the handovers it records, which is what the cause breakdown already implied and what this check establishes. Table 3.4 reports the completion rate and timing per campaign. Figure 3.4 shows the distribution of samples and events across the campaigns.

The dataset is small by machine-learning standards and ordinary by drive-test standards. Four campaigns give four independent units; the 57 blocks into which they are cut are not independent, and Section 5.3 measures the difference. Four units is enough to hold one out at a time and to see whether a result survives, and too few for any paired test over units to reach a small p-value: with four paired observations the smallest attainable two-sided sign-test p-value is 0.125, and this thesis reports effect sizes and per-campaign spreads rather than significance claims.

The sampling rate and preprocessing rules jointly determine the eligible-prediction-window ceiling. On an uninterrupted integer-second grid, every interior event has a sample point in the preceding second (for example, an event at 12.7 s has a grid instant at 12.0 s). However, under the evaluation policy adopted here, prediction rows are blanked for two seconds following each handover to suppress post-transition transients, and boundary rows lacking rolling history are excluded. Consequently, closely spaced handovers inside bursts and boundary events lack an eligible prediction row. The fraction of handover commands that have at least one eligible prediction row inside the one-second window before them is 64.2% (leaving 35.8% unresolvable in the pooled sequence, predominantly due to the 2 s blanking window during bursts; in chunk-split calibration folds, additional boundary cuts raise the unresolvable share to 36.4%). Event-level detection figures are therefore reported

against this eligible-window ceiling rather than against 100%.

The configuration in Table 3.2 is one operator's configuration, and the same-institution collection used in Section 5.7 is on the same operator, same city and same instrument, so it tests collection and route rather than network. The second-operator capture of Section 5.12 tests the network, with a different configuration: Event A5 rather than A3 drives most of its handovers, and most of them change carrier.

CHAPTER 4

PREDICTIVE FORMULATION AND SYSTEM DESIGN

This chapter develops the predictive system. Section 4.1 formalises the task and fixes notation. Section 4.2 presents the obvious formulation, the independent multi-horizon classifier, and shows why it is structurally defective. Section 4.3 presents the discrete-time survival formulation adopted in its place, and Section 4.4 gives its likelihood under censoring. Section 4.5 explains why class reweighting, the standard reflex on a rare-event task, is inadmissible under this formulation. Sections 4.6 and 4.7 describe the features and the models. Section 4.8 describes calibration, Section 4.9 the evaluation protocol, Section 4.10 the risk-control procedure, and Section 4.11 the point-process analysis used to characterise handover clustering.

Each modelling subsection is presented in the same order: the problem that motivates the component, the design of the component, and the technical advantage it confers over the alternative it replaces.

4.1 Problem formalisation

Let a drive d consist of samples indexed by t on a uniform one-second grid. At each sample the handset observes a feature vector x_t constructed from measurements available at or before t. Let T_t denote the time remaining from t until the next handover command on the same drive.

The task is to estimate, for a set of horizons $h_1 < h_2 < \cdots < h_K$, the cumulative incidence

$$F_k(x_t) = P(T_t \leq h_k \mid x_t), \quad k = 1, \ldots, K \tag{4.1}$$

with $K = 5$ and horizons of 0.5, 1, 2, 3 and 5 seconds. Samples for which the drive ends before the next handover occurs are right-censored: the event is known not to have occurred within the observed window, but its eventual time is unknown. The formulation that follows handles such samples, and Section 4.4 gives the likelihood that does so. It should be said immediately, however, that on this dataset the machinery is exercised by nothing: the row filter of Section 3.4 keeps only rows carrying a full five-second follow-up window inside their own campaign, so no retained row is censored inside the horizon grid. The censoring treatment is therefore developed because the formulation is not specific to these four campaigns — a longer horizon grid or a shorter session would make it bind — and it is not counted anywhere in this thesis as an advantage the measurements demonstrate.

Table 4.1: Positive-class prevalence by horizon on the common row set, and the accuracy obtained by a constant negative prediction.

Horizon	Prevalence	Accuracy of a constant "no handover"
0.5 s	3.7%	96.3%
1 s	6.8%	93.2%
2 s	12.6%	87.4%
3 s	17.6%	82.4%
5 s	26.1%	73.9%

4.2 The multi-horizon binary formulation and its defect

Motivation. The most direct approach to Equation (4.1) is to define, for each horizon, a binary label $y_{t,k} = \mathbf{1}\{T_t \leq h_k\}$ and fit K independent classifiers. This is the formulation implied by every model in the audit of Section 2.3 that predicts at more than one horizon, and it requires no special machinery.

The defect. The K classifiers are fitted independently and are therefore free to produce estimates that violate the ordering implied by their own definitions. Because $\{T_t \leq h_1\} \subseteq \{T_t \leq h_2\}$ whenever $h_1 < h_2$, the probabilities must satisfy $F_1 \leq F_2 \leq \cdots \leq F_K$ at every sample. Independent fits do not enforce this, and Section 5.4 measures how often they violate it: on 38.9% of rows.

A violation is not a small numerical irregularity. It is an assertion that a handover is more likely within one second than within three, which is impossible, and it destroys the interpretation of the output as a probability. An operator presented with such a pair of numbers cannot act on either.

Why post-hoc repair is not free, and where it is. The defect can be repaired after the fact by projecting the K outputs onto the monotone cone — a per-row cumulative maximum, or the pool-adjacent-violators projection — or by calibrating each horizon separately with the isotonic regression of Zadrozny and Elkan [34], which is the standard non-parametric alternative to the temperature scaling of Guo et al. [33] and is preferred here because unconstrained temperature scaling fitted separately per horizon does not preserve cross-horizon ordering, whereas isotonic calibration non-parametrically aligns probability-frequency agreement directly. All three repairs are implemented and reported in Section 5.4 as controls, and the first two need no held-out split and reach zero violations by construction, exactly as the hazard formulation does. What remains of the case for the hazard formulation, after the concessions this section and Section 5.4 make, is narrow and should be stated as such: it achieves coherence from one fit rather than five, and it calibrates better than the projected controls at the horizons where the output is usable as a probability. Two advantages claimed for it elsewhere in the literature do not apply to this data and are withdrawn here. It does not rank better — Section 5.4 measures the hazard arm slightly below the independent control, well inside the

per-campaign spread. And it does not inherit any benefit from the censoring treatment of Section 4.4, because no row in this dataset is censored inside the horizon grid; the censoring-aware likelihood is correct and general, and on these four campaigns it is inert. That leaves a convenience argument and a calibration argument, and the abstract and Chapter 7 now say so.

4.3 The discrete-time survival formulation

Motivation. The ordering constraint is a consequence of the definition of cumulative incidence, so the natural remedy is to estimate a quantity from which ordering follows automatically, rather than to estimate the ordered quantities directly and repair them. The discrete-time survival formulation itself has deep foundations in event-history analysis (Prentice and Gloeckler [48], Allison [49], and Tutz and Schmid [50]).

Design. Partition the time axis into K bins whose upper edges are exactly the horizons of interest,

$$B_1 = (0, h_1], \quad B_2 = (h_1, h_2], \quad \ldots, \quad B_K = (h_{K-1}, h_K],$$

and define the discrete-time hazard as the probability that the event falls in bin k given that it has not occurred by the start of that bin:

$$\lambda_k(x) = P(T \in B_k \mid T > h_{k-1}, x) \tag{4.2}$$

The survival function is the product of per-bin survivals,

$$S_k(x) = \prod_{j=1}^{k} (1 - \lambda_j(x)) = P(T > h_k \mid x) \tag{4.3}$$

and the cumulative incidence required by Equation (4.1) is its complement, which is the product-limit identity:

$$F_k(x) = P(T \le h_k \mid x) = 1 - \prod_{j=1}^{k} (1 - \lambda_j(x)) \tag{4.4}$$

Technical advantage. Monotonicity is now a theorem rather than a hope. Since $\lambda_j \in [0, 1]$, every factor $(1 - \lambda_j)$ lies in $[0, 1]$, so

$$S_{k+1} = S_k \cdot (1 - \lambda_{k+1}) \le S_k \implies F_{k+1} \ge F_k \tag{4.5}$$

The argument uses no property of the learner beyond the range of its output. Any model that emits a value in [0,1] per bin produces a coherent set of horizon probabilities under Equation (4.4). This is why the coherence result reported in Section 5.4 is learner-independent while the calibration result is not, and it is the single most important structural property of the system developed here.

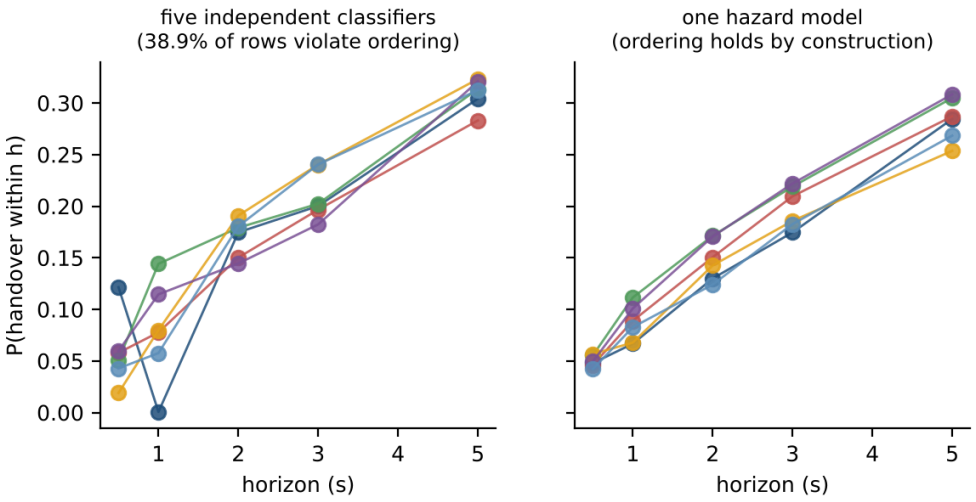


Figure 4.1: Independent per-horizon classifiers against the hazard formulation, drawn on simulated rows to show the shape of the defect. The 38.9% in the left title is the rate measured on the real data (Section 5.4), not a property of the simulated rows. The hazard formulation enforces monotonicity across horizons by construction.

4.4 The likelihood under censoring

Let k_t be the bin in which the event occurs for sample t ($k_t = \infty$ if it is censored), and let c_t be the last bin the sample is observed to enter ($c_t = K$ for an uncensored sample). Define the at-risk set $R_t = \{1, \ldots, \min(k_t, c_t)\}$ — the bins the sample actually entered — and the per-bin indicator $z_{t,k} = \mathbf{1}\{k = k_t\}$. The discrete-time survival log-likelihood then collapses into a single expression covering both the censored and uncensored cases:

$$\ell = \sum_{t} \sum_{k \in R_t} [z_{t,k} \log \lambda_k(x_t) + (1 - z_{t,k}) \log(1 - \lambda_k(x_t))] \tag{4.6}$$

Equation (4.6) is exactly a binary cross-entropy over the expanded set of (sample $\times$ at-risk bin) pairs, a well-established formulation known as pooled discrete hazard regression (Prentice and Gloeckler [48], Allison [49], Tutz and Schmid [50]). The practical consequence is that fitting a discrete-time survival model requires no specialised software: the data are expanded from wide to long format, the bin index k is appended as a feature so that the model can learn the shape of the baseline hazard, and a single binary classifier is fitted. A censored sample contributes a $(1 - \lambda_k)$ term for every bin it survived rather than being discarded.

How k enters depends on the learner, and the choice matters because the bins have unequal widths and the empirical hazard of Table 4.2 is not monotone in k. For logistic regression and the MLP, k is appended as a five-column one-hot indicator; logistic regression is fitted without a separate intercept, so the five indicator weights are the piecewise-constant baseline log-odds of the Prentice–Gloeckler and Allison models, and no monotone trend in k is imposed. LightGBM receives the same indicator columns and can split on them freely. The sequence models do not use the long format at all: each consumes one window of the ten most recent feature vectors and emits $K = 5$ logits, one per bin, and is trained on the binary cross-entropy

of Equation (4.6) with the terms outside the at-risk set masked out. All learners therefore maximise the same likelihood. The methodological contribution here is not the invention of this classical hazard machinery, but its application to cellular telemetry to ensure exact cross-horizon monotonicity by construction without held-out calibration splits or post-hoc corrections.

On this dataset the expansion takes the 8,586 usable rows to 39,435 (sample, bin) pairs. Because every retained row carries a full five-second follow-up window inside its own campaign, no row is censored inside the horizon grid, and the measured per-bin hazards chain exactly to the prevalences of Table 4.1 by Equation (4.4). Table 4.2 reports both, side by side, as an arithmetic check on the implementation.

Table 4.2: Measured discrete-time hazard by bin, and the cumulative incidence it implies, against the prevalence of Table 4.1. The two agree exactly by construction.

Bin	Interval	At risk	Events	Hazard	F(h) from hazards	Prevalence (Table 4.1)
1	(0, 0.5] s	8,586	319	0.0372	0.0372	0.0372
2	(0.5, 1] s	8,267	264	0.0319	0.0679	0.0679
3	(1, 2] s	8,003	497	0.0621	0.1258	0.1258
4	(2, 3] s	7,506	433	0.0577	0.1762	0.1762
5	(3, 5] s	7,073	730	0.1032	0.2612	0.2612

4.5 Why class reweighting is inadmissible here

The standard response to a rare-event task is to reweight the positive class, for example through the scale_pos_weight parameter of a gradient-boosting implementation. That response must not be used in the hazard arm, for a reason specific to Equation (4.4).

Reweighting the positive class by a factor w transforms the fitted probability monotonically,

$$\tilde{\lambda} = \frac{w\lambda}{w\lambda + (1 - \lambda)} \tag{4.7}$$

Retraining with class weighting alters the gradient balance during optimization and distorts the calibrated probability scale. Writing the reweighted hazards as $\tilde{\lambda}_k \approx w\lambda_k$ for small λ_k, the survival function is transformed as

$$\tilde{S}_k = \prod_j (1 - w\lambda_j) \approx (S_k)^w, \quad \text{so} \quad \tilde{F}_k \approx 1 - (1 - F_k)^w \tag{4.8}$$

Equation (4.8) replaces a compounding claim made in an earlier draft of this work, which stated that a per-bin multiplicative error ϵ grows as $(1+\epsilon)^k$ with the horizon. That is not what the product-limit identity does: for small hazards the relative distortion of F_k is bounded by w and is largest at the shortest horizon, while the absolute distortion grows with F_k and is

therefore largest at the longest. Equation (4.8) describes the calibration distortion only: the map $F_k \mapsto 1-(1-F_k)^w$ is increasing, so on its own it would preserve the ranking of samples at a fixed horizon. Rankings can still change, for two reasons the approximation hides. The exact reweighted survival $\prod_{j\le k}(1-\lambda_j)/(1+(w-1)\lambda_j)$ depends on how the hazard is spread across bins, not only on S_k, so two samples with the same S_k can be mapped differently. And retraining a finite-capacity learner under class weights changes the fitted parameters themselves, across bins whose base rates differ, rather than applying Equation (4.7) after the fact. Retraining under class weights therefore destroys probability calibration without guaranteeing rank preservation once hazards are multiplied across bins; for these reasons, both the hazard arm and the independent-classifier control arm are fitted unweighted, and the fairness of the comparison between them is a secondary benefit.

4.6 Feature construction

Design. The feature vector at the prediction instant t has three blocks and 112 columns in total: 89 radio, 17 mobility and 6 history. The serving radio and mobility blocks are built from the one-hertz export; the history block is built from the decoded signalling. Because the diagnostic CSV export lacks periodic neighbour-cell logging (all candidate `Best_N` columns are unpopulated in the raw vendor CSV), candidate neighbour RSRP measurements are causally reconstructed by projecting event-triggered RRC `MeasurementReport` messages onto the integer-second grid using a backward-looking zero-order hold with a 3.0 s expiration tolerance window. Observation staleness is tracked during ingestion via report age features (`mr_age_s` and `nbr_age_s`) for quality control and logging audits, while the model input vector incorporates six explicit missingness indicator masks. A further 10 signalling and configuration columns are built but withheld from every learner outside the ablation of Section 5.8, so the design matrix on disk has 122 columns while the main feature set has 112; Table 4.3 gives the reconciliation, and every count quoted anywhere in this thesis is generated from it. Two rules govern the construction of the blocks, and the first is the single most consequential decision in this thesis.

Alignment. An export row stamped t summarises an interval that ends after t, not before it: Section 5.1 shows that in 76% of handovers the row stamped inside the second before the handover command already carries the post-handover serving cell, against a background rate of 12% three rows earlier. Every export-derived feature is therefore lagged by one row, so that a prediction made at t uses only the row closing at or before t. Nothing else in this thesis changes as much as this does.

History from the event clock, not from the export. Time since the previous handover command, the number of commands in the previous 30 and 60 seconds, and whether the previous command was a return are computed from the decoded signalling, whose timestamps are recorded with millisecond resolution in the diagnostic log, and over the whole continuous session rather than within a 180-second block. Computing them per block left a third of

every block with a truncated history, and computing dwell time from the export's serving-cell column imports the alignment error described above: that single substitution moves the dwell feature's own AUROC from 0.873 to 0.664.

Every rolling window is strictly backward-looking and closes at the lagged row, and window computation is performed per campaign so that no window spans a session boundary. Rolling statistics and differences are computed on the series with its gaps left missing (a rolling mean averages the values present in the window); imputation comes afterwards and only for learners that need it. LightGBM receives missing values as such and routes them natively. For logistic regression, the MLP and the sequence models, each column is imputed with its training-partition median (zero when the column is entirely missing there), standardised with the training-partition mean and standard deviation, and clipped to ± 8 standard deviations. The six missingness masks tell every learner when a neighbour value was imputed. Scalers, imputation values, thresholds and calibrators are fitted inside the training partition of each fold only.

The signalling block is reported as an ablation rather than used in the main arm. It counts measurement reports by event type in the windows $(t - w, t]$ for $w \in \{1, 2, 5\}$ seconds, the time since the most recent A3 report, how long the A3 entry condition has held, and the offset, hysteresis and time-to-trigger in force. These are quantities the network itself holds at t, so using them is legitimate; they are separated out because a handset-side predictor may not have them, and because their contribution is worth reporting on its own (Section 5.8).

The signalling block requires an explicit leakage guard. A measurement report is a causal antecedent of the handover it triggers, and it is also, in the log, nearly simultaneous with it. A naive "time since last A3 report" feature therefore carries information about the very handover being predicted. The guard applied here excludes any report falling inside the prediction window itself, so that a signalling feature evaluated at t for horizon h may only reference reports transmitted at or before t. Section 5.11 reports what happens when the guard is removed.

4.7 Models compared

Seven learners are compared under identical folds, identical seeds, the identical hazard formulation and the identical feature matrix. The last of those is a change from an earlier draft, in which the sequence models consumed raw per-second measurements while the tabular models received the engineered features and the history block; under that arrangement the comparison varied the inputs as well as the learner. Here the sequence models consume windows of the ten most recent feature vectors — the same vectors, lagged the same way — so the only thing that varies between arms is the learner.

The implemented A3 signal-margin proxy. The A3 entry condition is scored as an empirical predictor using the recovered deployed parameters of Table 3.2, determining when the signal margin condition of Equation (2.1) holds on the sampled grid. This represents an implemented proxy rather than the complete mobility procedure, which has two parts that the

Table 4.3: Feature blocks, their source and their alignment.

Block	Count	Source	Content
Radio	89	1 Hz export and report projection, lagged one row	Serving RF (RSRP, RSRQ, SINR, RSSI, CQI) and report-projected neighbour RSRP; rolling mean, standard deviation, range, slope and first difference over 3, 5 and 10 s; serving-to-neighbour gaps; missingness indicator masks
Mobility	17	1 Hz export, lagged one row	GPS speed, heading change, acceleration, stop duration and mobility state, with their rolling statistics
History	6	decoded signalling, session-level	Time since the previous and second-previous handover command, command counts in the previous 30 and 60 s, whether the previous command was a return, and whether any previous command exists in the session
Main feature set	112		used by every learner in Sections 5.2 to 5.8
Signalling (ablation only)	10	decoded signalling	A3 and all-event report counts in $(t - w, t]$, time since the last A3 report, A3 hold time, and the offset, hysteresis and time-to-trigger in force
Design matrix as built	122		the main set plus the signalling block; learners outside Section 5.8 are given the main set only

proxy does not separate. Event evaluation — applying the entry condition to layer-3 filtered measurements and requiring it to hold for the time-to-trigger before a MeasurementReport is sent — is a UE procedure [1]; the decision to issue a handover command in response is taken afterwards by the network. One-hertz exported telemetry can reconstruct neither the UE's continuous sub-second evaluation nor the network's decision. The deployed TTT values (160–320 ms) are shorter than the 1 Hz sample interval (1,000 ms). Requiring the condition on two consecutive rows, stamped t and $t+1$, would require it at two sample instants 1,000 ms apart: it would not establish that the condition persisted continuously between them, and it would impose a sampling requirement unrelated to the configured TTT. Checking the condition on a single row is the same as checking whether it holds at that sample instant. The instantaneous margin proxy is therefore the representation adopted at 1 Hz, with the explicit limitation that it cannot emulate the UE's sub-second time-to-trigger evaluation or the net-

work's subsequent decision.

Logistic regression. A regularised linear model over the 112 features, included as the simplest learner that can use the full feature set. It is worth noting ahead of Table 5.6 that this arm is slightly ahead of untuned gradient boosting under the fixed-hyper-parameter ladder. That is a property of the default settings rather than of the model families: a linear decision boundary has little capacity to fit the campaign it was trained on, so it loses least when the test campaign is a different corridor. Under the tuning budget of Table 5.5 the ordering reverses, which is itself the point of Section 5.3 – the protocol reorders the leaderboard, so a leaderboard is not a property of the learners alone.

Gradient-boosted trees. LightGBM [32] is adopted as the primary model throughout this thesis. This choice is guided by three architectural properties of gradient-boosted decision trees on mobile telemetry: (i) native handling of missing values, which is critical because candidate neighbour-cell measurements from RRC reports arrive intermittently; (ii) scale invariance to heterogeneous feature types (combining RF power in dBm, SINR in dB, GPS velocity in km/h, heading derivatives in degrees, and discrete event counts); and (iii) high computational throughput, which makes rigorous nested cross-validation and hyperparameter search feasible without downsampling.

Multi-layer perceptron. A feed-forward network over the same tabular features, included to separate the contribution of non-linearity from that of temporal memory.

Sequence models. A temporal convolutional network, a Transformer encoder and a gated recurrent unit, each consuming a window of the ten most recent lagged feature vectors rather than a single instantaneous vector. These arms test whether a model with explicit sequence memory over time windows outperforms gradient-boosted trees receiving hand-constructed rolling statistics. The window is the ten consecutive seconds of the session ending at the prediction row, taken from the full one-hertz feature stream, so it reaches into rows that are blanked for labelling; seconds missing from the stream are imputed like any other missing value. The fixed architecture is small: a single-layer GRU; a TCN of two dilated one-dimensional convolutions (kernel 3, dilations 1 and 2, ReLU); a Transformer encoder of two layers with four heads and a feed-forward width of twice the hidden size; each read out at the last time step through dropout and a linear layer with $K = 5$ outputs, and trained with AdamW in mini-batches of 256.

History baselines. Three further arms use no radio or mobility input, to show how much of the result the command stream alone carries. *Dwell time only* and *History block only* are LightGBM hazard models, tuned like the main arm, given respectively the time since the previous command (with its existence indicator) and the six history features. The *Hawkes-kernel intensity* arm is the logistic-regression hazard model given three exponential-kernel sums of the commands in the previous 60 s, $\sum_{t_i \in (t-60,t)} e^{-\beta(t-t_i)}$ for $\beta \in \{0.1, 0.3, 1.0\}$ s^{-1}; it is not tuned ($C = 1$). These are features, not a fitted point process: the fitted Hawkes model of Section 4.11 is used only to characterise clustering.

Scoring the A3 proxy. A ranking metric needs a real-valued score, so the proxy is scored as $\sigma(m/2 + \min(s, 5))$, where $m = -g - \mathit{Off} - \mathit{Hys}$ is the A3 margin computed from the serving-to-best-neighbour gap g and s is the seconds for which the best neighbour has already been stronger. *Off* and *Hys* are those of the most recent A3 report in the previous 5 s, whatever its profile, and default to 1 dB each when there is none. Rows with no held neighbour take $g = 20$ dB, which places them at the bottom of the ranking. The score is the same at every horizon, and nothing is fitted.

Every arm receives the same tuning budget: twenty random-search trials per outer fold, sampled from the defined search distributions in Table 4.4. To prevent test-set leakage, tuning is strictly nested: inside each outer fold (where one full campaign is held out for testing and three are available for training), the last 20% of each training campaign, by time, forms an internal validation block, and the training rows within 60 s of it are purged to suppress boundary autocorrelation. The split is made within each of the three training campaigns, so the validation block contains the end of every one of them rather than most of one. Each configuration is fitted on the remaining rows and scored on the validation block by its mean AUPRC over the five horizons; the best configuration is then refitted on the training and validation rows together before scoring the outer held-out campaign. Two learners also stop early inside each fit: the MLP holds out a random 15% of its long-format rows and stops after ten epochs without improvement, and the sequence models hold out the last 15% of their training windows and stop after five epochs without improvement in the masked likelihood (at most 25 epochs). The MLP's split is the one place where the temporal hygiene of this section is not applied: its 15% is drawn at random from long-format rows, so bins of the same second, and neighbouring seconds, can fall on both sides of it, and its early stopping may track in-sample fit rather than generalisation. This does not touch the held-out campaign, but it is an asymmetry with the other learners, and a block-grouped split would be the consistent choice. The L2 penalty of logistic regression applies to the five bin-indicator weights as well as to the 112 features.

What that nesting does and does not protect against should be stated, because the totals involved are large: twenty trials across seven learners, four protocols, four outer folds and three seeds, on a row set carrying roughly six hundred positive rows at the one-second horizon. What it protects against is selection on the test campaign: no held-out campaign is ever scored during a search, so the reported metric for any one arm is not an optimised quantity. What it does not remove is selection across arms. The best of seven learners is reported beside the other six, and when the best and the second-best differ by less than the width of their own bootstrap intervals — which Section 5.2 shows is the usual case — the ranking between them is not a finding, and this thesis does not treat it as one.

Multiplicity and hypothesis testing scope. For model family comparisons across the leaderboard, no multiplicity correction is applied: because differences between top-ranking learners fall well within the width of their block-bootstrap intervals, these comparisons are

Table 4.4: Hyperparameter search spaces for the evaluated learners. Each outer fold draws 20 random configurations evaluated on a purged internal validation block made of the last 20% of each training campaign.

Learner	Hyperparameter	Search range and distribution
LightGBM	`learning_rate`	Log-uniform $[0.01, 0.2]$
	`num_leaves`	Integer uniform $[7, 63]$
	`min_data_in_leaf`	Integer uniform $[10, 200]$
	`feature_fraction`	Uniform $[0.4, 1.0]$
	`lambda_l2`	Log-uniform $[10^{-3}, 10.0]$
	`n_estimators`	Integer uniform $[100, 800]$
Logistic Regression	Inverse regularization C	Log-uniform $[10^{-3}, 10.0]$ (L2, L-BFGS, max_iter=3000)
MLP	Hidden units (2 layers)	Categorical $\{32, 64, 128, 256\}$
	L2 penalty α	Log-uniform $[10^{-5}, 10^{-1}]$
	Initial learning rate	Log-uniform $[10^{-4}, 10^{-2}]$ (Adam, early stopping)
Sequence models (GRU, TCN, Trans.)	Hidden dimension	Categorical $\{16, 32, 64\}$ (lookback window $W = 10$)
	Learning rate	Log-uniform $[3 \times 10^{-4}, 3 \times 10^{-3}]$
	Dropout	Uniform $[0.0, 0.4]$
	Weight decay	Log-uniform $[10^{-6}, 10^{-3}]$ (AdamW, 25 epochs)

treated descriptively rather than as formal null-hypothesis rejections, and the reader is given the full table and the intervals instead. Conversely, for formal statistical goodness-of-fit hypothesis testing — specifically the Kolmogorov–Smirnov residual tests evaluating the self-exciting Hawkes process across campaigns in Section 5.10 — multiplicity control is strictly enforced via the Holm–Bonferroni procedure across the four campaigns. Every comparative claim in Chapter 5 is stated as a paired difference with an interval rather than as an ungrounded ordering of point estimates.

4.8 Calibration

Ranking quality is insufficient for the task stated in Section 1.3, which requires the asserted probability to correspond to an empirical frequency. Three calibration treatments are evaluated.

Temperature scaling [33] fits a single scalar on a held-out partition, dividing the logits before the sigmoid. It is the least expressive treatment and cannot alter the ranking.

Isotonic regression [34] fits a non-decreasing step function per horizon. It is more expressive and can correct a wider class of miscalibration, but it consumes a held-out split and, applied per horizon, is free to reorder the horizons relative to one another.

The monotone projection repairs an incoherent set of horizon outputs by projecting them

onto the monotone cone. It is implemented for the control arm only, since the hazard formulation makes it unnecessary.

Calibration quality is reported as the expected calibration error, computed as the weighted mean absolute difference between the predicted probability and the empirical frequency across equal-mass bins, with the Brier score reported alongside it and converted to a Brier skill score against a constant forecast at the prevalence, $\mathrm{BSS} = 1 - \mathrm{BS}/[p(1-p)]$. ECE is an average over bins; with equal-mass bins on a rare event, few bins lie in the high-probability region, so a small ECE does not by itself show that high asserted probabilities are reliable.

4.9 Evaluation protocol

Motivation. The protocol is the component most likely to invalidate a result on data of this kind, for two reasons that are usually conflated. Adjacent one-second samples within a session are near-duplicates, so a random row split places nearly the same row on both sides of it. But 180-second blocks of one continuous session are not independent either: they share cells, traffic, weather, device state and driver. Section 5.3 measures both.

Design. The evaluation protocol has five elements, and a ladder of four splitting rules is reported rather than one.

Splitting ladder. Four splitting rules are compared: leave-one-campaign-out, which holds out a whole continuous session; blocked cross-validation with a 60-second purge following Roberts et al. [35]; random 180-second blocks of one campaign, which is the protocol this work used before the audit; and random rows, which is the protocol most of the comparator literature uses. Section 5.3 reports all four.

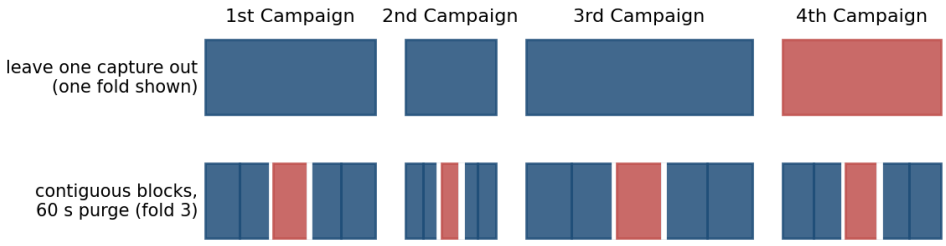


Figure 4.2: The splitting ladder. Leave-one-campaign-out holds out a whole continuous session; blocked splitting holds out time blocks; random-row splitting holds out individual samples, allowing near-duplicate rows across splits.

Repetition and uncertainty. Stochastic learners are run over three seeds and their predictions averaged. Intervals on pooled out-of-fold metrics come from a bootstrap that resamples whole 180-second blocks within each campaign; that interval is conditional on these four sessions and does not include between-day variation, so the per-campaign spread is always reported beside it. The 180 s block length was inherited from the segmentation of Section 3.4 rather than chosen to match the dependence of the data, and the evidence available suggests

it is too short: random blocks of 240 s and 300 s still inflate one-second AUPRC by 8.2% and 10.0% relative to campaign hold-out in the single-seed sensitivity runs (Table A.5), so dependence extends beyond one block. These intervals are therefore labelled within-session block-bootstrap intervals throughout and treated as descriptive and anti-conservative; they are not confidence intervals for a new session. The per-campaign values are an observed range over four units, not a prediction interval for a fifth. Comparisons between learners are reported as per-campaign differences over four paired observations, with the exact sign-test p-value quoted only to show its own resolution limit (0.125 at best). No significance claim is made from four units.

Fold hygiene. The feature scaler, the imputation values, the hyper-parameters, the isotonic calibrators of Section 5.4 and the 5% false-positive decision threshold of Appendix A are fitted inside the training or internal validation partition of each fold only, and the validation partition is separated from the training partition by the same 60-second purge. (That threshold is set on the validation block using a default-parameter LightGBM fitted on the fold's training rows, and applied to the tuned predictions, so the realised false-positive rate on the test campaign is close to, not exactly, 5%.) The conformal threshold of Section 4.10 is calibrated differently: on out-of-fold predictions of the leave-one-campaign-out models, either across pooled blocks or on a whole held-out campaign, as that section describes.

Metrics. Four families are reported. Ranking quality is reported as AUPRC, always with the prevalence floor and the lift over it, together with AUROC. Calibration is reported as expected calibration error over fifteen equal-mass bins and Brier score. Coherence is reported as the fraction of rows on which the horizon ordering is violated, with the magnitude of the violations. Event-level cost is reported as the fraction of handover commands detected, the number of false-alarm episodes per hour and per kilometre, and the distribution of lead times. A command at τ is detected at horizon h if some usable row in $[\tau - h, \tau)$ alarms, and its lead time is τ minus the first alarming row in that window. An alarm episode is a maximal run of alarming rows one second apart; it is a false-alarm episode if no handover command falls in (run start, run end $+ h$]. No cooldown is applied, so a run of alarms ahead of a handover counts once. Reporting all four families is deliberate: Wagner et al. [30] show that no single aggregate metric for event detection satisfies all desirable properties.

4.10 Distribution-free risk control

Motivation. A probability is not yet an operating point. An operator needs a decision rule: issue an alarm when the predicted probability exceeds a threshold. A conventional threshold minimizes a cost or targets a point on a receiver operating characteristic, but offers no guarantee on future performance. Conformal risk control [21], which extends the risk-controlling prediction sets of Bates et al. [22] from a coverage guarantee to a guarantee on any bounded monotone loss, ensures that the expected risk on a new test unit does not exceed a chosen level α — provided the calibration and test units are exchangeable and the loss is bounded

and monotone in the threshold. Whether that proviso holds is the empirical question of Section 5.5; observed losses falling below α on a few held-out units do not establish it.

The exchangeable unit and the loss. The risk is defined per driving unit, not per sample. For a unit d and decision threshold $\lambda \in [0, 1]$—where an alarm is issued whenever $\hat{p}_t \geq \lambda$—the row-level miss loss is the fraction of that unit's positive rows falling below the alarm threshold,

$$R_d(\lambda) = \frac{|\{t \in d : y_t = 1 \text{ and } \hat{p}_t < \lambda\}|}{\max(|\{t \in d : y_t = 1\}|, 1)} \tag{4.9}$$

which is bounded in $[0, 1]$ with $B = 1$. As the decision threshold λ increases, the alarm rule becomes stricter, alarms become rarer, and the miss loss $R_d(\lambda)$ is non-decreasing in λ. Under Theorem 1 of Angelopoulos et al. [21], expected risk control $\mathbb{E}[R_{n+1}(\hat{\lambda})] \leq \alpha$ is achieved by selecting the most conservative (largest) threshold whose finite-sample empirical bound satisfies the target:

$$\hat{\lambda} = \sup \left\{ \lambda \in [0, 1] : \frac{n\hat{R}(\lambda) + B}{n + 1} \leq \alpha \right\} \tag{4.10}$$

with $\lambda = 0$ as the fallback if no threshold satisfies the inequality. For the row-level loss that fallback achieves zero miss loss by alarming on every row, so $\inf_\lambda R_d(\lambda) = R_d(0) = 0$, as Theorem 1 of [21] requires. This is a statement about rows, and it is not the statement Section 1.3 promises. A network operator misses a *handover*, not a row, and a handover occupies several rows of its lead window. The event-level loss is therefore also carried through: writing E_d for the handover commands whose lead window lies in unit d, a command at τ is caught at horizon h if some usable row in $[\tau - h, \tau)$ alarms, and

$$R_d^{\text{ev}}(\lambda) = \frac{|\{\tau \in E_d : \max_{t \in [\tau - h, \tau)} \hat{p}_t < \lambda\}|}{\max(|E_d|, 1)} \tag{4.11}$$

where the maximum runs over the usable rows of unit d in the window and is taken to be $-\infty$ when there are none. A command with no usable row in its window is therefore counted as missed at every threshold rather than dropped from the denominator, which is what makes this loss comparable with the detection rate of Appendix A rather than with the row-level miss rate. It also means that the fallback of Equation (4.10) does not reach zero for this loss: $R_d^{\text{ev}}(0)$ equals the share of unresolvable commands in unit d, so $\inf_\lambda R_d^{\text{ev}}(\lambda) > 0$ whenever a unit contains one, the boundary condition of Theorem 1 of [21] fails, and no threshold satisfies the empirical bound for targets $\alpha < (n\hat{R}^{\text{ev}}(0) + 1)/(n + 1)$ unless the blanking policy is changed. Section 5.5 reports both, because they differ and because the difference is the interesting part.

$\hat{R}(\lambda)$ averages the chosen loss over calibration units, unweighted, so for the event-level loss it is the mean of per-block miss fractions rather than the pooled fraction of missed commands; a block with no command contributes nothing to the event-level average. Each com-

mand is assigned to the block of the last usable row before it (or, if it has none, of the next row), and its lead window uses all usable rows of its campaign in the evaluated set, so a window is not cut at a block boundary within a campaign. The share of unresolvable commands rises from 35.8% to 36.4% in the chunk-split folds because rows removed at the edges of the calibration and test sets leave some windows empty. The choice of unit is the substantive part of the claim, and it is tested rather than asserted. Section 5.5 reports the procedure twice: once in an exchangeable reference regime over the pooled 57 blocks, and once across campaigns.

Crucially, the observations across the 57 blocks are populated from *out-of-fold predictions* generated by the four leave-one-campaign-out models: for each campaign $c \in \{1, 2, 3, 4\}$, the predictor was fitted and tuned on the remaining three campaigns, producing predictions on campaign c that are strictly out-of-fold. In the exchangeable pooled experiment, the 57 blocks carrying these out-of-fold predictions are repeatedly split, 200 times, into calibration ($n = 28$) and test units ($n = 29$) by a random half-and-half partition. Because every prediction was generated by an estimator that never observed that campaign during fitting or hyperparameter tuning, the calibration-test split over out-of-fold predictions evaluates threshold calibration without data leakage. In contrast, cross-campaign deployment evaluates across distinct physical corridors where exchangeability fails under distribution shift.

Estimand and regime. Two qualifications fix what these experiments can show. First, the pooled-block regime answers a finite-pool question: if a block is drawn from the same 57 observed blocks, does a threshold calibrated on the others control its expected row-level miss loss? It does not establish validity for a new physical session, because the blocks of one campaign are dependent and a new session is not drawn from this pool. Second, the pooled out-of-fold scores come from four different fitted models, one per held-out campaign. Calibrating a single threshold over them therefore calibrates the procedure "fit on three campaigns, score the fourth" rather than any one deployed predictor; a predictor refitted on all four campaigns would need a calibration set of its own, which this corpus cannot supply without giving up a campaign. For these reasons every conformal result in Section 5.5 is reported as an empirical calibration diagnostic of that procedure, and no operating guarantee for a deployed predictor is claimed.

$$\alpha \geq \frac{1}{n+1} \iff n \geq \frac{1}{\alpha} - 1 \tag{4.12}$$

Equation (4.12) gives the finite-sample feasibility condition for the empirical bound of [21]: for the term $(n\hat{R}(\lambda) + B)/(n + 1)$ to reach α when empirical loss is zero ($\hat{R} = 0$), the target must satisfy $\alpha \geq 1/(n + 1)$. This is not an impossibility theorem for risk control itself (since the trivial fallback $\lambda = 0$ guarantees zero loss by alarming on all eligible rows), but rather the sample-size floor required for non-trivial threshold selection: a study wishing to promise a non-trivial 5% miss rate without defaulting to continuous alarm needs at least 19 calibration units, and one promising 2% needs 49. Because the four campaigns contain

15, 8, 20, and 14 blocks (mean $n = 14.25$), the calibration size in cross-campaign rotations is fold-specific ($n \in \{8, 14, 15, 20\}$), yielding feasibility floors from 0.048 to 0.111. Only rotations calibrating on the 20-block campaign can express $\alpha = 0.05$ non-trivially.

A second floor binds the event-level loss under the evaluation protocol, arising from pre-processing and blanking rules rather than from calibration sample size. Under the evaluation policy, a 2 s post-handover blanking window is enforced to suppress post-transition transients. A command that follows another within 2 s has no eligible prediction rows in its 1-second lead window $[\tau - 1, \tau)$ and cannot be alarmed. Alarming on every eligible row ($\lambda = 0$) cannot recover events that possess no eligible rows under this policy; continuous alarm only bypasses that floor if the blanking policy itself is abandoned. Across the corpus, 36.4% of one-second lead windows are in that position (against 35.8% in the unbroken sequence, and 12.0% at three seconds). Because the burst density varies across corridors, per-campaign floors at one second range from 31% on the dense-urban campaign to 47% on the urban loop. These floors belong to the grid together with the blanking policy, not to every one-hertz forecaster: a shorter blanking window lowers them, at the price of scoring rows immediately after a handover.

4.11 Point-process analysis of handover arrivals

Motivation. The ping-pong behaviour of Section 5.10 suggests that handovers do not arrive independently. Quantifying that clustering requires a model of the arrival process itself rather than of the per-sample probability.

Design. The handover arrival stream of each campaign is modelled as a self-exciting Hawkes process [23], [24] with conditional intensity

$$\mu(t) = \mu_0 + \sum_{t_i < t} \alpha \cdot \exp(-\beta(t - t_i)) \tag{4.13}$$

in which each arrival raises the intensity of subsequent arrivals by α and the excitation decays at rate β. The branching ratio α/β is the expected number of direct offspring per event under this kernel, and summarises the strength of the clustering. Each campaign is treated as a separate realisation: the excitation sum and the compensator $\Lambda(t) = \int_0^t \mu(u)\, du$ restart at the campaign's first command, so arrivals do not excite across sessions, and the time-rescaling residuals $\Lambda(t_i) - \Lambda(t_{i-1})$ are computed within each campaign ($957 - 4 = 953$ residuals in all). Section 5.10 reports that the Ogata residual test rejects the exponential kernel on this data, so that reading is withdrawn there and the ratio is used only as a comparable summary statistic.

Validation. A fit alone is not evidence of self-excitation. Following the networking precedent of Price-Williams and Heard [26], the fit is validated by the residual analysis of Ogata [25]: the time axis is rescaled by the fitted compensator, and the transformed inter-arrival times are tested against a unit-rate exponential distribution by a Kolmogorov–Smirnov

test on held-out drives. Section 5.10 reports both the branching ratio and the outcome of that test, including the respect in which the test rejects the fitted kernel.

CHAPTER 5

RESULTS AND DISCUSSION

All figures in this chapter are computed out of fold under the leave-one-campaign-out protocol of Section 4.9 unless stated otherwise, from features lagged as described in Section 4.6.

The numbers in this chapter come from the corrected measurement-report parser of Section 3.3, which recovers neighbour entries that carry RSRP without RSRQ. Every stage-22 experiment was re-run on the frozen four campaigns with the corrected parser: the main leave-one-campaign-out comparison (Tables 5.4, 5.5 and 5.10), the lag ablation (Table 5.2), coherence (Tables 5.7 and 5.8), cross-campaign risk control (Table 5.9), single-feature AUROC (Table 5.12), the burst strata (Table 5.13), the event-level operating points (Table A.2), the column-subset decomposition (Table 5.3), the splitting ladder (Table 5.6), the exchangeable-pool risk control (Table 5.9) and the window sensitivities (Table A.5). The three learners that do not use neighbour features (dwell, history, Hawkes) reproduce their earlier values exactly, so every change is attributable to the parser. Every model table in this chapter and in Appendix A therefore comes from one consistent run. Signalling-only quantities (A3 conversion, ping-pong, Hawkes, the alignment audit) do not depend on neighbour parsing and are unchanged. Section 5.12 adds a blind test on a capture from a second operator.

5.1 The alignment audit and what it costs

The first result of this chapter is not a prediction score. It is a property of the measurement export that determines what any prediction score on this data means.

The audit is simple. For each handover command at time τ, take the export row whose second contains τ — that is, the row stamped at $\lfloor \tau \rfloor$, strictly before the command — and ask whether its serving-cell identity is already the target of that command. If the row summarises the second before its timestamp, the answer should be no more often than chance.

Table 5.1: The alignment audit: share of handovers whose target cell already serves in export rows around the command.

Handovers considered	**n**	**row** $t-3$	**row** $t-2$	**row** $t-1$	**row** $t+0$	**row** $t+1$
all handovers	938	23.7%	21.7%	15.6%	75.9%	86.2%
isolated (previous > 8 s earlier)	291	12.1%	12.1%	11.7%	74.6%	89.7%

Table 5.1 also bears on whether the one-row lag leaves residual contamination, and it shows none. Leakage from the aggregation rule would appear as a step at the last row before the command. Instead, the share of rows already showing the target cell falls from row $t-3$ to row $t-1$ in both subsets: from 23.7% to 15.6% across all handovers, and from 12.1% to

11.7% on isolated ones. It then jumps only at row $t+0$. The higher pooled level is a baseline effect rather than a lag effect: with 26.3% of commands being ping-pong returns within 15 s and a median inter-command gap of 3.7 s, the next target cell has often served within the last few seconds. The model-side check points the same way. On quiet rows (no command in the previous 10 s), whose 238 positives all precede a command with no other command in the preceding 10 s, LightGBM reaches AUPRC 0.129 against a 4.3% floor (3.0× lift, AUROC 0.770). On in-burst rows it reaches 0.198 against 11.2% (1.8×, AUROC 0.714) (Section 5.8, Table 5.13). The ordering holds for burst cutoffs from 5 to 20 s (Table A.5). If residual burst contamination were carrying the result, lift would be highest inside bursts. It is lowest there, which is inconsistent with residual leakage driving the pooled 0.169.

Table 5.2: What the one-row lag costs, with everything else held fixed. Both rows use the same fixed hyper-parameters (ARM B), folds, seeds and learner, so their levels differ from the tuned arm of Table 5.4. Corrected-parser run.

Feature alignment	AUPRC 1 s	AUROC 1 s	AUPRC 5 s	Dwell feature AUROC 1 s
Export row stamped t used at t (v1)	0.604	0.910	0.560	0.873
Export row lagged one sample (this work)	0.156	0.744	0.435	0.664
Change	-74%	-0.166	-22%	-0.209

Table 5.3: Where the leak actually lives. One common row set of usable rows, identical folds, seeds and learner; only the set of columns shifted by one row changes. The serving-cell radio scalars are measured on whichever cell the export treats as serving, so after the end-of-second flip they describe the target rather than the source. Corrected-parser run; the first and last rows reproduce Table 5.2 to within 0.001.

What is lagged by one row	Columns	AUPRC 1 s	AUROC 1 s	AUPRC 5 s
nothing (v1 alignment)	0	0.604	0.911	0.556
neighbour and gap columns only	31	0.601	0.905	0.555
serving-cell radio scalars only	56	0.182	0.772	0.443
every serving-relative column (GPS left at row t)	92	0.156	0.743	0.435
every export column (v2, as reported throughout)	109	0.157	0.744	0.435
control: Event A3 rule, same arms	-	0.114	-	-

The consequence for prediction is immediate. A model asked at time t for the probability of a handover in $(t, t+1]$ and given the row stamped t is being shown, most of the time, the answer. Table 5.2 quantifies it on the same model, the same folds, the same hyper-parameters and the same features: the one-second AUPRC falls from 0.604 to 0.156 and AUROC from 0.910 to 0.744 when every export feature is lagged one row, a 3.9-fold inflation. At five seconds, where one second of misalignment matters less, the same change costs 0.560 to 0.435. (An earlier draft set the unlagged fixed-parameter arm against the tuned lagged arm; the comparison is now like for like.) Section 5.12 repeats the measurement blind on a second operator and finds a 4.3-fold inflation. The single-feature effect is starker: serving dwell time computed from the export's serving-cell column reaches AUROC 0.873 on its own, and the

same quantity computed from signalling timestamps reaches 0.664.

Three points are worth stating plainly. First, this is not a defect of the instrument in the sense of a fault: a one-hertz export has to summarise a second, and a row that carries the state at the end of its second is a defensible choice. It is a defect of any analysis that assumes the opposite without checking. Second, the error is invisible to every check that does not involve the event clock — the features are causal in appearance, no future label is used, and grouped splitting does not remove it, because the leak is inside the row rather than between rows. In the taxonomy of Kaufman et al. [36] it is leakage in training examples rather than leakage in feature construction: no illegitimate feature was built, and no future row was consulted; the legitimate feature simply describes an interval that already contains the event. That class is the one their paper identifies as hardest to catch, because every syntactic check passes. Kapoor and Narayanan [56] show how often leakage of this general kind propagates into published machine-learning-based science; what is specific here is the mechanism — a row that summarises an interval ending after its own timestamp — and the fact that it can be detected against an event clock. Third, it is cheap to detect, and the audit above is the recommendation this thesis makes to anyone building a predictor on a drive-test export.

Why the row behaves this way is not a mystery, and stating the reason turns the observation into a prediction that can be falsified. A handover command is issued at an arbitrary instant τ inside the second $[t, t+1)$. Write $\delta = \tau - \lfloor \tau \rfloor$ for where inside the second the command falls, x for the empirical diagnostic timestamp interval between the command arrival (`RRCConnectionReconfiguration`) and UE completion (`RRCConnectionReconfigurationComplete`), and

$$m = 1 - \delta - x, \quad \text{with } \delta, x \text{ and } m \text{ all in seconds} \tag{5.1}$$

for the margin by which that change beats the close of the second. Crucially, x represents an empirical protocol proxy derived from diagnostic completion timestamps in the decoded baseband log; it should not be conflated with the modem's internal serving-cell register commit, nor with the physical user-plane interruption time (which 3GPP TS 36.133 defines across physical-layer synchronisation and which diagnostic drive-test traces cannot independently establish). Baseband logging and export queuing latency inherently bounds the precision of this proxy. If the exporter writes one row per second carrying the state as of the end of the second, the row stamped t shows the target cell when $m > 0$ and the source cell when $m < 0$. Nothing about the network enters: the defect is what end-of-second aggregation does to an event whose execution is two orders of magnitude faster than the grid.

On our own captures, decoded signalling records protocol completion timestamps for each handover (65% of commands record a non-zero diagnostic completion interval; the ninetieth percentile is 99 ms and the largest is 105 ms). An in-sample diagnostic check on the 15 handovers whose diagnostic completion interval crosses the second boundary ($m < 0$) observes

zero contamination on the row stamped before the command (0/15; 95% Wilson score interval [0.0%, 20.4%]). The sample is small, so contamination up to 20.4% remains compatible with it; the 4,712-handover NUWiNS corpus below shows that end-of-second aggregation is sufficient to produce the effect, but not that this exporter applies that rule. Of the 923 handovers with $m > 0$, the row shows the target cell 77.1% of the time. The same gradient appears in δ directly: on handovers isolated from any other by two seconds, contamination runs at 86.1% for commands in the first quarter of the second and falls to 67.0% for commands in the last quarter, where the margin is smallest. Panels (d) and (e) of Figure 5.1 show both.

The residual matters for honesty about the size of the effect. In 22.5% of handovers the margin is positive and the row stamped t nevertheless shows the source cell; in 84% of those the row stamped $t + 1$ carries the target instead. That pattern is consistent with the exporter refreshing the serving-cell field one row late rather than with a second mechanism, and Table A.6 resolves the residual one handover at a time. Of the 211 handovers whose margin is positive and whose row t nevertheless shows the source, 178 — 84% — have the target cell in row $t + 1$, arriving exactly one row late, and Table A.6 lists the ten residual handovers with the smallest margins, unfiltered, with the cell identity in each row so the pattern can be read rather than taken on trust: nine of the ten show the target in row $t + 1$, and one (the fourth) moves to a third cell instead, as 16% of the residual do. The competing explanation, that the residual is simply the small-margin cases where the geometry was marginal, does not survive the same table: the median margin of the residual is 0.477 s against 0.522 s for the contaminated handovers, which is no separation at all. Our export therefore shows an attenuated form of the defect, and the uncontaminated figure for it is not zero but about 24%.

Two independent datasets test the account. The first is the control. The 4G/5G Irish drive-test traces of Raca et al. [37] are 59 driving traces logged at one hertz with a different tool, G-NetTrack Pro, on a different operator. That tool samples the modem rather than aggregating a second, so the account predicts no contamination at the level an aggregate produces, and there is none at that level: 29.4% of 1,175 cell changes have row t looking like the new cell, against 75.9% of handovers in our export and 93.2% under an end-of-second rule. That figure is not zero, and it is not claimed to be: with a ping-pong rate above a quarter the cell serving three rows earlier is often the cell about to serve again, so a sampled log has an elevated background for reasons unrelated to aggregation. The discriminating quantity is the shape rather than the level. The prediction test agrees. Using the row stamped t rather than the row before it is worth +38% AUPRC on that dataset, but the step from $t - 2$ to $t - 1$ is worth +57%, a larger gain. A contaminated final row shows up as an anomalous jump at the last step; a clean log shows the smooth decay with staleness that this dataset has.

The second dataset does not replicate the defect on a third instrument; it does something narrower and more useful, by letting the aggregation rule be set rather than inferred. The NUWiNS multi-carrier corpus [40], collected with XCAL on two cross-United-States routes over three tier-one operators and released as raw exports at ten hertz [41], carries the han-

dover event clock and the serving-cell register at native rate. Downsampling it to one hertz by keeping the last sample of each second reproduces the defect: over 4,712 cell-changing handovers in 12 files and 3 operators, the row stamped t carries the target cell 93.2% of the time (92.8% on the 3,160 isolated handovers), and the row stamped $t-1$ carries it 0.1% of the time. Keeping the first sample of each second instead collapses contamination to 4.1% on the same handovers. One rule produces the defect and the other removes it, on data the network never saw differently. What this establishes is that end-of-second aggregation is sufficient to produce the artefact at the magnitude observed, and how large it is when present. It does not establish that XCAL's own one-hertz exporter aggregates that way, which remains inferred from its consequences and which only a native-rate re-export of these captures would settle; Section 5.14 carries that as a residual risk.

The tail of the second then measures something the one-hertz grid was never supposed to resolve. Within a bin of δ, the row stamped t is contaminated precisely when the serving-cell register updates before the second closes, so the contamination share in that bin estimates the probability that the update delay is shorter than the time remaining. Over 903 handovers arriving in the last fifth of a second, contamination is 98.2% when 75 to 100 ms remain, 43.9% when 50 to 75 ms remain and 0% when less than 50 ms remain. Under the proposed aggregation model, the cliff places an inferred logging characteristic at roughly 60 ms, which is consistent with the native ten-hertz view of the same events, where the target is already serving at $\tau + 100$ ms in 100% of handovers. It should not be read as a measured latency of any one component: the native sampling phase, timestamp rounding, diagnostic buffering and exporter commit behaviour could each contribute, and a one-hertz export cannot separate them. Panel (f) of Figure 5.1 shows the curve. Read with that caution, the alignment defect is not only a hazard to be removed; it also gives an indirect estimate of a sub-second logging characteristic from a one-hertz export.

Three conclusions follow. The alignment error is consistent with a property of the aggregation rule rather than of the network: end-of-second downsampling reproduces it on native-rate XCAL data from three further operators on another continent, start-of-second downsampling removes it, and a tool that samples instead of aggregating shows no aggregation step. That this thesis's own exporter applies end-of-second aggregation is the most economical explanation of what is observed, not an observed fact. It is predictable, which means it can be audited in a few lines against any event clock, and the audit reports a mechanism rather than an anomaly. And it is consequential, because none of the 22 models audited in Section 2.3 reports such a check, and neither did the first version of this work.

One objection has to be answered before the repair can be evaluated, because it affects how the audit is interpreted. Lagging every export column removes the contaminated serving-cell assignment, but it also discards one second of radio and kinematic history that the aggregation rule cannot plausibly corrupt. On that reading, the fall from 0.604 to 0.157 (Table 5.3) reflects both the elimination of leaked future information and the penalty of one-second feature ageing.

Table 5.3 separates these components across feature subsets. The four arms share one row set, one set of folds, one learner and one set of seeds; only the set of columns shifted by one row changes.

The reduction is concentrated in the serving-cell radio scalars. Lagging the 31 neighbour and gap columns alone—the A3 condition itself, and the columns any analyst would suspect first—leaves the one-second AUPRC at 0.601, which is under 1% of the total fall. Lagging the 56 serving-relative radio columns alone takes it to 0.182, accounting for 94.4% of the reduction. And lagging the 17 mobility features as well—the GPS speed, acceleration and bearing that no aggregation rule can contaminate—shifts performance by less than 0.1%. Within the serving radio block, the removal of target-cell identity and the one-second ageing of signal measurements are confounded on a one-hertz grid; separating them cleanly would require causally timestamped native-rate logging. The one-row lag is therefore a conservative safeguard against look-ahead leakage under assumed export semantics, eliminating post-handover assignment contamination at the price of feature ageing by up to one sample period.

Why the radio scalars and not the cell identity is worth stating, because it is the part that transfers. A column named serving_rsrp is not a property of a cell; it is the reference signal power of whichever cell the export currently treats as serving. When the assignment flips inside the second, that column silently changes its referent and reports the target's power, which is by construction higher than the source's — an ordinary-looking radio scalar that is in fact the answer. The same is true of the RSRQ, SINR and CQI families and of every moving average and slope derived from them. The neighbour gap moves too, but it was already near its trigger threshold, so the flip adds little to it. The practical recommendation is therefore narrower and more useful than lag everything: on any export whose rows are aggregates, the columns to distrust are the ones defined relative to the serving-cell assignment, and they are not the ones that look like identifiers.

Everything reported after this section uses the lagged features. The unlagged numbers are retained in Table 5.2 and in the ablation of Section 5.11 as a measurement of the effect, never as a result. Table 5.3 is reported under fixed hyper-parameters so that the five arms are comparable with one another; its levels are not comparable with Table 5.4, which reports the tuned arm.

Figure 5.2 leaves the sample level entirely and plots the quantity an operator pays: at each horizon, the share of handover commands caught at the 5% false-positive operating point, the share the one-hertz grid can resolve at all, and the false-alarm episodes raised per hour of driving.

5.2 The primary prediction result

For predicting a handover within the next second (horizon $h_1 = 1$ s, with an empirical median lead time of approximately 0.50 s before the command), the model reaches pooled AUPRC 0.169 [within-session block-bootstrap 95% interval 0.150, 0.196; observed range over held-



Figure 5.1: The alignment audit and its mechanism. (a) A handover command at τ inside the second $[t, t+1)$. (b) The audit: share of handovers whose target cell already serves in the rows around the command, this work. (c) Aggregated rows against sampled rows: this work, the Irish G-NetTrack traces and NUWiNS under an end-of-second rule. (d) Contamination against the execution margin m. (e) Contamination against the command's position δ in the second, in both XCAL corpora. (f) The contamination cliff near the end of the second, from which a logging characteristic of roughly 60 ms is inferred (not a measured latency).

out campaigns 0.125–0.223] against a prevalence floor of 6.8%, a lift of 2.5x, at AUROC 0.773 and an expected calibration error of 0.014. At five seconds it reaches AUPRC 0.454 [within-session interval 0.411, 0.509; observed range 0.385–0.531] against a 26.1% floor, a lift of 1.7x (median lead time 3.36 s). An A3-derived baseline—defined by evaluating the continuous A3 signal margin (neighbour RSRP minus serving RSRP, offset, and hysteresis) at the 1 Hz sample instants—scores AUPRC 0.114 at one second (1.7x) and AUROC 0.685. Because 1 Hz sampled exports cannot reconstruct the UE's sub-second time-to-trigger evaluation (160–320 ms on this network), this baseline evaluates an instantaneous signal-margin proxy; it models neither the UE-side reporting procedure nor the network's handover decision that follows a report. Checking persistence across two consecutive rows would test the condition at two instants 1,000 ms apart, which neither establishes continuous sub-second persistence nor corresponds to the configured TTT. Evaluating whether the margin holds at

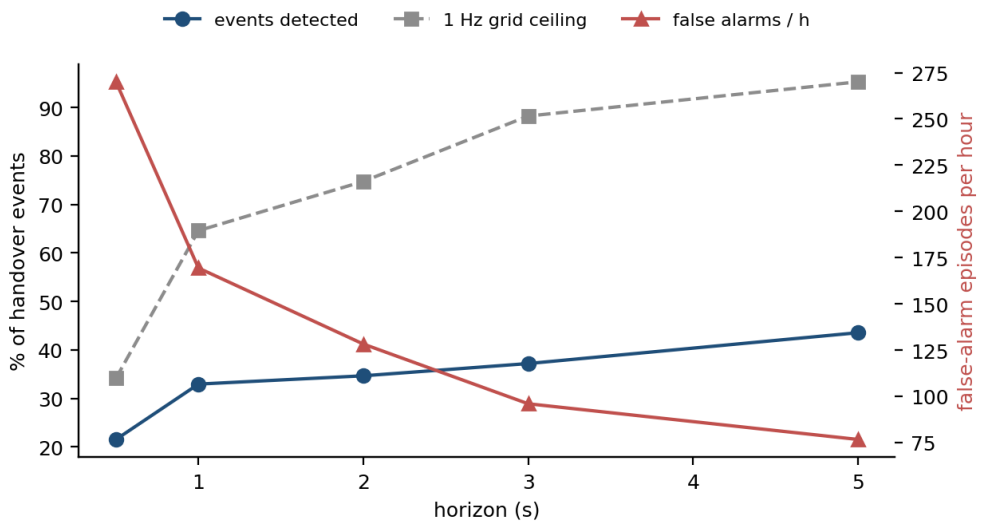


Figure 5.2: Event-level cost of the warning at the 5% false-positive operating point, pooled over the four held-out campaigns: share of handover commands detected, share resolvable on the one-hertz grid, and false-alarm episodes per hour, by horizon. Lead times and per-kilometre rates are in Table A.2.

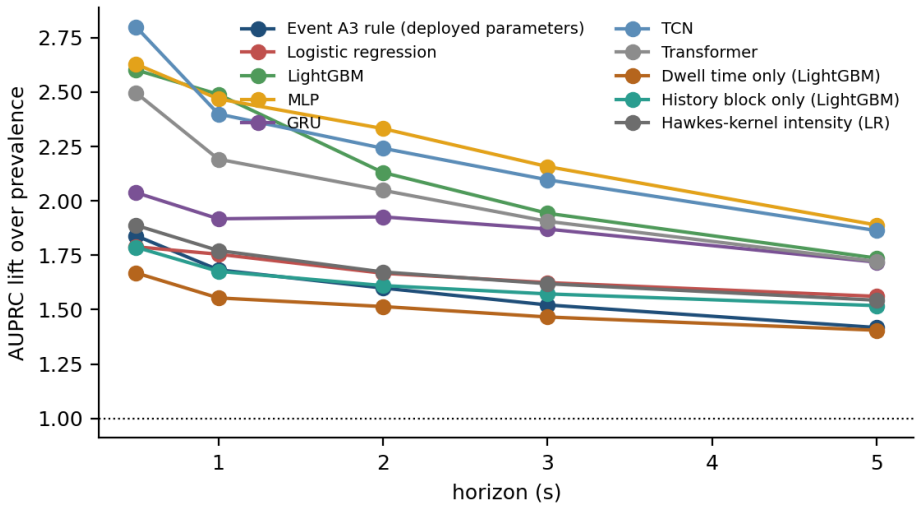


Figure 5.3: AUPRC lift over the prevalence floor by horizon across learners, out of fold under leave-one-campaign-out. LightGBM, MLP, and TCN achieve comparable performance within overlapping bootstrap intervals (pooled 1 s AUPRC 0.169, 0.168, and 0.163 respectively, corrected parser), all above the prevalence floor and the A3 signal-margin proxy. The comparison is descriptive: no significance test is made over four campaigns.

the current sample row is therefore the representation adopted, and its limitation as an instantaneous proxy is explicitly acknowledged. On these evaluated rows, the learned multi-horizon hazard model outperforms this specified A3-derived baseline, while scoring materially below its pre-audit unlagged level; both observations belong in the same paragraph.

Table 5.4: Out-of-fold performance of LightGBM under the hazard formulation, leave-one-campaign-out. ARM A (tuned): nested random search of twenty trials per outer fold, averaged over three seeds. Intervals are within-session 180 s block-bootstrap percentile intervals, conditional on these four sessions and anti-conservative (Section 4.9); they are not confidence intervals for a new session. Observed range over the four held-out campaigns: [0.125, 0.223] at 1 s and [0.385, 0.531] at 5 s. Corrected-parser run. BSS: Brier skill against a constant forecast at the pooled prevalence, $1 - \mathrm{BS}/[p(1-p)]$, computed from the rounded Brier scores.

Horizon	Prevalence	AUPRC [95% within-session interval]	Lift	AUROC	ECE	Brier	BSS
0.5 s	0.037	0.097 [0.079, 0.126]	2.6×	0.776	0.008	0.035	2%
1 s	0.068	0.169 [0.150, 0.196]	2.5×	0.773	0.014	0.060	5%
2 s	0.126	0.268 [0.240, 0.304]	2.1×	0.755	0.024	0.103	6%
3 s	0.176	0.342 [0.308, 0.387]	1.9×	0.744	0.035	0.135	7%
5 s	0.261	0.454 [0.411, 0.509]	1.7×	0.734	0.052	0.176	9%

Table 5.5: Every learner under identical folds, features, formulation and tuning budget. ARM A (tuned): nested random search of twenty trials per outer fold, averaged over three seeds. Corrected-parser run; the dwell, history and Hawkes arms use no neighbour features and are unchanged from the original run.

Learner	AUPRC 1 s [95% within-session interval]	Lift 1 s	AUROC 1 s	AUPRC 5 s	Lift 5 s
LightGBM	0.169 [0.150, 0.196]	2.5×	0.773	0.454	1.7×
MLP	0.168 [0.149, 0.192]	2.5×	0.762	0.493	1.9×
TCN	0.163 [0.145, 0.189]	2.4×	0.756	0.487	1.9×
Transformer	0.149 [0.129, 0.179]	2.2×	0.729	0.450	1.7×
GRU	0.130 [0.117, 0.149]	1.9×	0.702	0.449	1.7×
Hawkes-kernel intensity (LR)	0.120 [0.103, 0.138]	1.8×	0.666	0.403	1.5×
Logistic regression	0.119 [0.103, 0.143]	1.8×	0.624	0.408	1.6×
Event A3 rule (deployed parameters)	0.114 [0.101, 0.132]	1.7×	0.685	0.370	1.4×
History block only (LightGBM)	0.114 [0.098, 0.134]	1.7×	0.652	0.397	1.5×
Dwell time only (LightGBM)	0.105 [0.091, 0.120]	1.6×	0.637	0.367	1.4×
Prevalence floor	0.068	1.0×	0.500	0.261	1.0×

Two features of Table 5.4 require comment. First, the lift falls with the horizon while the AUPRC rises, because the prevalence floor rises faster than the score does; reading AUPRC across rows without the floor is precisely the error Section 2.3 documents in the comparator literature. Second, the calibration error rises with the horizon, so the five-second output should be read as a ranking rather than as a probability. Against the original-parser run, every neighbour-using learner moved down by 0.000 to 0.011 at one second; the ordering at one second is unchanged.

The interval on each AUPRC in Table 5.4 comes from a block bootstrap over 180-second blocks and is conditional on these four sessions. It measures variation within the corridors driven, not variation between corridors; because Section 5.3 demonstrates that contiguous

180 s blocks within a drive exhibit persistent temporal autocorrelation, these within-session bootstrap intervals are descriptive and anti-conservative. Across the four physical sessions, the values observed on held-out campaigns range over [0.125, 0.223] at 1 s and [0.385, 0.531] at 5 s (Table 5.10). The blind second-operator capture of Section 5.12, at 0.137, falls inside the one-second range. That range is the more honest description of between-session variation, but it is an observed range over four units, not a prediction interval: a fifth campaign could fall outside it. Nor is the 180 s block length justified by the dependence it is meant to capture — Table A.5 shows inflation persisting for random blocks of 240–300 s — so the within-session intervals are labelled as such and read descriptively. No interval anywhere in this thesis is used to declare a difference significant.

That applies directly to Table 5.5, which is a leaderboard and should not be read as one. LightGBM is first at 0.169, MLP second at 0.168 and TCN third at 0.163 at one second, and the intervals of MLP, TCN and the Transformer overlap the leader's. At five seconds the order is different: MLP (0.493) and TCN (0.487) lead LightGBM (0.454). The ordering of the top arms is not a finding of this thesis, and none of its conclusions depends on it. Section 5.12 adds a stronger reason for that caution: on a blind capture from a second operator, logistic regression ranks first and LightGBM's lead disappears, so "primary model" names the learner this thesis carries through its analyses on the development operator, not a learner shown to be best. What the table is for is the comparison of levels against the prevalence floor and against the implemented A3 signal-margin proxy, which is a gap of a different order from the gaps between learners. The same caution applies to the tuning: twenty trials over seven learners, four folds and three seeds is a large search relative to roughly six hundred positive rows at one second, and the protocol of Section 4.7 protects the held-out campaign from it without removing the researcher's freedom to have chosen the search in the first place. The alignment audit and the feature lag were themselves decided after all four campaigns were in hand; Section 3.2 records the freeze, and the honest description of everything after it is exploratory development rather than confirmatory evaluation.

5.3 Splitting protocol: what each unit is worth

The result in this section concerns the protocol rather than the predictor, and it is the one with the widest implication for the comparator literature.

The experiment is a controlled substitution. Everything is held fixed — the data, the lagged features, the formulation, the hyper-parameters, the seeds — and only the splitting rule changes, across the four-step ladder of Section 4.9. Table 5.6 reports the result at one second and Figure 5.5 across the ladder.

Three readings follow. Random-row splitting inflates the one-second AUPRC of the primary model by 49% over leave-one-campaign-out, and of the recurrent arm by 25%. Splitting by random 180-second blocks — grouped splitting, and the protocol this work used before the audit — still inflates it by 12%, because blocks of one continuous session are not indepen-

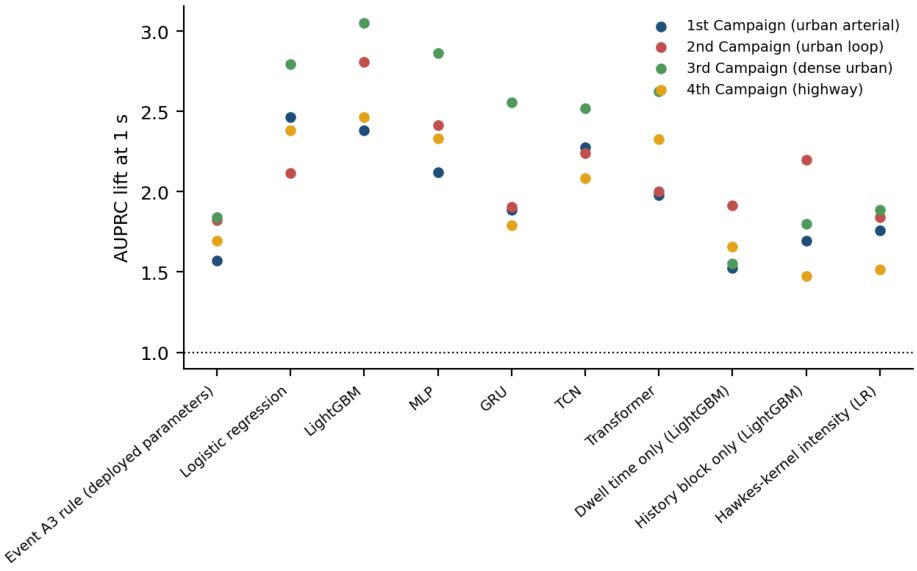


Figure 5.4: One-second AUPRC lift by learner, with each held-out campaign drawn separately. The spread reflects genuine inter-campaign variability in route and speed.

dent of one another. Blocking contiguously in time with a 60-second purge removes about half of that: it sits 6% above the leave-one-campaign-out figure (0.166 vs 0.156 for LightGBM). The single-seed sweep of Table A.5 agrees: across purges of 0 to 180 s the blocked protocol stays 4–11% above campaign hold-out, so a purge narrows the gap without closing it. Blocked splitting with a purge is therefore the better of the within-session protocols when only one session is available, but it remains optimistic. (An earlier, original-parser version of this sweep showed the gap closing at purges of 90 s and longer; with the corrected parser it does not.) However, as Section 5.14 notes, contiguous blocks within the same session still share route geography, vehicle kinematics, and time-of-day channel conditions, so a temporal purge reduces boundary leakage but does not fully eliminate shared intra-session macro-environmental covariates.

Table 5.6: One-second AUPRC under four splitting protocols. ARM B (fixed hyper-parameters): one configuration per learner, held fixed across all four protocols so that the splitting rule is the only thing that varies. Corrected-parser run.

Learner	Leave one campaign out	Blocks + 60 s purge	Random 180 s blocks	Random rows	Inflation, random rows vs LOCO
lr	0.165	0.163	0.168	0.178	+8%
lgbm	0.156	0.166	0.176	0.233	+49%
tcn	0.154	0.146	0.158	0.174	+13%
mlp	0.150	0.151	0.159	0.163	+8%
transformer	0.126	0.144	0.157	0.168	+34%
gru	0.124	0.118	0.131	0.155	+25%
a3	0.114	0.114	0.114	0.114	+0%

The mechanism is straightforward once stated. Two samples one second apart on the same drive are near-duplicates. Two 180-second blocks of the same session are not duplicates, but they share cells, traffic and trajectory, so a model can recognise the session rather than the situation. The implemented A3 signal-margin proxy, which has no learned parameters to

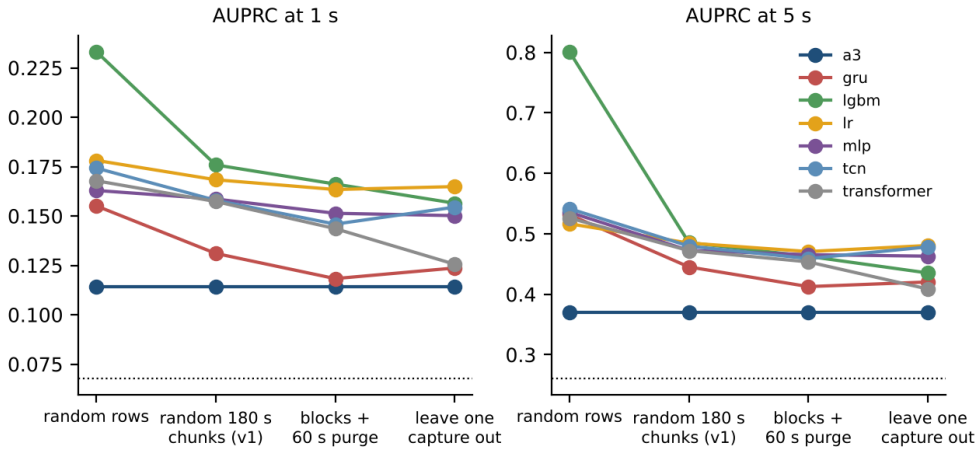


Figure 5.5: The splitting ladder, at one and five seconds (corrected parser). Almost every learner rises from campaign hold-out to random rows, unevenly by architecture; random-row cross-validation inflates LightGBM the most.

over-fit with, moves by +0% across the whole ladder and is the control that shows the effect is over-fitting rather than a difference in test-set difficulty.

The consequence is sharper than the familiar statement that a leaky protocol raises scores: because the inflation is architecture-dependent, it reorders the leaderboard rather than raising it uniformly. Under leave-one-campaign-out with fixed hyper-parameters the leading learner at one second is regularised logistic regression (0.165), the same learner that ranks first on the blind second-operator capture of Section 5.12; under random-row splitting it is LightGBM, whose inflation is 49% (to 0.233) against 8% for the linear model. The sequence models are inflated unevenly too: 34% for the Transformer and 25% for the gated recurrent unit, against 8% for the MLP. At five seconds the reordering is larger still: LightGBM is fourth under campaign hold-out and first under random rows, inflated by 84%, while logistic regression falls from first to sixth. A study using the standard protocol of the comparator literature would therefore report a completely different winner on the same data. This is the answer to objective O2, and it is why the protocol is presented in this thesis as a contribution rather than as housekeeping.

5.4 Coherence and calibration

The hazard formulation produces no horizon-ordering violations, which follows from Equation (4.5). The independent-classifier arm violates ordering on 38.9% of rows, with a mean violation of 0.025 and a maximum of 0.434 where it occurs. Every violation figure in this thesis is the rate of a single fitted system, which is what a deployment would meet; Table 5.7 reports beside it the rate of the three-seed ensemble, which is lower (27.9% for the same arm) because averaging predictions smooths violations away rather than removing their cause.

The honest comparison, however, is not against the raw control. A per-row cumulative maximum removes every violation at no cost, needs no held-out split, and is two lines of code; the pool-adjacent-violators projection does the same. Both are reported in Table 5.7, and an

Table 5.7: Horizon coherence and calibration for the hazard formulation and its four controls. ARM C (coherence): fixed hyper-parameters, with an extra calibration block held out of every fold so the isotonic arms can be fitted. That block is the fold's internal validation block of Section 4.7 (the last 20% of each training campaign, after a 60 s purge); every arm, hazard included, is fitted on the remaining training rows only and is not refitted on the block, which is used solely to fit the isotonic maps. Corrected-parser run.

Arm	Rows violating ordering (one fit)	Rows violating (3-seed ensemble)	Mean violation	Largest violation	ECE 1 s [95% within-session interval]	ECE 5 s	AUPRC 1 s	Needs a held-out split
independent	38.9%	27.9%	0.025	0.434	0.043 [0.038, 0.054]	0.134	0.155	no
independent + cumulative max	0.0%	0.0%	0.000	0.000	0.043 [0.037, 0.053]	0.132	0.156	no
independent + PAV	0.0%	0.0%	0.000	0.000	0.043 [0.038, 0.053]	0.133	0.155	no
independent + isotonic	44.4%	32.1%	0.032	0.581	0.016 [0.013, 0.023]	0.074	0.153	yes
independent + isotonic + PAV	0.0%	0.0%	0.000	0.000	0.015 [0.012, 0.023]	0.075	0.151	yes
hazard	0.0%	0.0%	0.000	0.000	0.037 [0.030, 0.044]	0.113	0.151	no

Table 5.8: Paired differences between the hazard arm and each control at one second. A positive ECE difference favours the control, since a lower calibration error is better. The pooled column is the metric computed on the out-of-fold predictions of all four campaigns concatenated, minus the same for the control; its interval resamples 180 s blocks within campaigns. The per-campaign columns compute the metric on each held-out campaign alone. Both use the three-seed averaged predictions. A campaign is counted as favouring the hazard arm when its unrounded difference is positive for AUPRC or negative for ECE.

Control arm	Metric	Hazard − control (pooled)	Block-bootstrap interval of the pooled difference (descriptive)	1st Campaign	2nd Campaign	3rd Campaign	4th Campaign	Campaigns favouring hazard
independent	AUPRC 1 s	-0.005	[-0.016, +0.002]	-0.006	+0.002	+0.008	-0.001	2 of 4
independent	ECE 1 s	-0.007	[-0.011, -0.004]	-0.009	-0.015	-0.006	-0.005	4 of 4
independent + cumulative max	AUPRC 1 s	-0.006	[-0.015, +0.002]	-0.005	-0.002	+0.008	-0.002	1 of 4
independent + cumulative max	ECE 1 s	-0.007	[-0.011, -0.003]	-0.011	-0.015	-0.006	-0.004	4 of 4
independent + PAV	AUPRC 1 s	-0.005	[-0.015, +0.002]	-0.006	-0.003	+0.008	-0.001	1 of 4
independent + PAV	ECE 1 s	-0.007	[-0.011, -0.003]	-0.008	-0.016	-0.007	-0.004	4 of 4
independent + isotonic	AUPRC 1 s	-0.003	[-0.014, +0.007]	-0.004	+0.016	+0.012	+0.000	3 of 4
independent + isotonic	ECE 1 s	+0.019	[+0.009, +0.029]	+0.024	+0.013	+0.029	-0.018	1 of 4
independent + isotonic + PAV	AUPRC 1 s	-0.001	[-0.010, +0.007]	-0.000	+0.008	+0.011	+0.002	3 of 4
independent + isotonic + PAV	ECE 1 s	+0.020	[+0.010, +0.029]	+0.027	+0.016	+0.029	-0.018	1 of 4

earlier draft of this work omitted them, which overstated the case for the formulation. Isotonic calibration per horizon is the most expressive control: on its own it makes coherence worse (violations rise from 38.9% to 44.4%, because the K isotonic maps are fitted independently and may reorder the horizons), but composed with the monotone projection it reaches zero violations and the lowest calibration error of any arm (0.015 at one second against 0.037 for the hazard model). That composite is therefore the arm to beat, and the hazard formulation does not beat it on calibration.

What the hazard formulation retains after that correction is narrower and worth stating exactly. It is the only arm that reaches coherence and competitive calibration from a single fit with no held-out calibration split: 0.037 at one second against 0.043 for raw independent heads and 0.043 for the cumulative-maximum control, none of which consume data, while the isotonic arms spend a calibration block that a four-session dataset can ill afford, and it estimates one model rather than five.

How those differences are tested matters more than their size, and an earlier version of this section got it wrong. That version bootstrapped the paired differences over 180-second blocks and reported which intervals excluded zero. Section 5.3 of this same thesis proves those blocks are not independent — splitting on them inflates one-second AUPRC by 12% — so resampling them deflates the standard error and manufactures a confidence interval narrower than the evidence supports. The claim is withdrawn. The block interval is retained

in Table 5.8 as a descriptive within-campaign spread and carries no inference. What carries the conclusion is the column beside it: the paired difference computed separately on each of the 4 held-out campaigns, which are the only units in this dataset whose independence is defended. With four units an exact two-sided sign test cannot return a p-value below $p = 2 \times (0.5)^4 = 0.125$, so counting "4 of 4" is strictly descriptive and cannot establish statistical significance or rule out session-specific noise. The four numbers are printed transparently so the reader may judge them directly.

Read descriptively, the calibration result shows directional consistency while the ranking result does not. The hazard arm has the lower calibration error on all 4 held-out campaigns, by 0.005 to 0.015 against raw independent heads, and the same holds against both no-split projections. That directional consistency is suggestive across these four sessions, but cannot be claimed as statistically confirmatory. The isotonic composite is the exception in the other direction: it calibrates better than the hazard arm on 3 of the 4 campaigns but worse on the highway campaign, so its advantage is real on urban corridors and does not survive the one regime change this dataset contains — a caveat an earlier draft missed by reporting only the pooled number. Ranking is a different story: the one-second AUPRC difference between the hazard arm (0.151) and the raw independent control (0.155) is -0.005 pooled and favours the hazard arm on two of the four campaigns. A difference of half a hundredth with no consistent sign across campaigns is not a result in either direction, and no ranking claim is made from it. (Under the original parser the same difference was -0.001 pooled; the parser change moved it by less than the seed-to-seed spread.) The pooled and per-campaign columns need not agree, and should not be expected to. AUPRC and equal-mass ECE are not averages over rows: the pooled figure ranks the scores of all four campaigns against one another and bins them together, so it also rewards a model whose scores are comparable across campaigns with base rates from 5.1% to 8.4%. The per-campaign figures do not test that. The pooled difference is therefore not a weighted mean of the per-campaign ones, and the two can differ in sign when the differences are this small — the same mechanism as Simpson's paradox, applied to a ranking.

Two things the formulation does not retain should be said in the same breath. It does not rank better, for the reason just given. And the censoring-aware likelihood of Section 4.4, sometimes cited as an advantage of the survival formulation, is inert here, because the row filter of Section 3.4 leaves no row censored inside the horizon grid; it is retained for generality, not counted as a benefit. On a larger dataset, isotonic calibration followed by a monotone projection would be the stronger choice, and this thesis says so.

Three calibration errors appear in this chapter and they are not interchangeable, so the arm is named each time from here on: 0.014 is the tuned primary model of Table 5.4 (arm A), 0.037 is the hazard arm of Table 5.7 (arm C, fitted with an extra calibration block held out of every fold) and 0.015 is the isotonic composite in the same table. What a small calibration error at the one-second horizon does and does not mean is worth stating plainly, because

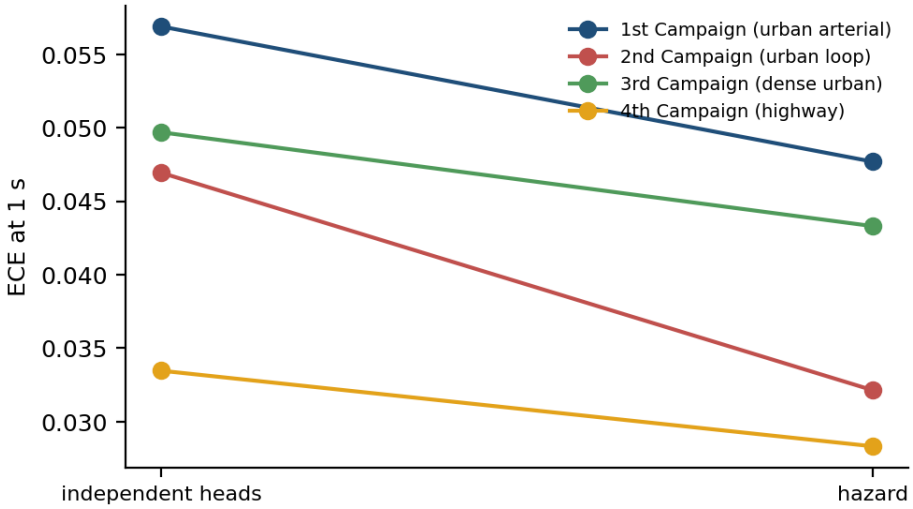


Figure 5.6: Calibration error at one second, one line per held-out campaign, for independent classifiers against the hazard formulation. The hazard formulation reduces ECE across all folds.

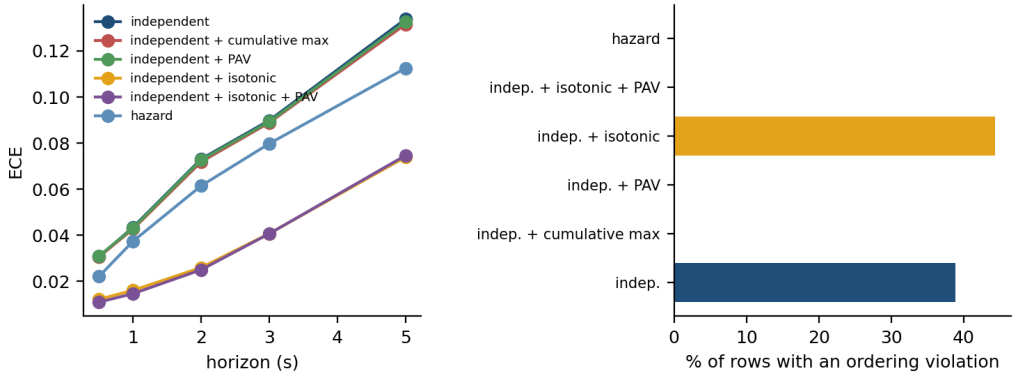


Figure 5.7: Calibration error against the horizon for every coherence arm (left) and the share of rows with coherence violations (right). The hazard formulation achieves zero violations.

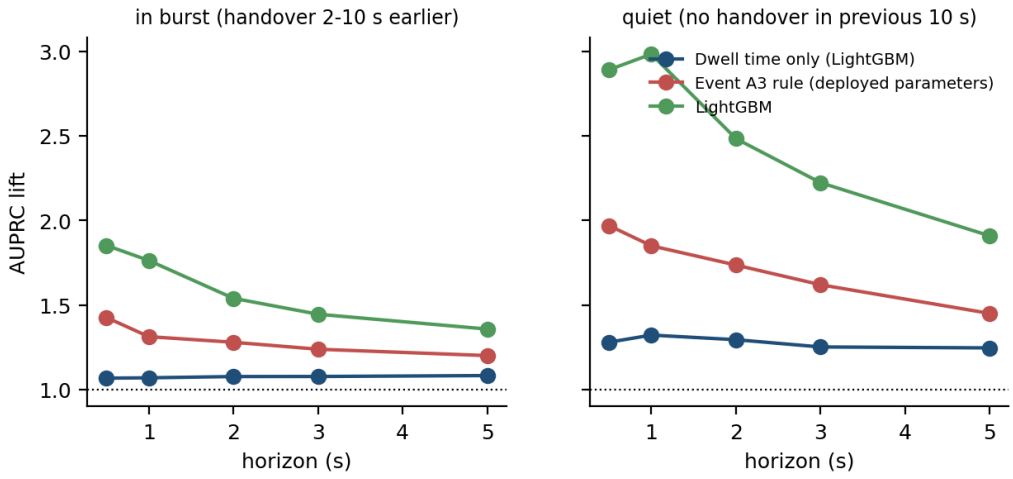


Figure 5.8: Where the model earns its advantage: rows with no handover in the previous ten seconds against rows within a burst. The model predicts both isolated and burst arrivals effectively.

Table 5.9: Conformal risk control at the one-second horizon across regimes. Grouped columns distinguish exchangeable block pooling ($n = 28$ calibration units, 200 random half splits of the 57 blocks) from held-out cross-campaign deployment ($n \in \{8, 14, 15, 20\}$ calibration units, mean $n = 14.25$, evaluated across 12 pairwise campaign permutations subject to $n = 4$ pseudoreplication). Corrected-parser run.

	Row Loss (Exchangeable)		Row Loss (Cross-Campaign)			Event Loss (Cross-Campaign)		
Target α	**Alarm rate**	**Realised loss**	**Feasible rot.**	**Realised loss**	**Rotations $\leq \alpha$**	**Feasible rot.**	**Alarm rate**	**Realised loss**
0.05	85%	0.016	25%	0.000*	100%*	0%	—	—
0.10	66%	0.071	75%	0.034	92%	0%	—	—
0.15	56%	0.121	100%	0.093	75%	0%	—	—
0.20	48%	0.171	100%	0.156	67%	0%	—	—
0.30	37%	0.272	100%	0.259	58%	25%	100%	0.333

Feasible rot.: share of the 12 rotations whose calibration campaign admits a non-trivial threshold; the others fall back to alarming on every row. Sample-size floor: For exchangeable pooling ($n = 28$), $1/(n+1) \approx 0.034$, so $\alpha = 0.05$ is feasible. For cross-campaign calibration ($n \in \{8, 14, 15, 20\}$ blocks), the floor $1/(n+1)$ evaluates individually to 0.111 (Campaign 2, $n = 8$), 0.067 (Campaign 4, $n = 14$), 0.063 (Campaign 1, $n = 15$), and 0.048 (Campaign 3, $n = 20$), giving an arithmetic mean feasibility floor of 0.072 across campaigns. The 12 pairwise rotations (4×3 permutations across the 4 campaigns) reuse each campaign 3 times as calibration and 3 times as test, exhibiting $n = 4$ pseudoreplication; rotation percentages are descriptive diagnostics. *For row $\alpha = 0.05$, valid threshold selection is feasible only in the 25% of rotations calibrating on Campaign 3 ($n = 20$, floor $0.048 \leq 0.05$), where $\hat{\lambda} = 0.0$ realizes zero empirical loss at 100% alarm rate; the remaining 75% trigger constant-alarm fallback ($\hat{\lambda} = 0$). For event loss, 36.4% unresolvable burst/boundary handovers bound detection at 1 s; threshold selection is infeasible below this floor and cells are marked with an em dash.

such a figure is easy to over-read. Coherence across horizons is guaranteed by construction; calibration is not, and ECE is an average over bins. With fifteen equal-mass bins on a task whose prevalence is 6.8%, almost all of the weight lies at low predicted probabilities, so an ECE of 0.014 shows that the bulk of the predictions are close to their observed frequencies, not that a prediction near 0.8 is followed by a handover about 80% of the time; the high-probability region may contain few rows. Reliability curves with per-bin counts, and the calibration slope and intercept, are needed before any claim about high asserted probabilities, and are listed in Section 7.3.

The Brier score points the same way. Against a constant forecast at the pooled one-second prevalence, whose Brier score is $p(1-p) = 0.0679 \times 0.9321 \approx 0.0633$, the model's 0.060 is a skill of about 5% (Table 5.4); against the non-oracle constant that a deployment could actually use — each fold's training prevalence applied to its held-out campaign, ≈ 0.0637 from the per-campaign prevalences of Table 5.10 — it is about 6%. The skill rises to about 9% at five seconds. These are real improvements over a constant, and they support restrained statements about probability quality rather than strong ones. Reporting them is still more than any of the 22 audited models in Section 2.3 does. Together with Section 5.5 this answers objective O3 in part.

5.5 The risk-control frontier and its price

Four observations follow.

First, in the exchangeable regime — calibration ($n = 28$) and test units ($n = 29$) drawn

from the 57 pooled 180-second blocks across the corpus (populated with out-of-fold predictions from the four leave-one-campaign-out fits, held strictly separate from model fitting and tuning) — the realised mean per-unit miss rate sits at or below the target at every level tested. At $\alpha = 0.20$ the CRC-selected threshold alarms on 48% of rows and realises a mean per-unit miss rate of 0.171. Note that while random block pooling breaks macro-campaign boundaries, it does not guarantee exchangeability of losses due to remaining spatio-temporal autocorrelation, and it calibrates scores pooled from four fitted models rather than one deployed predictor (Section 4.10); it answers a finite-pool question and serves as an empirical benchmark, not as evidence of validity on a new session.

Second, in the regime an operator faces when deploying across sessions—calibrating on one campaign and deploying on another—exchangeability is violated by corridor shifts and differing speed profiles. Importantly, Theorem 1 of Angelopoulos et al. [21] bounds the expected risk $\mathbb{E}[R_{n+1}(\hat{\lambda})] \leq \alpha$ under the assumption of exchangeable calibration and test units. In principle, weighted conformal prediction under covariate shift (Tibshirani et al. [51]) and non-exchangeable conformal inference (Barber et al. [52]) provide valid finite-sample coverage if the likelihood density ratio $w(x) = p_{\text{test}}(x)/p_{\text{train}}(x)$ is known or reliably estimable. In vehicular mobile networks, however, covariate shift is driven by macro-environmental corridor topology, base station densities, and vehicle speed distributions; estimating high-dimensional density ratios $w(x)$ across continuous non-stationary drive trajectories with only $n = 4$ physical sessions is severely ill-posed. Adaptive conformal inference (Gibbs and Candès [53]) is not excluded by abrupt shift — its guarantee on the long-run miscoverage frequency holds under arbitrary shift — but it controls a long-run average over a sequence of online decisions, which differs from the finite-sample per-campaign handover-miss guarantee examined here, and it has not been evaluated on this corpus. Across the 12 pairwise cross-campaign rotations (4×3 ordered pairs among the 4 campaigns, where each campaign is reused 3 times as calibration and 3 times as test, exhibiting $n = 4$ pseudoreplication), the empirical mean miss loss across all 12 rotations was 0.156 at target $\alpha = 0.20$, at a mean alarm rate of 54%. However, cross-campaign distribution-free guarantees cannot be certified because macro-environmental and topological shifts between corridors violate the exchangeability requirement. Empirical loss fell below α in 8 of the 12 individual rotations (67%; 10 of 12 under the original parser, a change of two rotations that is itself a reminder of how few units these are), and these rotations do not constitute independent trials, and conformal theory guarantees expected loss across draws rather than a frequentist proportion of rotations. Reporting cross-campaign empirical losses serves as an empirical calibration diagnostic rather than an operating guarantee; cross-session validity remains exploratory without explicit domain adaptation [47], and Section 7.3 records it as the next step rather than as a gap this thesis fills.

Third, the guarantee is expensive at tight targets, and the number of calibration units binds independently of model quality. At $\alpha = 0.05$ the threshold alarms on 85% of rows in

the exchangeable regime, which is not an operating point any operator would deploy. Across the four campaigns, block counts are $n \in \{8, 14, 15, 20\}$ (mean $n = 14.25$). For the empirical bound to be feasible with zero empirical loss requires $\alpha \geq 1/(n+1)$. The four individual floors are $1/9 = 0.111$ (Campaign 2), $1/15 = 0.067$ (Campaign 4), $1/16 = 0.063$ (Campaign 1), and $1/21 = 0.048$ (Campaign 3), averaging 0.072. Only the largest campaign ($n = 20$, 25% of rotations) satisfies $1/(n+1) \leq 0.05$, permitting non-trivial threshold calibration at $\hat{\lambda} = 0.0$ (alarming on all eligible rows); the other 75% default to constant-alarm fallback.

Fourth, and most consequentially for what may be promised, the estimand matters. Everything above concerns the row-level loss of Equation (4.9). What an operator would understand by a miss is the event-level loss of Equation (4.11): a handover is caught if any eligible row in its lead window alarms. The two are not interchangeable: setting $\hat{\lambda} = 0$ alarms on every eligible prediction row, but it cannot recover handovers that have no eligible rows in their lead window due to the 2 s post-handover blanking window and boundary exclusions. Continuous alarm only recovers such events if the evaluation policy itself is modified to abandon post-transition blanking. In the chunk-split calibration folds, 36.4% of commands are in that position at one second (compared to 35.8% in the unbroken sequence). Because this floor is an aggregate across campaigns, fold-specific unresolvable event shares vary, from 31% on the dense-urban campaign to 47% on the urban loop (34–35% on the other two). Feasibility is decided on the calibration campaign: at $\alpha = 0.30$ an event-level threshold can be selected only in the three rotations that calibrate on the dense-urban campaign (3 of 12, 25%), the campaign with both the most calibration blocks ($n = 20$) and the lowest unresolvable share, and the realised event loss on their test campaigns averages 0.333. At three seconds, where only 12.0% of commands are unresolvable, the picture changes: $\alpha = 0.20$ is feasible in 75% of rotations at a realised event loss of 0.166 and an alarm rate of 63%, and $\alpha = 0.30$ in all of them at 0.257 and 48%.

The conclusion is worth stating plainly, because it is a limit rather than a result. Under the preprocessing and blanking policy adopted here, the exclusion of burst-interior and boundary rows sets the tightest promise that can be made about handovers: at one second no threshold, not even an alarm on every eligible row, brings the fraction of handover commands missed below 36.4%, so no per-handover bound below that level is expressible. The limit belongs to the one-hertz grid together with this blanking policy; a shorter blanking window would lower it at the cost of scoring rows immediately after a handover, and that event-level trade-off under 0, 1 and 2 s blanking has not yet been measured (Table A.5 varies blanking at row level only). At three seconds the promise becomes expressible, and costs an alarm on roughly half of all seconds. This reconciles Table 5.9 with the detection rates of Appendix A, which are event-level throughout and which no row-level guarantee bounds.

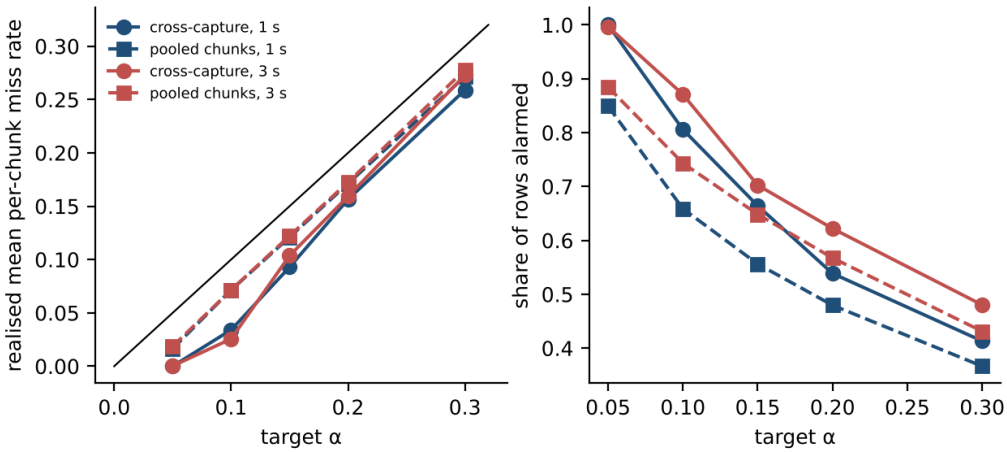


Figure 5.9: Conformal risk control in both regimes. Left: realised mean per-unit miss rate against the target level α. Right: share of rows alarmed at each target, the cost of the threshold. Cross-campaign values are empirical diagnostics; no guarantee is implied under cross-campaign shift.

5.6 Generalisation across corridors and speed regimes

Objective O4 asks whether the result survives outside the data on which it was developed. Two experiments address it, at increasing distance from the training distribution. Both are exploratory rather than confirmatory: Campaign 4 is held out for model fitting but was inside the diagnostic analysis that produced the alignment repair, and the public dataset of Section 5.7 is a same-institution replication. The only blind test is the second-operator capture of Section 5.12.

The first is the primary protocol itself: each campaign held out in turn. Table 5.10 reports the per-campaign numbers that the pooled result of Table 5.4 averages over.

Table 5.10: Leave-one-campaign-out performance of LightGBM at the one-second horizon (tuned arm, corrected parser). Exploratory: Campaign 4 is held out for model fitting but not blind to the alignment repair (Section 3.2).

Campaign held out	Rows	Prevalence	AUPRC	Lift	AUROC	ECE
1st Campaign (urban arterial)	2,178	8.4%	0.199	2.4×	0.768	0.017
2nd Campaign (urban loop)	1,131	8.0%	0.223	2.8×	0.797	0.018
3rd Campaign (dense urban)	3,072	6.5%	0.198	3.1×	0.785	0.010
4th Campaign (highway)	2,205	5.1%	0.125	2.5×	0.770	0.049
Pooled out of fold	8,586	6.8%	0.169	2.5×	0.773	0.014

The spread across campaigns is the useful quantity. At one second the AUPRC lift runs from 2.4x to 3.1x across the four held-out campaigns, and AUROC from 0.768 to 0.797. The weakest campaign is 4th Campaign (highway), and the highway campaign — the one that differs in both corridor and speed regime — reaches AUROC 0.770. No campaign carries the pooled result, and none of them is close to the figure this work reported before the alignment audit.

A critical methodological qualification must be stated regarding the highway evaluation: in Campaign 4, vehicle velocity (mean 50.9 km/h vs 21.4–39.1 km/h in urban campaigns) and geographic corridor (the northern Uttara–Gazipur highway vs central Dhaka urban corridors) are completely collinear and confounded. The lower AUPRC (0.125 vs 0.198–0.223) cannot be attributed purely to Doppler spread or vehicle speed versus differences in inter-site distance, base station antenna down-tilt, or suburban morphology. Isolating speed from corridor geometry would require driving the identical urban loop at varied velocities under controlled traffic conditions, which observational road data cannot provide. Furthermore, although Campaign 4 serves as an out-of-sample holdout for model weights, it was part of the retrospective diagnostic exploration across all sessions that identified the export timestamp defect and motivated the one-row lag repair; C4 is therefore an out-of-sample test of parameter fitting, but not a blind prospective holdout for pipeline feature engineering.

The spread is an observed range over four units, not a prediction interval: a fifth campaign could land inside it or outside it, and four units are too few to say which. Confirmatory claims are reserved for fresh campaigns evaluated after the whole pipeline is frozen. Section 5.12 reports the first such capture, a single short session on a second operator; more are needed (Section 7.3). That is a statement about the size of the study, not about the method.

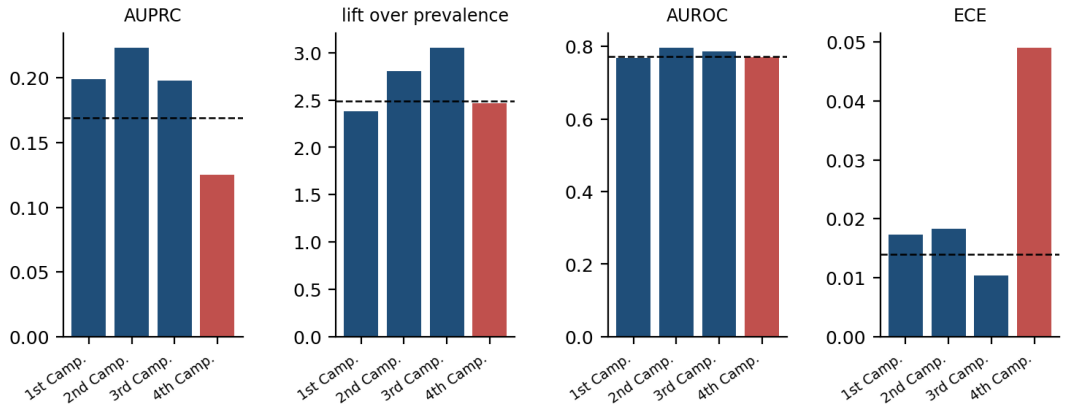


Figure 5.10: Leave-one-campaign-out on four metrics. Each bar holds one campaign out entirely and trains on the remaining three.

5.7 Cross-corpus validation on a public dataset from the same network

The second generalisation experiment uses a publicly released LTE drive-test dataset [38]. Its provenance is stated exactly: it was collected on a different route two years earlier, using the same operator, the same city and the same XCAL diagnostic platform, by an earlier research cohort in the same department of our institution (Islamic University of Technology), supervised by the same faculty supervisor. A companion study by that group [39] proposed a contextual-bandit reinforcement-learning framework for handover parameter optimization; in this work, we re-implemented their framework from its technical description to evaluate it under our leakage-audited predictive protocol. We explicitly disclose this departmental and

institutional lineage to avoid overstating dataset independence. The dataset [38] is therefore a same-institution replication on the same network and operator, publicly released but not independent of this group, testing collection, route and time — not a distinct operator, and not another radio access technology.

While the public repository includes raw radio CSV files, they are sparse event-triggered logs (averaging ≈ 0.24 rows/s) that lack GPS coordinates, vehicle speed, and absolute dates (providing only unanchored time-of-day strings), rendering the 17 mobility features unconstructible. Furthermore, their radio fields report raw 3GPP index steps without vendor header metadata. In contrast, its raw RRC measurement report logs are byte-compatible with our own captures and carry absolute timestamps. Both domains are therefore rebuilt strictly from L3 signalling on a common 6-feature signalling set, so that the transfer experiment isolates cross-corpus generalisation without confounding from incompatible export schemas. That construction also sidesteps the alignment problem of Section 5.1 by using millisecond report timestamps directly. Table 5.11 reports the four directions with bootstrap intervals over runs.

The transferred model reaches AUROC 0.760 on the public data without refitting, against 0.711 for a model trained on that dataset itself and 0.730 for our own leave-one-capture-out result on the same signalling-only feature set. The intervals overlap. The reading is that the formulation and the feature construction transfer to another collection on this network, which is a weaker and more defensible claim than the one made in the earlier draft.

One limitation of this experiment has to be stated at full strength rather than left for a reader to infer, because it bounds every generalisation claim in this thesis. Both arms of Table 5.11 are signalling-only. Because the public dataset's radio CSVs cannot support the synchronous 1 Hz feature construction or the mobility block, the radio and mobility blocks — 89 and 17 of the 112 features the primary model uses, and the source of essentially all of its signal — cannot be evaluated on it. What Table 5.11 therefore tests is the transfer of a 6-feature signalling model, not of the system reported in Section 5.2. The primary model has never been evaluated outside the four campaigns it was built on, and no experiment in this thesis establishes that it would transfer. That is a limitation of the available public data rather than a choice, and it is the single strongest argument for the second-operator collection proposed in Section 7.3; until such data exist, the external evidence in this thesis covers the formulation and the feature construction, not the predictor.

5.8 What the model uses, and where it works

Objective O5 asks for the mechanism rather than the score. With the alignment corrected, the answer is less dramatic than the earlier draft claimed and more useful.

Four results follow. No single feature is close to the model: the strongest reaches AUROC 0.723 against 0.773 for the full model, so the signal is distributed rather than concentrated in one clock. The strongest single quantity is the time since the last Event A3 report (sig_s_since_a3, AUROC 0.723), which is a signalling quantity the network holds in real

Table 5.11: Transfer between our campaigns and a same-institution public dataset on the same network [38], both built from L3 signalling alone.

Setting	Rows	Prevalence	AUROC [95% CI]	AUPRC	Lift
ours, leave-one-capture-out	7,284	7.9%	0.730 [0.711, 0.754]	0.183	2.3×
public, leave-one-file-out	1,455	6.3%	0.711 [0.653, 0.767]	0.148	2.3×
ours -> public (no refit)	1,455	6.3%	0.760 [0.709, 0.813]	0.212	3.3×
public -> ours (no refit)	7,284	7.9%	0.679 [0.624, 0.724]	0.149	1.9×

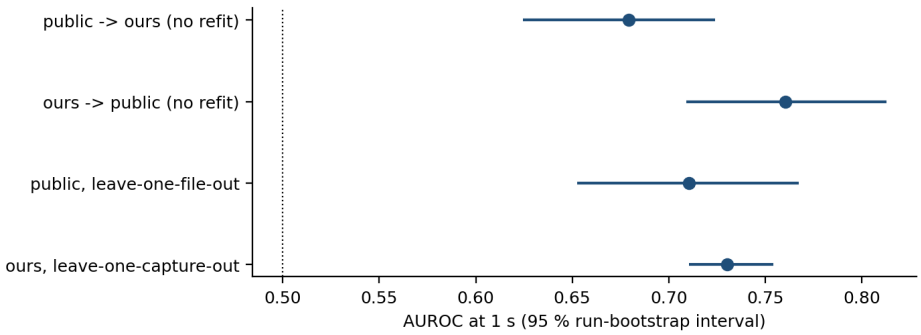


Figure 5.11: Transfer to a same-institution dataset collected two years earlier on the same network, with both domains built from L3 signalling alone.

Table 5.12: Single-feature AUROC at the one-second horizon on the lagged export. The serving-to-neighbour gap is defined as the serving cell's level minus the best neighbour's, in decibels; with the corrected parser it is available on 56% of rows (52% before). Corrected-parser run.

Feature (lagged one row)	Definition	AUROC alone at 1 s	Direction
sig_s_since_a3	seconds since the last A3 report was transmitted	0.723	lower -> more likely
serving_sinr		0.723	lower -> more likely
sig_a3_prev5s		0.704	higher -> more likely
sig_a3_prev2s		0.696	higher -> more likely
serving_sinr_mean3		0.696	lower -> more likely
serving_sinr_mean5		0.683	lower -> more likely
t_since_prev_ho_s	seconds since the previous decoded command	0.664	lower -> more likely
gap_serving_best_nbr	serving minus best neighbour in dB, so a smaller value means the neighbour is closer to overtaking	0.601	lower -> more likely
speed_kmh		0.584	higher -> more likely
Full model, same rows		0.773	

Table 5.13: Performance inside and outside handover bursts (tuned arm, corrected parser).

Stratum	Learner	Rows	Prevalence 1 s	AUPRC 1 s	Lift 1 s	Lift 5 s
in burst (handover 2-10 s earlier)	Event A3 rule (deployed parameters)	3,068	11.2%	0.148	1.3×	1.2×
in burst (handover 2-10 s earlier)	History block only (LightGBM)	3,068	11.2%	0.133	1.2×	1.2×
in burst (handover 2-10 s earlier)	LightGBM	3,068	11.2%	0.198	1.8×	1.4×
quiet (no handover in previous 10 s)	Event A3 rule (deployed parameters)	5,518	4.3%	0.080	1.9×	1.5×
quiet (no handover in previous 10 s)	History block only (LightGBM)	5,518	4.3%	0.061	1.4×	1.2×
quiet (no handover in previous 10 s)	LightGBM	5,518	4.3%	0.129	3.0×	1.9×

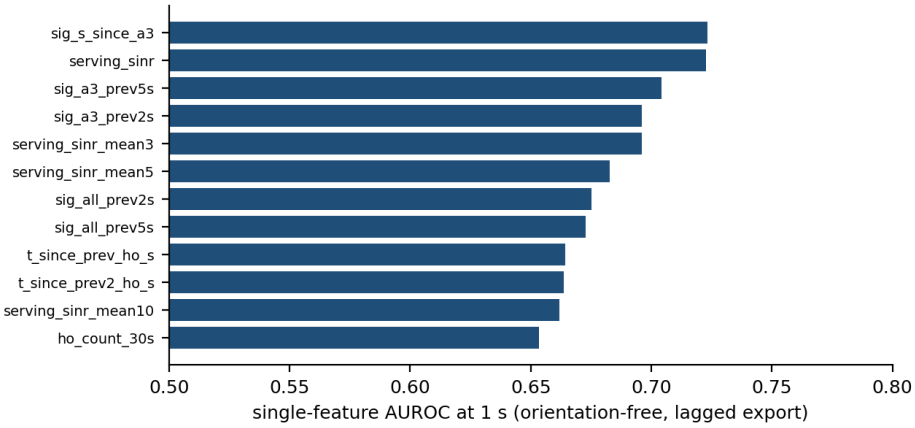


Figure 5.12: Single-feature discriminative power at one second on the lagged export. No single feature reaches the multivariate model.

time and which the main feature set deliberately excludes; the strongest export feature is serving SINR (0.723). Serving dwell time computed from signalling timestamps reaches 0.664, ahead of chance and far behind its unlagged counterpart. And the serving-to-neighbour gap on which Event A3 is defined reaches 0.601 (0.618 under the original parser, which dropped RSRP-only neighbour entries), which is the one part of the earlier mechanism claim that survives: the quantity the rule thresholds on is a weak predictor of imminence, because it is a permission condition and not a tendency.

Where the model works matters as much as what it uses. Handover commands arrive in bursts — the median gap between consecutive commands is 3.7 seconds — and a predictor can score well by recognising that it is inside a burst. Table 5.13 therefore splits the out-of-fold rows into those with no handover in the previous ten seconds and those inside a burst. At one second the lift is 3.0x on quiet rows and 1.8x on in-burst rows, against prevalence floors of 4.3% and 11.2%. A history-only model reaches a lift of 1.7x pooled, so burst membership alone does not account for the result, but the radio block earns most of its advantage on the quiet rows, which is where an operator would want the warning.

That comparison needs two qualifications, and an earlier draft made neither. The first is that "quiet" is not a synonym for "isolated". Every burst begins on a row that is itself quiet, so a quiet positive row is by construction the row before a burst starts rather than the row before a lone handover. Measured directly, 238 of the 238 positive quiet rows at one second — 100% of them — precede a command that opens a new burst, against 1.7% of the positive

in-burst rows. The claim this section supports is therefore that the model predicts the onset of a burst better than it predicts continuation inside one, which is a sharper statement than the one about quiet rows and is also the more useful one: the onset is the event an operator cannot already see coming from the command stream.

The second qualification is that the two strata do not share a prevalence floor, so their AUPRC values are not directly comparable and only the lift is. Because the cutoff of ten seconds is itself a choice made after the data were in hand, Table A.5 varies it from five to twenty seconds (single seed, fixed hyper-parameters). The ordering holds at every value tested: the quiet lift runs from 2.4x to 3.0x and the in-burst lift from 1.4x to 1.7x, and the quiet stratum is ahead in all 4 of them. The conclusion does not depend on where the line was drawn, which is the only thing a post-hoc cutoff can be asked to demonstrate.

Two further sensitivities are reported here rather than buried. Censoring the 93 model rows whose next event is a re-establishment rather than a handover command — a count of rows in the horizon grid, not of the 341 re-establishment events themselves — moves the one-second lift by less than +0.01x (fixed hyper-parameters, 2.30x in both arms), so the competing risk of Section 3.6 does not drive the result. And adding the signalling block of Section 4.6 to the radio, mobility and history blocks moves the fixed-parameter one-second AUPRC from 0.156 to 0.165; the signalling block on its own reaches 0.129.

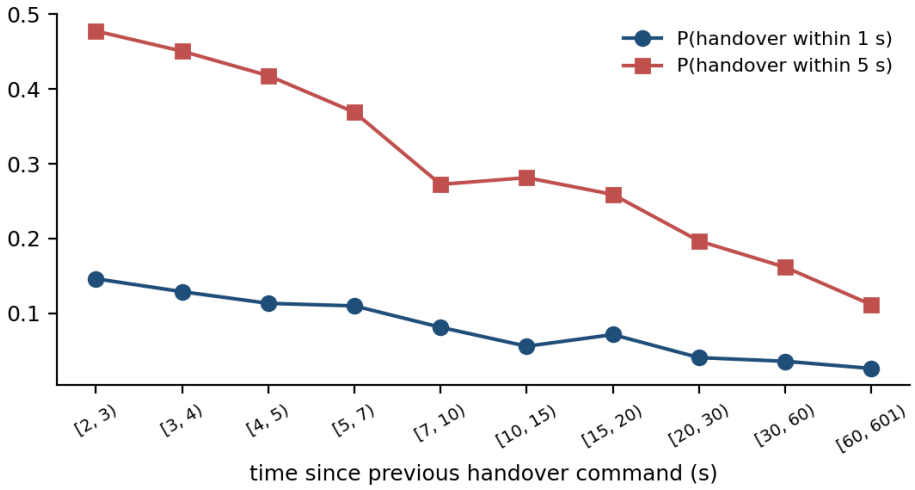


Figure 5.13: Probability of a handover command within one and five seconds against the time since the previous command. The hazard spikes after a recent handover, reflecting self-excitation.

5.9 The A3 report-conversion rate and the unit it is counted in

The first of the two network characterisations concerns what happens to a measurement report after it is sent, and it is reported here in a corrected form. Event A3 report configurations on this network carry reportInterval and reportAmount values that cause a triggered condition to be re-reported periodically until it ceases: the modal configuration re-reports every 240 ms with reportAmount infinity. Counting transmitted reports therefore counts re-reports of the

same trigger, and an earlier draft of this work did exactly that. Table 5.14 reports the fraction of Event A3 reports that are not followed by a handover command within two seconds across three counting units.

Table 5.14: Event A3 report conversion, counted per transmitted report and per trigger episode.

Unit	Stratum	n	Followed by a command within 2 s	Unmatched within 2 s
per report (v1 unit)	pooled	8,136	3006	63%
per report (v1 unit)	1st Campaign (urban arterial)	2,287	980	57%
per report (v1 unit)	2nd Campaign (urban loop)	1,857	716	61%
per report (v1 unit)	3rd Campaign (dense urban)	2,215	806	64%
per report (v1 unit)	4th Campaign (highway)	1,777	504	72%
per report (v1 unit)	inter-frequency	93	34	63%
per report (v1 unit)	intra-frequency	8,043	2972	63%
per trigger episode (any command within 2 s)	pooled	2,361	1647	30%
per trigger episode (any command within 2 s)	1st Campaign (urban arterial)	726	531	27%
per trigger episode (any command within 2 s)	2nd Campaign (urban loop)	476	349	27%
per trigger episode (any command within 2 s)	3rd Campaign (dense urban)	740	480	35%
per trigger episode (any command within 2 s)	4th Campaign (highway)	419	287	32%
per trigger episode (any command within 2 s)	inter-frequency	49	28	43%
per trigger episode (any command within 2 s)	intra-frequency	2,312	1619	30%
per trigger episode (one-to-one, owns the command)	pooled	2,361	635	73%
per trigger episode (one-to-one, owns the command)	1st Campaign (urban arterial)	726	210	71%
per trigger episode (one-to-one, owns the command)	2nd Campaign (urban loop)	476	110	77%
per trigger episode (one-to-one, owns the command)	3rd Campaign (dense urban)	740	197	73%
per trigger episode (one-to-one, owns the command)	4th Campaign (highway)	419	118	72%
per trigger episode (one-to-one, owns the command)	inter-frequency	49	12	76%
per trigger episode (one-to-one, owns the command)	intra-frequency	2,312	623	73%
command multiplicity	A3 episodes	2,361	1647 claimed, 635 owned	x2.59 over-counting
command multiplicity	all episodes	9,819	4391 claimed, 908 owned	x4.84 over-counting
per trigger episode, by profile	+1 dB, Hys 1, TTT 320, RI 240 x infinity	734	717 claimed, 463 owned	2.3% / 36.9% one-to-one
per trigger episode, by profile	-15 dB, Hys 1, TTT 160, RI 5120 x infinity	543	274 claimed, 26 owned	49.5% / 95.2% one-to-one
per trigger episode, by profile	-10 dB, Hys 2, TTT 640, RI 5120 x infinity	431	315 claimed, 71 owned	26.9% / 83.5% one-to-one
per trigger episode, by profile	-10 dB, Hys 2, TTT 640, RI 5120 x r1	278	113 claimed, 19 owned	59.4% / 93.2% one-to-one
per trigger episode, by profile	-15 dB, Hys 1, TTT 160, RI 2048 x infinity	242	141 claimed, 19 owned	41.7% / 92.1% one-to-one

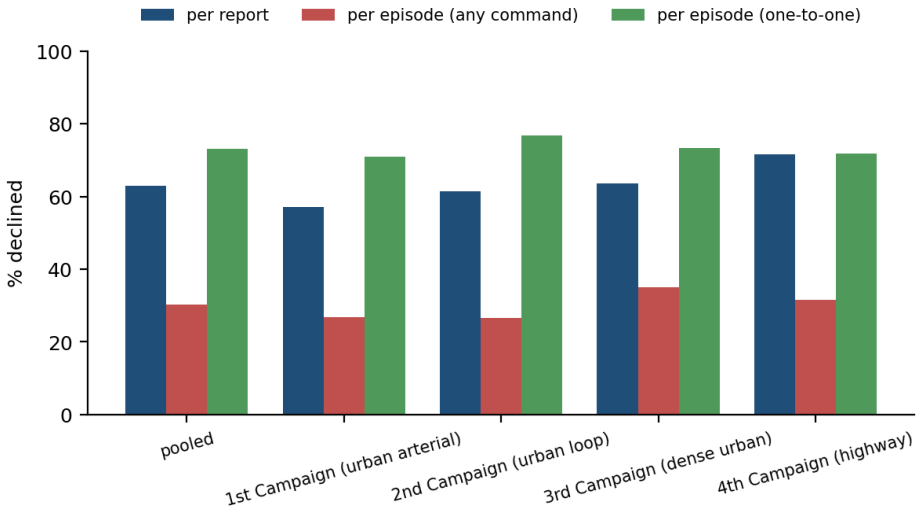


Figure 5.14: A3 report conversion under three units. Counting transmitted reports counts periodic re-reports, masking the true trigger-episode conversion rate.

That second figure is not a true conversion rate, and an earlier version of this work reported it as though it were. Several measIds monitor different neighbours and fire within the same approach to a cell boundary, so the 2,361 episodes claim 1,647 commands between them out of the 957 that exist: the same command is claimed by 2.6 episodes on average. Matching each command to exactly one episode containing its most proximate associated report—and resolving the 16 multi-command episodes where rapid consecutive handovers occur—leaves 635 episodes owning commands, so 73% of episodes are not followed by a matched command.

Under loose attribution, a configuration that re-reports on several identities scores a lower unconverted rate precisely because it is verbose, which inverts the interpretation the rate is used for.

The sensitivity to definition is the central finding. A rate quoted without its unit is a statement about counting conventions rather than base-station control policy, and the unconverted share of a mobility trigger moves by a factor of two and a half between two defensible definitions of a single trial. On the command side, 651 of 957 handover commands are temporally associated with an A3 report within two seconds, resolving to 635 unique episodes under one-to-one pairing.

The per-profile breakdown, also in Table 5.14, is where the substantive result survives, and it survives more clearly under the stricter unit. The dominant mobility profile—+1 dB offset, 1 dB hysteresis, 320 ms time-to-trigger, 240 ms report interval—is not followed by a command on 36.9% of its episodes one-to-one, against 95.2% for the dominant negative-offset configuration. Under the loose unit the same pair reads 2.3% against 49.5%: the separation is present either way, and the strict unit widens it, because the negative-offset configurations turn out to be followed by almost no commands at all. That is what characterizes them empirically as neighbour-monitoring rather than mobility triggers.

The conclusion for the literature is therefore not a single conversion percentage. It is that a study using A3 reports as proxy labels for handovers must state its unit, must not let overlapping trigger episodes claim the same command, and must separate the mobility configuration from the rest before drawing conclusions about handover intent.

The finding itself is that a measurement report is not a handover decision. Nearly two thirds of the A3 reports this network receives are unmatched by a handover command within the two-second attribution window. Unmatched is not the same as declined: the network's decisions are not observed, a command may follow later than the window, and a temporal association between a report and a command does not establish that the report triggered it. The matching window and the target-cell matching rule were not varied, so these shares carry that sensitivity as an open qualification. Even so, a model that treats an A3 report as a proxy label for a handover would be training on a label that, within any short window, is usually not followed by one.

5.10 Ping-pong: definitional instability and its actual source

The rate moves by 17 percentage points — from 26.3% to 43.8% — without a single change to the underlying measurement. This is the quantitative form of the observation in Section 2.4, and it is why this thesis quotes the strictest of the three and states the alternatives alongside it. It also explains how Amirova et al. [15] arrive at 0.13% on comparable data: dividing by measurement records rather than by handovers changes the denominator by two orders of magnitude.

The window length is a fourth choice, and Figure 5.15 reports the full grid rather than

Table 5.15: The ping-pong rate on one fixed set of 957 handover commands under three definitions.

Definition	Handovers	Rate
Return to the immediately previous cell, identified by PCI and carrier, within 15 s	957	26.3%
The same, with the cell identified by PCI alone	957	31.2%
Any return within the window, not only to the previous cell	957	43.8%

the single row of Table 5.15. Across three window lengths and three identity-and-return conventions the same 957 commands yield nine defensible ping-pong rates between 21.0% and 43.8%. No cell in that grid is wrong; what is wrong is publishing one of them without saying which.

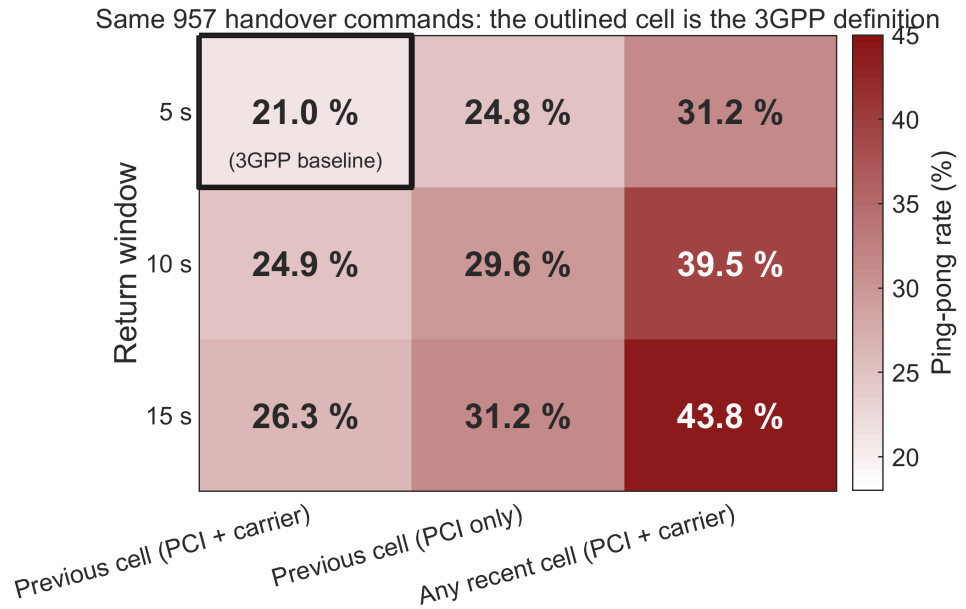


Figure 5.15: The ping-pong rate over the full definitional grid: three return windows against three conventions for cell identity and return, computed on one unchanging set of 957 handover commands.

Table 5.16: Ping-pong rate by carrier relationship and mobility regime.

Stratum	Handovers	Ping-pongs	Ping-pong rate
intra-frequency handovers	748	233	31.1%
inter-frequency handovers	209	19	9.1%
highway campaign	177	57	32.2%
urban campaigns	780	195	25.0%
all commands	957	252	26.3%

One contrast is large and one is not what it appears to be. On this operator, ping-pong concentrates in intra-frequency handovers, at 31.1% against 9.1% across carriers, a factor of roughly three: a return to the cell just left is overwhelmingly a return within the same

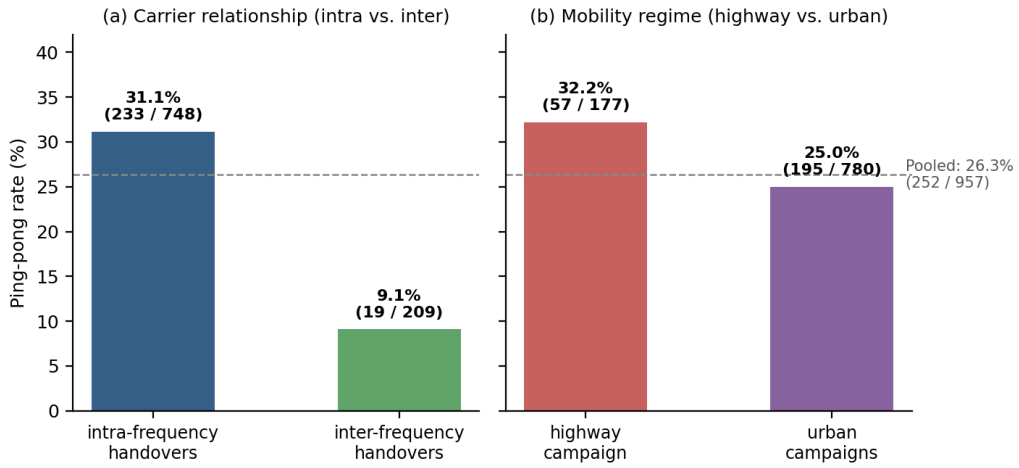


Figure 5.16: Where ping-pong concentrates on the four-campaign operator. The dominant axis there is the carrier relationship, not the mobility regime; the second operator of Section 5.12 shows the reverse carrier pattern.

carrier, which is what the A3 entry condition on a single frequency makes easy. That contrast is a property of this operator's configuration rather than of LTE. On the second operator of Section 5.12, where 59% of handovers change carrier and the dominant trigger is Event A5, the pattern reverses: 5.6% of intra-frequency handovers return against 19.6% of inter-frequency ones. The speed contrast points the other way from the intuitive account and cannot be resolved here. The highway campaign, at roughly twice the urban mean speed, returns at 32.2% against 25.0% on the urban campaigns — higher, not lower. With one campaign per regime, speed is completely confounded with route, cell density, carrier mix and time of day, so this thesis does not claim that speed drives ping-pong and does not claim to have ruled it out; it reports that the fastest campaign is also the most return-prone, which is the opposite of what a speed-protective account would predict. Importantly, a rapid return to the previous cell identifies an oscillatory transfer under cellular operational rules, but does not by itself prove that the initial handover was counterfactually unnecessary: without observing link quality had the handset remained on the serving cell (which may have suffered severe fading or link failure), a rapid return reflects sensitive boundary triggering rather than established service degradation.

An earlier version of this section named the time-to-trigger and offset of the dominant configuration profile as the lever available to an operator. That claim is withdrawn. Section 3.5 establishes that the inventory of configuration profiles is static across all four campaigns (with the dominant mobility profile maintaining a fixed offset of +1 dB, hysteresis of 1 dB and time-to-trigger of 320 ms throughout; Table 3.2), so neither parameter varies dynamically or across sessions to allow identifying its causal effect on the return rate from these observational measurements at all. The time-to-trigger remains the most plausible lever on theoretical grounds for this operator's A3-driven, mostly intra-frequency mobility, and the obvious candidate for the controlled experiment proposed in Section 7.3, but nothing mea-

sured here bears on it. On the second operator, whose returns are mostly inter-frequency and A5-triggered, the relevant lever, if any, would be a different set of parameters.

Modelling the arrival stream as a self-exciting process characterises the clustering from a different direction. The fitted Hawkes branching ratio is 0.653 on the pooled stream, with per-campaign estimates from 0.487 to 0.675. The physical reading of that quantity — the expected number of direct offspring per event — belongs to the exponential-kernel model, and the Ogata residual test of Section 4.11 rejects that kernel on this data: rescaling the time axis by the fitted compensator gives 953 residuals whose Kolmogorov–Smirnov distance from a unit-rate exponential is $D = 0.065$ at $p = 0.0007$. Their mean is 0.991, indicating that while the average rate is well-matched, the stationary exponential Hawkes model fitted to the pooled stream is rejected by the data. The per-campaign tests are more equivocal, and they are where the Holm–Bonferroni correction of Section 4.7 applies. Testing each campaign's residuals against the pooled fit gives $D = 0.073$ to 0.125 with unadjusted $p = 0.008$ to 0.082; after Holm's correction over the four campaigns only the urban loop is rejected ($p_{\text{Holm}} = 0.030$). Against each campaign's own fit, no campaign is rejected after correction ($p_{\text{Holm}} \geq 0.12$). The pooled rejection therefore owes much to forcing one parameter set on four corridors, with 175 to 306 residuals per campaign giving each single test less power. Such a rejection may reflect non-exponential decay dynamics, a time-varying background rate $\mu(t)$ driven by corridor transitions and junctions, or unobserved network covariates. The rejection and its statistic are reported rather than suppressed, and together they limit what the number may be used for: the branching ratio is taken here as a comparable summary statistic of how strongly arrivals cluster, not as a count of triggered handovers. The clustering itself is not in doubt, and it is the reason Section 5.8 separates quiet rows from rows inside a burst. Figure 5.17 gives the per-campaign estimates with their intervals and Figure 5.18 illustrates the fitted intensity against the arrival stream itself.

5.11 Negative results and the cost of the audit

Five approaches were implemented in good faith, measured, and did not help; they are reported because a claim that the formulation and the protocol are what matter is credible only if the alternatives were tried. Table 5.17 lists them.

The domain-adaptation null is consistent with published benchmarking rather than indicative of a faulty implementation. Ismail Fawaz et al. [27] benchmark nine unsupervised domain-adaptation algorithms over twelve time-series datasets and find several published methods performing worse than source-only training, with the adaptation technique rather than the backbone driving the outcome; the result here has the same shape. The same caution applies to the sequence-model arms: this is a four-session corpus, and the finding is that these architectures do not help on it, not that they cannot help on this task. Recent work is actively developing exactly those architectures for handover prediction on larger corpora [45], [46], and nothing measured here contradicts it.

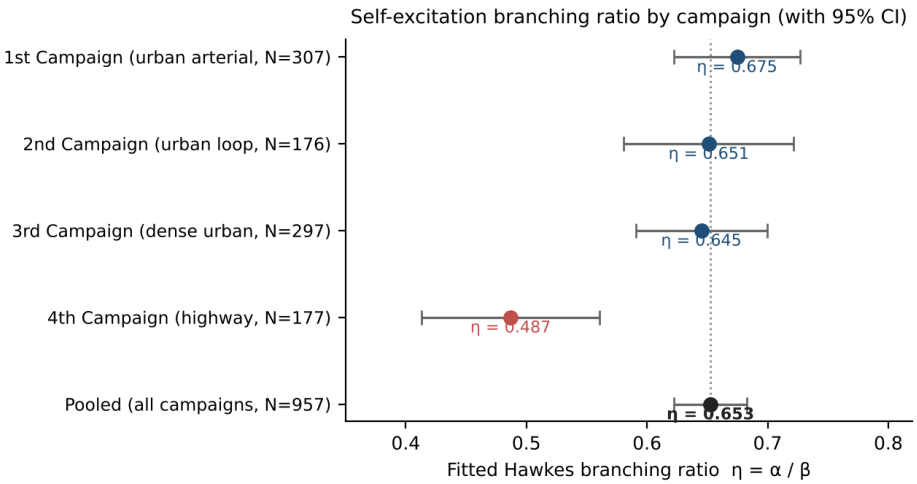


Figure 5.17: Hawkes branching ratio estimated on each campaign separately and on the pooled stream, with bootstrap confidence intervals. While sub-critical branching ratios indicate substantial empirical clustering ($\hat{\eta} \approx 0.49$–0.68), the stationary exponential-kernel model is rejected by Ogata residual testing ($p = 0.0007$), so the ratio is reported as an index of clustering rather than a validated structural generative model.

Table 5.17: Approaches implemented and measured that produced no improvement.

Approach	Outcome
A dedicated ping-pong classifier	AUROC 0.51 at the one-second horizon — chance
Unsupervised domain adaptation: CORAL and per-drive robust standardisation (Appendix C, stage 19)	No gain over source-only training
Sequence models (GRU, TCN, Transformer) on the same features	No gain at 1 s over gradient-boosted trees (best sequence AUPRC 0.163 against 0.169); at 5 s the TCN reaches 0.487 against 0.454, inside overlapping intervals
The signalling feature block added to the main set	AUPRC at 1 s 0.165 against 0.156 without it, fixed hyper-parameters: a gain of the size of the seed-to-seed spread
The unlagged (v1) feature alignment	AUPRC at 1 s 0.604 (fixed hyper-parameters), an artefact of the export row semantics (Section 5.1)

Table 5.18: Nine cases drawn at random (seed 20260919) from the out-of-fold predictions of the fixed-parameter LightGBM hazard model (seed 0) under leave-one-campaign-out, corrected parser. The alarm threshold, 0.187, gives a 5% false-positive rate on the pooled out-of-fold rows. Serving RSRP and SINR are synchronous 1 Hz measurements; Gap is serving RSRP minus the best held neighbour RSRP, so a negative gap means the neighbour is stronger (neighbours held under a 3.0 s zero-order hold, median report age ≈ 0.27 s; — means no neighbour was held). Each row gives the values at the scored instant only; Figure 5.20 shows the surrounding traces.

Case	Campaign	Score	To next command (s)	RSRP (dBm)	SINR (dB)	Gap (dB)	Speed (km/h)
true positive	1st Campaign	0.23	0.28	-89	-2	-2	0
true positive	1st Campaign	0.31	0.31	-89	0	-2	38
true positive	4th Campaign	0.36	0.03	-119	-2	-6	30
false alarm	4th Campaign	0.28	3.64	-107	-3	1	35
false alarm	4th Campaign	0.22	5.63	-102	-16	-19	62
false alarm	4th Campaign	0.19	4.28	-87	-2	0	79
missed handover	1st Campaign	0.08	0.82	-90	0	—	45
missed handover	1st Campaign	0.04	0.27	-83	-9	-1	2
missed handover	3rd Campaign	0.09	0.19	-91	-6	-1	24

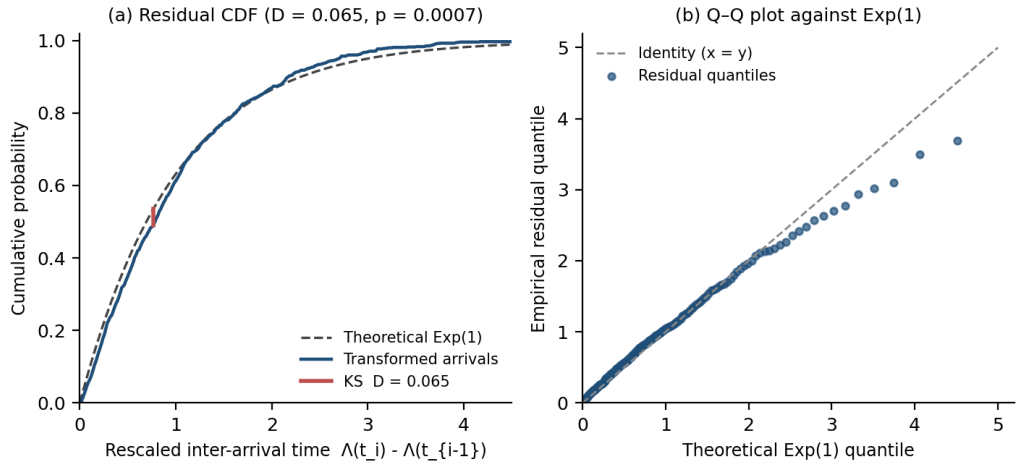


Figure 5.18: Self-excitation in the handover arrival stream. While the Ogata residual test formally rejects the stationary exponential kernel ($D = 0.065, p = 0.0007$, likely due to time-varying background rate and corridor transitions), the fitted intensity captures the strong empirical clustering of handovers into bursts.

Aggregate metrics can hide whether a model has learnt something or is exploiting an artefact, and this thesis reported none of the individual cases until a reviewer observed that the only trace it showed anywhere was the idealised schematic of Figure 1.1. Table 5.18 and Figure 5.20 correct that. Nine rows were drawn uniformly at random, under a fixed seed, from the out-of-fold predictions at the 5% false-positive operating point of Figure 5.2 and Table A.2: three the model scored above threshold with a command following within one second, three it scored above threshold with no command following within one second, and three it scored below threshold with a command following. Nothing was chosen for being illustrative, which is the point of showing them.

Three things are visible in them that no aggregate reports. The first is that real discrete feature traces do not match the idealized continuous trajectory of Figure 1.1: while serving

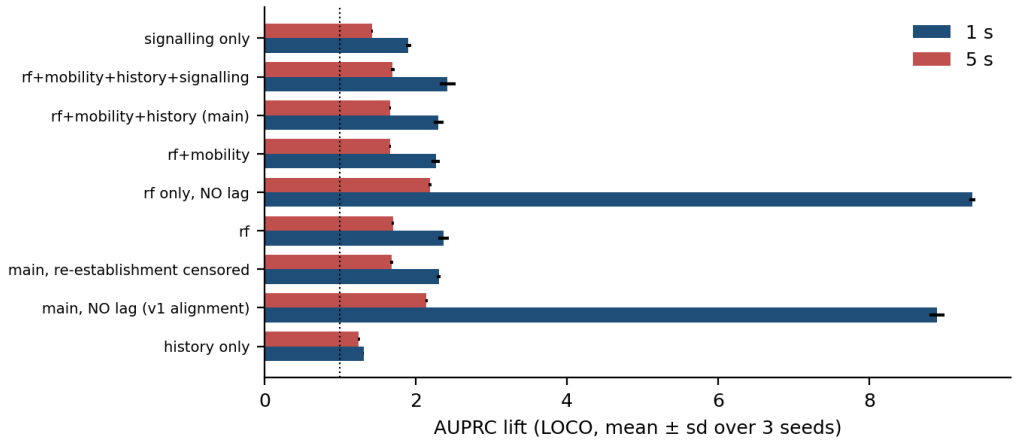


Figure 5.19: Feature-block, alignment and censoring ablations at one and five seconds, out of fold under leave-one-campaign-out.

RSRP is refreshed synchronously at 1 Hz, the best-neighbour curve is causally held from event-triggered reports (with a median report age of 0.27 s, bounded by the 3.0 s expiration window). The curves cross repeatedly in the feature matrix, dip several decibels between consecutive seconds, and recross without a handover command being decoded. Importantly, comparing fresh serving RSRP with older, held neighbour measurements reflects an observed feature pattern on the sample grid rather than a verified simultaneous physical A3 entry condition, which requires continuous Layer 3 filtered evaluations at the UE. The second is that the false alarms are early rather than spurious. In all three drawn false alarms a command does follow, 3.6 to 5.6 s after the scored row, outside the one-second horizon but inside the five-second one; at the scored instant the held neighbour is level with the serving cell or above it (gaps of $+1$, -19 and 0 dB). The model is reading a real approach to a handover and mistiming it. From a 1 Hz grid one cannot tell whether the UE's sub-second evaluation was still inside its time-to-trigger, or whether the margin briefly broke and re-formed before the command. The third is that the misses split into two kinds. One has no held neighbour at all at the scored row, so every neighbour-margin feature is missing. The other two show a neighbour level with the serving cell (gap -1 dB) — the same state the false alarms show — and the model scores them low. On this grid a level margin that ends in a command one second later and a level margin that ends in one five seconds later look alike, which is consistent with the modest ceiling reported throughout Chapter 5 rather than with a model that could be repaired by tuning.

These are nine cases and are offered as such. They cannot establish a rate and are not used to; the rates are in Table 5.4 and Appendix A. What they establish is that the model's errors have a structure a reader can inspect, which is a claim that could not be checked at all before they were shown.

The largest negative result in this thesis is the alignment audit of Section 5.1, and it is recorded here as such: the strongest configuration this project measured — AUPRC 0.604 at one second — was an artefact of a one-second misalignment between the measurement grid

and the event clock, and the corrected figure is 0.156 for the same fixed-parameter model and 0.169 for the tuned one. Retaining that result would have produced a more impressive thesis and a false one.

As recorded in Table 5.17, adding the ten explicit control-plane state features (A3 report counts, hold times, and deployed thresholds) to the main feature set changes the one-second AUPRC from 0.156 to 0.165 under fixed hyper-parameters, a gain no larger than the seed-to-seed spread of 0.004 to 0.007. An earlier draft compared the tuned main arm (then 0.179) with the fixed-parameter signalling arm and read the difference as a loss; compared like for like, the control-plane block adds little, and it is kept out of the main feature set for the reason given in Section 4.6 rather than because it hurts.

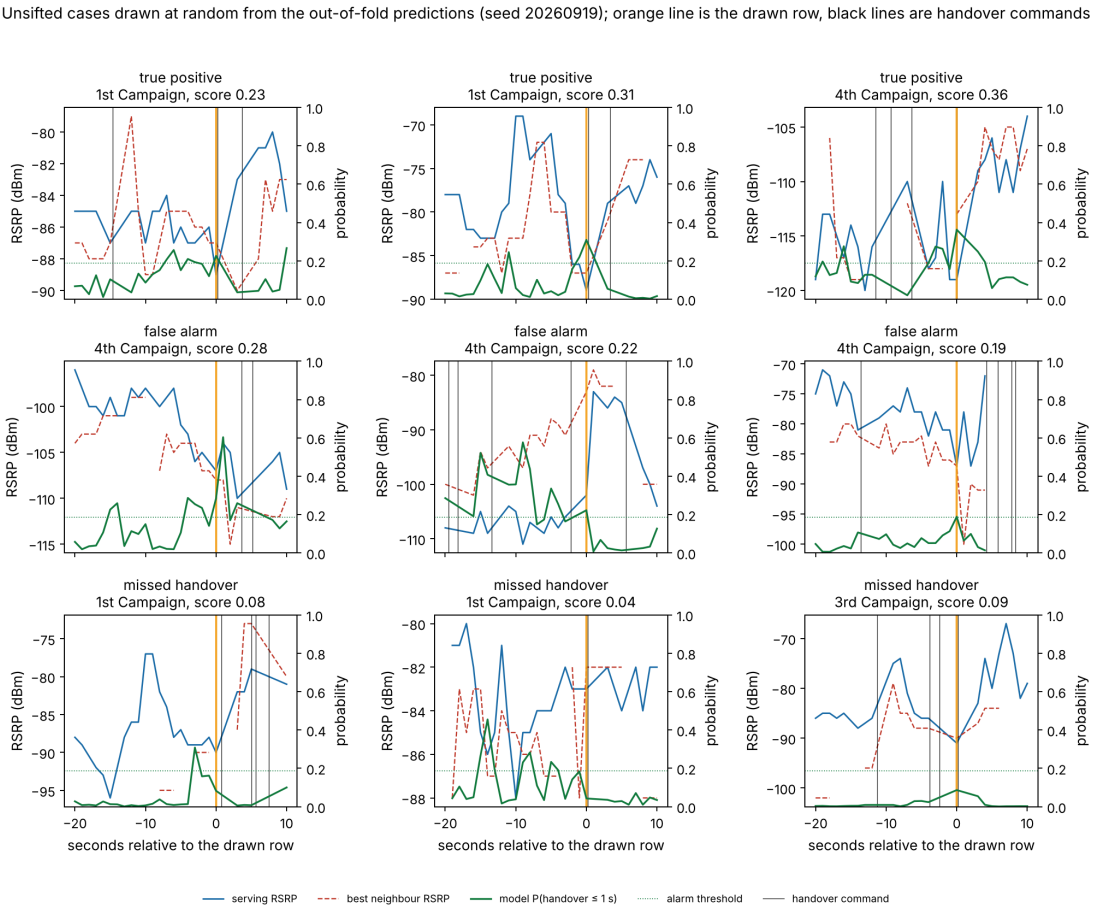


Figure 5.20: Nine cases drawn uniformly at random (seed 20260919) from the out-of-fold predictions at the 5% false-positive operating point (threshold 0.187, dotted green); the orange line marks the drawn row and the black lines mark decoded handover commands. Blue: serving RSRP (1 Hz export); Red: best neighbour RSRP (zero-order hold from decoded reports, with median observation age $\approx$ 0.27 s); Green: model probability of command within 1 s. Traces reflect observed feature patterns rather than continuous internal PHY/RRC state machines.

Finally, removing the signalling leakage guard of Section 4.6 raises the apparent score substantially, which is the expected behaviour of a leak and is the reason the guard exists. That configuration is not reported as a result anywhere in this thesis.

5.12 Blind test on a second operator

Every result above was developed on four campaigns from one operator, here called operator A. A fifth capture was recorded on 9 October 2026, after the modelling freeze and after the alignment audit. It was made with the same handset and XCAL build, on a different operator (operator B). It is therefore blind to every pipeline decision, the one-row lag included, which no earlier campaign is (Section 3.2). The capture is used for one purpose: to test whether the two methodological findings and the predictive level carry over. It is not pooled into the leave-one-campaign-out results, and nothing was re-tuned on it.

Table 5.19: The second-operator capture against the four development campaigns.

Property	Operator B, 9 Oct (blind)	Operator A, range over 4 campaigns
Route	Gazipur	Dhaka urban and Uttara–Gazipur highway
Duration, sessions	17.4 min, 1	24–60 min, 1 each
Usable rows	883	1,131–3,072
180 s blocks	6	8–20
Main band	Band 40 (TDD), about two-thirds of rows	Bands 1, 3, 28, 41
Handover commands (completed)	87 (87)	176–307 (all)
Inter-frequency share of handovers	59%	18–24%
Prevalence at 1 s	5.9%	5.1–8.4%
Dominant trigger	A5, TTT 480 ms (37% of handovers)	A3 +1 dB, TTT 320 ms (44–54%)
Neighbour entries without RSRQ	98%	25–36%

Protocol. The frozen protocol of Section 4.9 is applied unchanged: fitting and calibration rows come from the four operator-A campaigns, hyper-parameters are chosen by the same nested twenty-trial search, results are averaged over seeds 0–2, and the score is computed on the 9 October rows only. This is exactly the fold that leave-one-campaign-out would produce for a fifth campaign. The interval for each AUPRC, and for each difference against LightGBM, is a paired bootstrap over eighteen contiguous 60-second segments. It is a within-session interval and contains no between-day or between-operator variance.

The methodological findings replicate. Table 5.20 repeats the alignment audit and the lag ablation. At the command row the target cell already serves in 66.7% of operator-B handovers, against 4.6% one row earlier, and none of the 20 isolated handovers shows the target before row $t+0$. The step at $t+0$ is the same shape as on operator A, where the per-campaign range is 67–81% at row $t+0$ and 13–19% at row $t-1$. The unlagged model, trained on operator A, scores AUPRC 0.512 on operator B against 0.119 lagged: a 4.3-fold inflation, against 3.9-fold on operator A at the same fixed hyper-parameters. Neither quantity was tuned on this capture, and the lag was fixed before the capture existed. The central methodological claim of this thesis — that a one-hertz XCAL export carries the post-handover cell in the row before the command, and that ignoring it roughly quadruples the apparent skill — therefore holds on a second network.

Table 5.20: Confirmatory checks on the blind capture. Top: share of handovers whose target cell already serves in the export row at each offset. Bottom: fixed-parameter LightGBM trained on operator A, scored on operator B.

Alignment audit	**n**	**row** $t-3$	**row** $t-2$	**row** $t-1$	**row** $t+0$	**row** $t+1$
Operator B, all handovers	87	13.8%	8.0%	4.6%	66.7%	88.5%
Operator B, isolated (previous >8 s earlier)	20	0%	0%	0%	65.0%	95.0%
Operator A, all, per-campaign range	174–297	19–35%	15–30%	13–19%	67–81%	83–88%
Lag ablation	**n**	**AUPRC 1 s**	**Lift 1 s**	**AUROC 1 s**	**AUPRC 5 s**	
Export row stamped t used at t (v1)	884	0.512	8.7×	0.874	0.439	
Export row lagged one sample	883	0.119	2.0×	0.754	0.345	

The predictive level transfers; the ranking does not. Table 5.21 gives every learner on the blind capture. LightGBM, the model this thesis carries through its analyses, reaches AUPRC 0.137 at one second (lift 2.3×, AUROC 0.773). That is inside its operator-A per-campaign range of 0.125–0.223, and its AUROC is the same as the pooled operator-A figure. It is also no better than the deployed A3 proxy on AUPRC (0.159), although it ranks better (AUROC 0.773 against 0.691). The simplest learner does best. Logistic regression reaches 0.261 (4.4×, AUROC 0.842) and beats LightGBM in 99% of paired segment-bootstrap draws, with a difference of +0.124 [+0.025, +0.208]. The three models that use only handover history — dwell, history block and the Hawkes kernel — also score above LightGBM, though their intervals for the difference include zero. The sequence models and the MLP sit at LightGBM's level.

Table 5.21: Blind second-operator test at one second (883 rows, 52 positives, prevalence 5.9%). Trained on the four operator-A campaigns under the frozen protocol. Brackets: paired bootstrap over eighteen contiguous 60 s segments of the one session (within-session; no between-operator variance). The sequence models were not included in the bootstrap.

Learner	**AUPRC 1 s [interval]**	**Lift**	**AUROC**	**ECE**	**Δ AUPRC vs LightGBM [interval]**	**AUPRC 5 s**
Logistic regression	0.261 [0.137, 0.406]	4.4×	0.842	0.132	+0.124 [+0.025, +0.208]	0.552
Hawkes-kernel intensity (LR)	0.192 [0.126, 0.250]	3.3×	0.786	0.028	+0.055 [−0.024, +0.085]	0.526
Dwell time only (LightGBM)	0.190 [0.105, 0.281]	3.2×	0.792	0.030	+0.053 [−0.040, +0.115]	0.497
History block only (LightGBM)	0.175 [0.105, 0.277]	3.0×	0.760	0.041	+0.039 [−0.031, +0.101]	0.450
Event A3 rule (deployed parameters)	0.159 [0.093, 0.231]	2.7×	0.691	0.349	+0.023 [−0.045, +0.053]	0.355
GRU	0.144	2.4×	0.787	0.037	—	0.395
Transformer	0.143	2.4×	0.792	0.040	—	0.431
LightGBM	0.137 [0.099, 0.242]	2.3×	0.773	0.047	—	0.357
TCN	0.136	2.3×	0.775	0.046	—	0.427
MLP	0.135 [0.101, 0.218]	2.3×	0.767	0.058	−0.002 [−0.056, +0.044]	0.384
Prevalence floor	0.059	1.0×	0.500	—	—	0.200

Three readings follow, in order of confidence. First, the hazard formulation, the lagged features and the protocol carry over: every learned model trained on one operator ranks the next second of another operator's handovers at AUROC 0.76 to 0.84, well above the floor. Second, LightGBM's lead on operator A is a within-operator result. Section 5.2 already declined to treat the top of the leaderboard as a finding, and Section 5.3 showed that logistic regression leads under fixed hyper-parameters. On a network the pipeline never saw, the most flexible learners lose their advantage and the least flexible one gains. Third, a likely reason —

a post-hoc hypothesis, not a result — is that the neighbour and gap features do not mean the same thing on the two networks. Operator B reports RSRP only, configures different offsets and hands over mostly between carriers, so a serving-to-neighbour gap on one carrier says little about an inter-frequency A5 trigger. One observation is consistent with this. Under the original parser, which recovered almost none of operator B's neighbour entries (16 of 776), the same fixed-parameter LightGBM scored 0.176 on this capture; with the neighbour features present it scores 0.122. On operator B the neighbour block costs skill rather than adding it.

Calibration separates the learners in the other direction. Logistic regression ranks best but is poorly calibrated (ECE 0.132), while LightGBM and the history-only models stay calibrated (ECE ≤ 0.047). A single learner that both ranks and calibrates well across operators has not been shown.

Risk control on the blind capture. Calibrated on one operator-A campaign and tested on the operator-B session, the row-level procedure of Section 5.5 is feasible at $\alpha = 0.20$ in all four rotations. The realised mean per-unit miss loss is 0.03 to 0.08, well under the target, at alarm rates of 44–69%. The bound holds on this capture but is conservative, which is what one would expect when the calibration and test units are not exchangeable. The test session has six 180-second units, so this result is descriptive only. The per-handover estimand is no better placed than on operator A: 39% of operator-B handovers have no eligible row in their one-second lead window.

The network is different, and some characterisations are operator-specific. On operator B, Event A5 with a 480 ms time-to-trigger drives 37% of handovers, followed by A3 at +5 dB/320 ms (26%) and A3 at -10 dB/640 ms (9%). The operator-A mobility profile (A3 +1 dB/320 ms) is absent. The share of A3 trigger episodes not followed by a matched command, one-to-one, is 87% (operator A: 71–77%), so A3 reports are a weaker proxy for an imminent handover there. Ping-pong under the strict definition is 13.8% (operator A: 19.5–40.3% per campaign), and the carrier pattern of Section 5.10 reverses: 5.6% of 36 intra-frequency handovers return, against 19.6% of 51 inter-frequency ones. The fitted Hawkes branching ratio is 0.83 on this one short session (operator A: 0.49–0.68), with the same caveat on the kernel as in Section 5.10. These are the statistics of one 17-minute drive and are reported as a contrast, not as estimates for operator B. What they establish is narrower: the definitional findings of Sections 5.9 and 5.10 are about how rates are counted and hold on both networks, while the substantive patterns — which carrier relationship returns more, which trigger drives mobility — belong to an operator's configuration.

What this test can and cannot carry. It is one session of 17 minutes, with 52 positive rows at one second, on one route. It is the only data in this thesis that is blind to the alignment

repair, and on that question it is decisive, because the alignment and lag results are large and in the predicted direction. On model choice it is a single external data point, and it argues for caution rather than for a new primary model: selecting logistic regression now, on the strength of this capture, would repeat on a smaller scale the post-hoc selection that the capture has just exposed. A longer campaign on operator B, and a third operator, are listed in Section 7.3.

5.13 Discussion against the objectives

Objective O1 is met in a qualified form, as a predictive signal rather than a usable operating point: the next handover command is predicted within the next second (horizon $h_1 = 1$ s, with an empirical median lead time ≈ 0.50 s) at AUPRC 0.169 against a 6.8% floor (2.5x) and AUROC 0.773, from quantities a handset observes strictly before the prediction instant, against AUPRC 0.114 for the implemented A3 signal-margin proxy. On a blind capture from a second operator the same model reaches 0.137 (2.3x, AUROC 0.773), inside the four-campaign range, but there it does not beat the A3 proxy (0.159) and is beaten by logistic regression (0.261). The qualification is that this is a modest signal, well short of what an uncorrected protocol reports on the same data. Its operational usefulness is not established: the measured operating points (Appendix A) were not chosen against an explicit alarm budget or a minimum useful lead time for a particular network action, the forecast does not name the target cell, and the delays a live system would pay before a feature becomes available were not measured.

Objective O2 is met, and the answer is larger than the modelling differences the thesis reports. Between the leakiest and the soundest protocol the one-second AUPRC of the primary model moves by 49%, and a one-row misalignment moves it by 287% at fixed hyperparameters, measured from the lagged score up to the unlagged one (the same change is -74 % measured downward, as in Table 5.2; on the blind second-operator capture it is 330%) — both larger than the gap between the best and worst learner under a fixed protocol.

Objective O3 is met in part. Coherence is exact by construction, and the conformal formulation is reported with the alarm rate it costs, the feasibility floor set by the number of calibration units, and the measured behaviour in the non-exchangeable regime. The part that is not met is the estimand: the bound evaluated here is over rows, and the per-handover bound the objective asks for is not expressible at one second on this one-hertz grid under the post-handover blanking policy, because 36.4% of commands have no usable row in their lead window. It becomes expressible at three seconds, at an alarm rate near one second in two. The second part that is not met is exchangeability itself: calibrating on one campaign and evaluating on another represents the cross-environment deployment regime. The empirical mean miss loss across held-out campaigns was 0.156 at a target of $\alpha = 0.20$, with 8 of the 12 rotations at or below the target. Calibrated on one campaign of the development operator and tested on the second operator, the mean per-unit miss loss was 0.03 to 0.08 at the same target — conservative, at alarm rates of 44–69%. However, cross-campaign results cannot be

certified as formal distribution-free guarantees because macro-environmental shifts violate exchangeability, and the 12 pairwise campaign rotations exhibit pseudoreplication across $n = 4$ physical sessions. Both limitations are properties of the operational setting rather than of the procedure, and the objective is reported as partly met on that basis rather than restated to fit what was achieved.

Objective O4 is partially met with structural confounding: leave-one-campaign-out achieves AUROC between 0.768 and 0.797 across held-out sessions, including 0.770 on the highway corridor. However, vehicle speed and corridor geography are structurally collinear on Campaign 4, preventing unique attribution of the performance gap, and Campaign 4 was evaluated in retrospective diagnostic exploration of the alignment repair. Furthermore, cross-corpus transfer achieves AUROC 0.760 on a same-institution public dataset from the same operator and metropolitan area [38], but this transfer is evaluated on common L3 signalling features rather than the complete 112-feature multi-modal model due to schema limitations in the public release. The blind second-operator test of Section 5.12 is the first evaluation that is blind to every pipeline decision. It shows that the formulation's level transfers (AUROC 0.773, lift 2.3x) while the ranking of learners does not.

Objective O5 is met in corrected form. The single strongest lesson is methodological — the alignment audit — and the substantive one is that the quantity Event A3 thresholds on is a weak predictor of imminence (AUROC 0.601 alone), while no single available feature replaces it. Both methodological findings — the end-of-second flip and the lag inflation — replicate blind on the second operator.

5.14 Threats to validity

The methodological and empirical findings of this thesis are subject to threats across four standard dimensions of validity, which are explicitly addressed below.

5.14.1 Internal validity

Internal validity concerns whether the observed relationships between features, protocols, and event occurrences are genuine rather than artefacts of measurement or preprocessing.

- **Export timestamp alignment and feature ageing:** As documented in Section 5.1, periodic 1 Hz logging exports exhibit end-of-second aggregation semantics, corrupting pre-handover rows with post-handover serving-cell identities in 76% of cases. Applying a one-row lag removes this look-ahead contamination under the inferred export semantics (Table 5.1 shows no step at row $t-1$), but introduces one second of information ageing across uncorrupted radio and kinematic features. The 211 residual cases where $m > 0$ yet row t shows the source cell are consistent with a one-row exporter refresh delay rather than a failure of the aggregation model (Table A.6). The exporter's rule itself is inferred, not documented; only a native-rate re-export of these captures can confirm it and separate leak removal from feature ageing.

- **Residual burst contamination at row $t-1$:** Row $t-1$ shows the target cell in 11.7% of isolated handovers and 15.6% of all handovers. In both subsets this is below the subset's own earlier rows (12.1%/12.1% and 21.7%/23.7% at rows $t-2$/$t-3$), so there is no step at row $t-1$; the higher pooled level reflects ping-pong returns to recently serving cells. Model lift is also highest on quiet rows (3.0× vs 1.8× in bursts; Section 5.8, Table 5.13), which is inconsistent with residual burst contamination driving the result.

- **Protocol design leakage on Campaign 4:** Although Campaign 4 was held out during model parameter estimation and hyperparameter selection, it was examined during retrospective diagnostic exploration of the timestamp alignment defect; hence Campaign 4 is an out-of-sample test unit for learned model weights, but not a blind prospective holdout for pipeline feature engineering.

- **Preprocessing and blanking window policy:** To prevent post-handover transient states from contaminating consecutive predictions, an operational 2 s blanking window is enforced following each handover command. While necessary to avoid predicting trivial immediate successors, this policy mechanically sets the maximum achievable fraction of resolvable handovers at 64.2% at the 1 s horizon (leaving 35.8% unresolvable in continuous driving, rising to 36.4% under chunk boundary cuts).

- **Post-hoc parameter thresholds and repository freeze:** The 180 s block duration, 60 s temporal purge, 10 s burst cutoff, and 2 s blanking window were selected during protocol design and are not recorded in the pre-Campaign-4 freeze manifest (Section 3.2), which was itself not registered externally; they should be read as post-hoc choices. Sensitivity analysis across parameter ladders (Table A.5) confirms local numerical stability around chosen values.

5.14.2 External validity

External validity concerns the generalisability of these findings beyond the specific measurement corpus.

- **Instrumentation and single-device limits:** All four campaigns were recorded using a single commercial drive-test handset model and a single Accuver XCAL diagnostic firmware build. The NUWiNS experiment shows that end-of-second aggregation applied to native-rate XCAL data is sufficient to produce the artefact; it does not show that every XCAL one-hertz export, or this one, aggregates that way. Handset-specific baseband logging, timestamp rounding and diagnostic buffer commit latencies cannot be separated from one another without native-rate, multi-device, multi-firmware replication.

- **Carrier and operator coverage:** The four modelling campaigns were conducted on a single commercial LTE operator in Dhaka and Gazipur. One further 17-minute session

on a second operator (Section 5.12) shows that the alignment flip and the lag inflation replicate there, that the formulation's level transfers, and that the ranking of learners and the carrier pattern of ping-pong do not. One short session is a single external data point, not an estimate of cross-operator performance; heterogeneous small-cell deployments and other vendors remain untested.

- **Speed and corridor confounding:** In Campaign 4, vehicle velocity (50.9 km/h vs 21.4–39.1 km/h) and corridor topography (suburban highway vs dense urban corridors) are collinear. The performance variation observed in Section 5.6 cannot be uniquely separated into velocity-induced Doppler fading versus spatial base-station layout.

- **Schema limits in same-institution transfer and departmental lineage:** Same-institution validation on the Shafi et al. [38] dataset was restricted to the 6-feature L3 signalling block because the public release lacked synchronous 1 Hz radio metrics and GPS coordinates. Furthermore, that dataset was collected in the same metropolitan area on the same operator by an earlier research cohort in our department; transferability of the complete 112-feature multi-modal model to distinct operators and international markets is only partly addressed by the single second-operator session of Section 5.12.

5.14.3 Construct validity

Construct validity assesses whether the operationalized variables accurately reflect the underlying theoretical constructs.

- **Signal-margin baseline vs. the mobility procedure:** The implemented Event A3 baseline evaluates the radio margin condition $(M_n - M_s > \text{Off} + \text{Hys})$ on the discrete 1 Hz export grid. It serves as an empirical signal-margin proxy, not an emulation of the UE's sub-second time-to-trigger evaluation and reporting, nor of the network's subsequent handover decision.

- **Measurement-report parsing:** Neighbour features are decoded from MeasurementReport messages. TS 36.331 makes the RSRQ field of a neighbour entry optional, and the first parser required it; it dropped RSRP-only entries and mis-paired some others. The corrected parser (Section 3.3) is used throughout. It was fixed on the basis of the message grammar, without reference to any model output, and the four-campaign results moved by at most 0.011 AUPRC at one second.

- **Recorded timestamp vs. feature availability:** A feature is treated as available at the timestamp of the row it comes from. A live predictor would also pay acquisition, export, feature-processing and inference delays, none of which were measured, so the lead times reported here are upper bounds on what a deployed system would obtain.

- **Occurrence vs. target:** The forecast concerns whether any handover command occurs within the horizon, not which cell it will name. Target-cell preparation needs the target as well and a lead time long enough to complete preparation, so detection rates here bound only part of that use case.

- **Report–command attribution:** Reports are matched to commands within a 2 s window; "unmatched" does not mean the network declined a report, and temporal association does not establish that a report triggered a command. Sensitivity to the window and to target-cell matching was not measured.

- **Signalling commands vs. handover necessity:** Ground truth labels are derived from decoded `RRCConnectionReconfiguration` commands with confirmed completions. While this verifies that a physical handover took place, observational traces cannot establish counterfactual necessity—whether retaining the serving cell would have caused radio link failure or whether the handover was suboptimal.

5.14.4 Statistical conclusion validity

Statistical conclusion validity concerns the appropriateness of the statistical methods and sample sizes used to draw inferences.

- **Limited degrees of freedom ($n = 4$):** With four continuous drive campaigns, two-sided non-parametric sign tests are bounded by $p \geq 0.125$. Accordingly, all cross-campaign comparisons are reported as empirical effect sizes, spreads, and paired differences rather than claiming null-hypothesis rejection.

- **Temporal autocorrelation and block bootstrap intervals:** While 180 s block bootstrapping provides a descriptive measure of variance, contiguous blocks within the same drive share macro-environmental conditions. Consequently, within-session bootstrap intervals are anti-conservative, and true between-session uncertainty is represented by the LOCO spread [0.125, 0.223].

- **Finite-sample floors in conformal risk control:** Conformal risk control requires $n \geq 1/\alpha - 1$ calibration units to achieve non-trivial threshold selection. For small campaigns ($n \leq 20$), tight targets ($\alpha \leq 0.05$) mathematically exceed this feasibility floor, defaulting to constant alarm fallback. Furthermore, the 12 pairwise cross-campaign rotations exhibit pseudoreplication ($n = 4$ degrees of freedom) and violate exchangeability, rendering cross-campaign risk bounds descriptive diagnostics rather than certified operational guarantees.

- **Multiplicity control:** Model leaderboard comparisons among the seven learners are treated descriptively without multiplicity adjustment, as differences fall within the within-session bootstrap intervals and within the per-campaign spread. Conversely,

the per-campaign goodness-of-fit tests of the point-process model (Section 5.10) are corrected with the Holm–Bonferroni procedure across the four campaigns; the pooled test is a single test.

CHAPTER 6

DEMONSTRATION OF OUTCOME BASED EDUCATION

This chapter demonstrates how the work reported in Chapters 1 to 5 addresses the course outcomes, program outcomes, knowledge profiles and complex engineering problem and activity attributes required of a capstone project in EEE 4700/4800. Each claim made here refers to a specific section of the thesis rather than asserting an attribute in the abstract, and attributes that the work does not genuinely address are marked as such rather than claimed.

6.1 Introduction

Outcome Based Education requires that a capstone project be assessed against what the student is able to do at its conclusion rather than against the volume of work performed. The sections that follow map the activities of this project onto the defined outcomes, in the order set out in the departmental thesis format: course outcomes in Section 6.2, aspects of program outcomes in Section 6.3, knowledge profiles in Section 6.4, the use of complex engineering problems in Section 6.5, the socio-cultural, environmental and ethical impact in Section 6.6, and the attributes of complex engineering problem solving and complex engineering activities in Sections 6.7 and 6.8, and project management, resources and budget in Section 6.9.

6.2 Course outcomes addressed

Table 6.1 maps each course outcome to the part of this thesis that provides the evidence for it.

6.3 Aspects of program outcomes addressed

Table 6.2 lists the program outcomes this project addresses and where each is evidenced.

6.4 Knowledge profiles addressed

Table 6.3 lists the knowledge profiles the project draws on, with a pointer to the chapter where each is used.

6.5 Use of complex engineering problems

The problem addressed here satisfies the defining characteristics of a complex engineering problem in several independent respects.

It involves conflicting technical requirements whose trade-offs must be resolved quantitatively rather than asserted. The clearest is the warning trade-off of Section 5.5: an earlier and more sensitive warning is more useful to the network and also more expensive, and the

Table 6.1: Course outcomes addressed in this project.

CO	CO statement	PO	Addressed
CO1	Apply knowledge of electrical and electronic engineering to the solution of complex engineering problems	PO1	√
CO2	Identify a contemporary real-life problem related to electrical and electronic engineering by reviewing and analysing existing research works	PO2	√
CO3	Determine functional requirements of the problem considering feasibility and efficiency through analysis and synthesis of information	PO4	√
CO4	Select a suitable solution and determine its method considering professional ethics, codes and standards	PO8	√
CO5	Adopt modern engineering resources and tools for the solution of the problem	PO5	√
CO6	Prepare management plan and budgetary implications for the solution of the problem	PO11	√
CO7	Analyse the impact of the proposed solution on health, safety, culture and society	PO6	√
CO8	Analyse the impact of the proposed solution on environment and sustainability	PO7	√
CO9	Develop a viable solution considering health, safety, cultural, societal and environmental aspects	PO3	√
CO10	Work effectively as an individual and as a team member for the accomplishment of the solution	PO9	√
CO11	Prepare various technical reports, design documentation, and deliver effective presentations for demonstration of the solution	PO10	√
CO12	Recognise the need for continuing education and participation in professional societies and meetings	PO12	√

frontier from a 20% target miss rate at a 48% alarm rate to a 5% target at 85% is measured rather than assumed.

It is not amenable to a standard procedure. The evaluation protocol the comparator literature treats as standard is demonstrably wrong on this data, and Section 5.3 measures by how much; so is the implicit assumption that a measurement row describes the interval before its own timestamp, which Section 5.1 measures.

It spans several sub-disciplines: the LTE control plane, radio propagation, statistical survival analysis, distribution-free inference, point-process modelling and software engineering, and no single one of them is sufficient.

6.6 Socio-cultural, environmental and ethical impact

Environmental and sustainability. The environmental argument made here is confined to what was measured. Section 5.10 shows that 26.3% of the handovers observed returned to the cell just vacated within fifteen seconds (Table 5.15), and that the phenomenon is concentrated in intra-carrier transfers under one dominant configuration profile. Each such pair is signalling work, radio resource occupancy and processing load expended for no change in

Table 6.2: Aspects of program outcomes addressed, with the evidence supplied by the thesis.

PO	Aspect	Addressed	Evidence
PO1	Engineering knowledge	√	LTE mobility procedures and RRC signalling, Sections 2.1 and 3.3; survival analysis and conformal risk control, Chapter 4
PO2	Problem analysis	√	Problem statement and requirements, Section 1.3; protocol audit, Section 2.3
PO3	Public health	—	No health claim is made; see Section 6.6
PO3	Safety	√	Connection continuity and radio-link failure, Section 1.2
PO3	Societal	√	Section 6.6
PO3	Environmental	√	Measured unnecessary signalling, Section 5.10
PO4	Design of experiments	√	Grouped-drive rotation and seed repetition, Section 4.9
PO4	Analysis and interpretation of data	√	Chapter 5 in its entirety
PO4	Synthesis of information	√	Protocol audit of 22 models, Section 2.3
PO5	Modern tool usage	√	Drive-test instrumentation and analysis stack, Section 3.1 and Appendix C
PO6	Societal	√	Section 6.6
PO6	Legal	√	Data handling and operator anonymity, Section 6.6
PO7	Societal and environmental	√	Section 6.6
PO8	Professional ethics and responsibilities	√	Reporting of negative results, Section 5.11
PO8	Norms	√	Conformance to 3GPP TS 36.331 terminology throughout
PO9	Individual and team work	√	Three-member team sharing field campaigns, decoding, modelling and writing, Section 6.9
PO10	Comprehend and write effective reports	√	This report
PO10	Design documentation	√	Chapters 3 and 4
PO10	Make effective presentations	√	Progress and defence presentations
PO11	Engineering management principles	√	Section 6.9
PO11	Economic decision-making	√	Section 6.9
PO12	Life-long learning	√	Methods adopted from outside the field (discrete-time survival analysis, conformal risk control, point processes), Chapters 2 and 4

Table 6.3: Knowledge profiles addressed, with justification.

K	Justification
K3	A systematic, theory-based formulation of engineering fundamentals: the discrete-time survival formulation of Section 4.3, including the product-limit identity and its monotonicity proof, and the censoring-aware likelihood of Section 4.4
K4	Specialist knowledge at the forefront of the discipline: the decoding of RRC measurement configuration by replaying VarMeasConfig, whose measId bindings persist until removed, Section 3.3, and the recovery of deployed Event A3 parameters, Section 3.5
K5	Knowledge supporting engineering design in a practice area: the feature construction, model selection and calibration design of Sections 4.6 to 4.8, each chosen against a stated alternative
K6	Knowledge of engineering practice and technology: the conduct of four drive-test campaigns with commercial diagnostic instrumentation on a live operator network, Chapter 3
K7	Comprehension of the role of engineering in society: the impact analysis of Section 6.6, including the explicit refusal to make a health claim the measurements do not support
K8	Engagement with research literature: the screening of 108 publications and the protocol audit of 22 comparable models in Section 2.3, together with the methodological provenance recorded in Sections 2.5 to 2.8

serving cell. Identifying that fraction is a precondition for reducing it. This thesis does not claim a measured energy saving, because no such measurement was made, and Section 1.5 states why a causal benefit claim is not identified on data of this kind.

Ethical and legal. Four ethical considerations arose and were handled explicitly. Measurements were collected from the authors' own handsets and subscriptions, so no third-party user data was recorded and no subscriber identifiers appear in the dataset. The operator is not named, because the purpose of reporting its configuration is scientific rather than comparative. Negative results were reported rather than omitted, including the one in Section 5.1 that removed this project's most impressive number. And the dataset used for the independent check in Section 5.7 was produced in the same department as this work, which is stated in the text rather than left for a reader to discover.

Declaration of competing interests. The supervisor of this thesis is a co-author of the publicly released dataset used for the same-institution check of Section 5.7 [38] and of the comparator method discussed in Section 5.7 and reimplemented in stage 20 of Appendix C [39]. Neither the dataset nor the comparator was modified for this work; both were taken as released, and the results involving them are reported in the same form as every other result. The relationship is declared here because the word independent, applied to a collection produced by an overlapping group on the same operator with the same instrument, would otherwise carry more weight than the evidence supports. The authors declare no other competing interest.

Cultural. The measurement campaigns were conducted on public roads under ordinary traffic conditions, and the driving itself conformed to local traffic regulation. No aspect of the work required access to private premises or to restricted infrastructure.

6.7 Attributes of complex engineering problem solving addressed

Table 6.4: Attributes of complex engineering problem solving addressed.

P	Attribute	Addressed	How
P1	Depth of knowledge required	√	Survival formulation and its monotonicity proof, Section 4.3; conformal risk control, Section 4.10
P2	Range of conflicting requirements	√	The warning frontier of Section 5.5: lead time against false-alarm cost, measured across the whole curve
P3	Depth of analysis required	√	No standard procedure applies; the standard evaluation protocol is shown to be invalid here, Section 5.3
P4	Familiarity of issues	√	The deployed configuration was unknown before decoding, Section 3.5: the offset that drives most handovers (+1 dB) has the sign the literature assumes but a smaller magnitude, and the negative offsets belong to monitoring configurations that rarely lead to a handover
P5	Extent of applicable codes	√	3GPP TS 36.331 and TS 36.300 govern the signalling interpretation throughout Chapter 3
P6	Extent of stakeholder involvement	√	Operator, subscriber and regulator interests differ; Section 6.6 states which are and are not addressed
P7	Interdependence	√	Six sub-problems — decoding, labelling, feature design, formulation, evaluation and risk control — each of which constrains the others

6.8 Attributes of complex engineering activities addressed

Table 6.5: Attributes of complex engineering activities addressed.

A	Attribute	Addressed	How
A1	Range of resources	√	Commercial drive-test instrumentation, decoded control-plane logs, a gradient-boosting and deep-learning software stack, and 108 screened publications
A2	Level of interaction	√	Resolution of conflicting requirements between prediction horizon, calibration quality and alarm cost, Sections 5.4 and 5.5
A3	Innovation	√	The application of a discrete-time hazard formulation and a drive-level conformal risk bound to mobility prediction, neither of which appears in the audited literature
A4	Consequences for society and the environment	√	Section 6.6, bounded to what was measured
A5	Familiarity	√	The problem falls outside standards-governed practice: no 3GPP procedure specifies handover prediction, and no standard definition of ping-pong exists, as Section 5.10 demonstrates quantitatively

6.9 Project management, resources and budget

The project was conducted over two academic terms. The first term covered problem identification, the literature screening of Section 2.3 and the construction of the decoding pipeline; the second covered the four measurement campaigns of 2026, the formulation and modelling of Chapter 4, the evaluation of Chapter 5 and the preparation of this report. The alignment audit of Section 5.1 was performed during the final revision, after all four campaigns and after the first complete draft, and it required every experiment in Chapter 5 to be re-run.

Table 6.6: Resource and cost summary for the project.

Item	Justification	Cost (BDT)
Drive-test instrumentation	Departmental equipment, used under supervision	Nil (departmental)
Mobile data and subscription	Four measurement campaigns on a live network	3,000
Vehicle fuel and hire	Approximately 96 km across four campaigns, including the highway run	9,000
Computation	Model training and evaluation on personal and departmental hardware	Nil
Report production and binding	Three hard-bound copies as required	3,000
Total		15,000

CHAPTER 7

CONCLUSIONS

7.1 Summary of findings

This thesis set out to determine whether the next LTE handover can be forecast from the measurements a handset already reports, how far in advance, and with what guarantee. On four drive-test campaigns conducted on a live commercial network in Dhaka and Gazipur — four continuous sessions, 10,260 samples and 957 handover commands whose ground truth is decoded from RRC signalling (yielding 938 quality-controlled commands inside retained session blocks, and 934–937 horizon-evaluable events under lead boundaries) — the answer is a qualified yes, and the qualification is the first finding.

The measurement grid must be aligned to the event clock before anything else is believed. In 76% of handovers the one-hertz export row stamped before the handover command already carries the post-handover serving cell. Lagging every export feature by one row moves the one-second AUPRC of the same fixed-parameter model from 0.604 to 0.156, and the dwell feature's own AUROC from 0.873 to 0.664. While Campaign 4 serves as an out-of-sample holdout for model weights, it was part of the retrospective diagnostic exploration that derived the one-row lag repair; C4 is therefore an out-of-sample test for model parameter fitting, but not a blind holdout for pipeline feature engineering. A fifth capture on a second operator, recorded after the freeze, closes that gap for the central claim: there the row at the command already carries the target in 67% of handovers against 5% one row earlier, and the unlagged model is inflated 4.3-fold. The audit costs three lines of code and is the recommendation this thesis makes most strongly.

The next handover is predictable, modestly. Gradient-boosted trees under a discrete-time hazard formulation reach pooled AUPRC 0.169 across campaigns (observed range over held-out campaigns [0.125, 0.223], not a prediction interval for a new campaign; within-session block-bootstrap 95% interval [0.150, 0.196], anti-conservative) at one second against a 6.8% floor (2.5x, AUROC 0.773) and AUPRC 0.454 at five seconds (1.7x, LOCO spread [0.385, 0.531]), against AUPRC 0.114 for the implemented A3 signal-margin proxy scored as a predictor. On the blind second-operator capture the same model reaches 0.137 (2.3x, AUROC 0.773), inside the four-campaign range. There, however, logistic regression reaches 0.261 and the history-only models also score higher, so the lead of the gradient-boosted trees is a within-operator result and "primary model" is a label for the learner carried through the analyses, not a finding. Objective O4 is partially met with structural confounding, because vehicle speed and corridor geography are collinear on the highway campaign, and cross-corpus evaluation on a same-institution public dataset collected in the same metropolitan area on the same

operator by an earlier research cohort in our department [38] was constrained to common L3 signalling features.

The unit of splitting is worth more than the model. Against leave-one-campaign-out, random-row splitting inflates the one-second AUPRC by 49% and random 180-second blocks of the same session by 12%; the inflation is architecture-dependent, so a careless protocol reorders the leaderboard rather than merely raising it.

A single hazard model makes the five horizons mutually consistent, and the correct comparison for that property is a monotone projection rather than raw independent classifiers. Independent heads violate ordering on 38.9% of rows; a cumulative-maximum projection, isotonic calibration composed with that projection, and the hazard model all reach zero. The hazard model calibrates better than the projections that cost nothing (0.037 against 0.043 at one second) and worse than the isotonic composite (0.015), which spends a calibration split this dataset cannot spare.

Conformal risk control evaluated over rows yields an empirical miss rate at or below the target within pooled 180-second blocks (0.171 at $\alpha = 0.20$, at a 48% alarm rate); calibrating on one campaign and evaluating on another, the pooled empirical mean miss loss across the 12 pairwise campaign rotations was 0.156 at target $\alpha = 0.20$, with 8 of the 12 rotations at or below the target; tested on the second operator, the per-unit miss loss was 0.03 to 0.08 at the same target, a conservative result. However, cross-campaign distribution-free guarantees cannot be certified because macro-environmental shifts violate exchangeability, and the 12 pairwise permutations exhibit pseudoreplication over $n = 4$ physical sessions. Over handovers, which is what the objective promised, no useful bound is expressible at one second, because 36.4% of commands have no usable row in their lead window (under the two-second post-handover blanking policy and boundary exclusions) and are missed whatever the threshold. The grid together with the evaluation policy, rather than model discrimination alone, is the binding constraint on what a study like this may promise. All of these conformal results are empirical calibration diagnostics of a procedure whose scores come from four fitted models; none is an operating guarantee for a deployed predictor.

Two properties of the deployed network were characterised, both corrected against an earlier draft, and a third quantity was shown to have no unit-free value at all. The share of A3 reports with no matched command within a two-second attribution window reads 63% per transmitted report, 30% per trigger episode when overlapping episodes may claim the same command, and 73% once each command is matched to exactly one episode; what survives every unit is the separation between the mobility configuration and the monitoring configurations (37% unmatched against 95%). Unmatched is a counting statement, not an observation that the network declined a report. Weighted by handovers rather than by configurations, 75% of A3-triggered handovers fire under a positive offset. And the ping-pong rate on one fixed set of 957 handover commands moves from 26.3% to 43.8% depending on three definitional choices that published work routinely leaves unstated. The definitional findings hold on the

second operator too; the substantive patterns do not. There Event A5 drives most handovers, 87% of A3 trigger episodes go unmatched one-to-one, and ping-pong is more frequent across carriers than within one — the reverse of the first operator.

Underlying all of these is the most compact statement of what this work found: on drive-test data, the protocol and the timestamp semantics decide the result before the model does. The second-operator test adds a corollary: the timestamp semantics travel with the instrument, while the best model and the network's habits stay with the operator.

7.2 Limitations

The limits of the present study are restated here because they determine what follows from it.

The evidence base is four continuous sessions, one operator, four days and one radio access technology, with one further 17-minute session on a second operator. The deployed configuration is stable across all four campaigns, so the cross-regime transfer of Section 5.6 rests on a single configuration set. The second-operator session tests what a second corridor cannot, and it shows that the model ranking and the carrier pattern of ping-pong change with the network. One short session cannot say by how much: it has 52 positive rows at one second and six 180-second blocks, and its intervals are within-session. Neighbour features depend on a parser fix made after the four campaigns were analysed (Section 3.3); every model analysis was re-run with it, and it changed the headline four-campaign results by at most 0.011 AUPRC. All captures were conducted using a single commercial drive-test handset running a single Accuver XCAL firmware build; the NUWiNS experiment shows that end-of-second aggregation is sufficient to produce the artefact, not that this exporter applies it, and handset-specific baseband logging, timestamp rounding or diagnostic buffer commit latencies cannot be separated without native-rate, multi-device, multi-firmware replication. The operating points of Appendix A show predictive signal but not usefulness for a particular network action: they were not set against an explicit alarm budget or minimum useful lead time, the forecast does not name the target cell, and the delay before a feature becomes available to a live predictor was not measured. The combination of the one-hertz export and the two-second post-handover blanking policy caps event-level detection and fixes the information cut-off one sample period before the prediction instant, so every figure here is conservative by that margin and exact semantics await a native-rate export.

Declaration of competing interests. The supervisor of this thesis is a co-author of the same-institution public dataset used for cross-corpus validation in Section 5.7 [38] and of the comparator reinforcement-learning study discussed in Section 5.7 [39]. Neither the dataset nor the comparator was modified for this work; both were taken as released, and results are reported under our leakage-audited protocol. We declare no other competing financial or personal interests.

7.3 Recommendations for future work

Seven items follow from those limits, in the order the limits dictate.

A native-rate export and a published alignment statement. The single most valuable follow-up is a re-export of the same captures at the message rate rather than at one hertz. Section 5.1 infers the write rule from its consequences and shows on a native-rate corpus collected elsewhere that the inferred rule is sufficient to produce the effect; a native-rate export of these captures, compared under documented start-, end- and instantaneous-sampling rules, would replace inference with observation, separate leak removal from the one-second ageing penalty, and allow horizons below one second to be modelled rather than approximated.

Multi-device and multi-firmware cross-instrumentation. The alignment artifact in Section 5.1 was audited on a single commercial handset running a single Accuver XCAL firmware release. The NUWiNS experiment shows that end-of-second aggregation is sufficient to produce the artefact on XCAL data; multi-device replication across distinct modem chipsets (e.g., Qualcomm Snapdragon, MediaTek Dimensity, and Samsung Exynos) and diagnostic loggers is needed to separate baseband commit latencies from tool-specific write buffering.

More sessions and prospective public pre-registration. Eight to ten independent sessions across days, times and routes would make a paired test over units informative, tighten the conformal feasibility floor and give the block bootstrap something to be conditional on. Furthermore, future campaigns should be registered in an external public registry prior to data acquisition to ensure that holdouts are blind to both model fitting and pipeline feature engineering, addressing the retrospective lag-repair limitation identified in Campaign 4.

A longer second-operator campaign and a third operator. The 17-minute session of Section 5.12 shows that the methodological findings transfer and that the learner ranking does not. It is too short to say which learner should be preferred across operators, or why. A multi-session campaign on the second operator, and a third operator, would turn that single data point into a between-operator comparison. They would also test the hypothesis that neighbour-gap features do not transfer between an A3-driven intra-frequency network and an A5-driven inter-frequency one, and they would break the single-configuration limitation, since the offsets reported in Section 3.5 are one operator's choice rather than a property of LTE. Any learner chosen for cross-operator use should be selected on such data before it is evaluated, not on the session that exposed the problem.

An experimental test of the time-to-trigger. Section 5.10 withdraws the claim that the time-to-trigger is the lever governing ping-pong, because the configuration set does not vary anywhere in this data and the claim is therefore unsupported rather than merely uncertain.

It remains the most plausible lever on theoretical grounds, and varying it under controlled conditions while measuring the resulting return rate is the direct test. Such an experiment would also supply the treatment variation that the causal design above requires, and would break the confounding between speed, route and cell density that four campaigns cannot.

Operational and calibration evidence the present tables do not contain. Four analyses would turn the predictive signal into an operational argument, and all of them can be computed from the existing out-of-fold predictions: (i) recall evaluated against an explicit alarm budget and a minimum useful lead time for a named action such as target-cell preparation, with acquisition, export, feature-processing and inference delays measured or modelled; (ii) reliability curves with per-bin counts, calibration slope and intercept, and fold-specific constant-probability baselines; (iii) the event-level detection–false-alarm trade-off under 0, 1 and 2 s post-handover blanking; and (iv) the sensitivity of the A3 report–command attribution to the matching window and to a target-cell matching rule. An online adaptive conformal variant [53], evaluated sequentially across campaigns, would test a long-run guarantee that the present design cannot.

Release of code, data, and verifiable checksums. The protocol and alignment arguments of Sections 5.1 and 5.3 are claims about how results in this field should be produced, and such claims are only worth making if others can reproduce and contest them — Table 2.1 grades 22 published models on exactly that property, so this work's own two rows in it have to say something specific. The alignment audit, the four evaluation protocols, the feature construction and the table-generation pipeline are kept in the project repository and are to be released as a versioned open-source archive with this thesis, together with the derived row-level frames, the decoded event streams, and the stage scripts. Appendix C documents the pipeline execution DAG, artifact layout, git provenance, and SHA-256 verification hashes for all derived frames. The raw vendor logs are not redistributable, which is stated rather than elided; the derived frames are sufficient to reproduce every number in Chapter 5.

7.4 Concluding remark

The contribution of this work is not a new learning algorithm, and it is worth closing by saying so directly. Gradient boosting, discrete-time survival analysis and conformal risk control are all established methods, and this thesis is explicit throughout about where each came from. What it offers is measurement-grade ground truth decoded from signalling, a formulation that makes multi-horizon outputs coherent by construction, an evaluation protocol whose unit is defended and whose alternatives are measured, a conformal calibration diagnostic whose exchangeability assumption is examined in both regimes rather than assumed, and an alignment audit that removed this project's own most impressive result. On the evidence of the protocol audit in Section 2.3, the parts of that list the field most consistently omits are the

least novel ones.

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APPENDIX A

FULL PER-HORIZON RESULTS

Table A.1: Per-horizon AUPRC for every learner under leave-one-campaign-out, against the prevalence floor (tuned arm, corrected parser).

Learner	0.5 s	1 s	2 s	3 s	5 s
LightGBM	0.097	0.169	0.268	0.342	0.454
MLP	0.098	0.168	0.293	0.380	0.493
TCN	0.104	0.163	0.282	0.369	0.487
Transformer	0.093	0.149	0.258	0.336	0.450
GRU	0.076	0.130	0.242	0.330	0.449
Hawkes-kernel intensity (LR)	0.070	0.120	0.210	0.285	0.403
Logistic regression	0.066	0.119	0.210	0.286	0.408
Event A3 rule (deployed parameters)	0.068	0.114	0.201	0.268	0.370
History block only (LightGBM)	0.066	0.114	0.203	0.277	0.397
Dwell time only (LightGBM)	0.062	0.105	0.190	0.258	0.367
Prevalence floor	0.037	0.068	0.126	0.176	0.261

Table A.2: Event-level operating characteristics of the primary model at the 5% false-positive operating point. Eligibility rule: a command at τ is counted if it lies between the held-out campaign's first usable prediction row and h seconds after its last usable row ($t_{\min} \le \tau \le t_{\max} + h$), so the count rises with the horizon (934 to 937 of the 938 quality-controlled commands). "Resolvable" means at least one usable row lies in $[\tau - h, \tau)$; unresolvable commands are counted as missed. At 1 s, 604 of the 935 eligible commands (64.6%) are resolvable. Sections 3.6 and 5.5 quote 64.2% (602 of 938) from a separate count over all 938 quality-controlled commands on the continuous sequence; the two differ in denominator and by two commands, so they do not sum with each other's complement. The operating threshold is set per fold at a 5% row-level false-positive rate on that fold's calibration block.

Horizon	Events	Resolvable on the 1 Hz grid	Detected	Detection rate	False alarms / h	False alarms / km	Median lead time
0.5 s	934	34.2%	201	21.5%	270.0	8.22	0.23 s
1 s	935	64.6%	308	32.9%	169.4	5.16	0.50 s
2 s	935	74.8%	324	34.7%	128.3	3.91	1.18 s
3 s	936	88.2%	348	37.2%	96.0	2.92	1.96 s
5 s	937	95.3%	408	43.5%	76.7	2.34	3.36 s

Table A.3: The alignment audit of Section 5.1 across three datasets.

Property	**This work**	**Irish 4G/5G (MMSys 2020)**	**US tier-1 (NUWiNS PAM 2025)**
Instrument	XCAL, 1 Hz export produced by the tool	G-NetTrack Pro, 1 Hz application log	XCAL, native 10 Hz export downsampled here
Network and region	One operator, Dhaka and Gazipur	One operator, Ireland (Raca et al., MMSys 2020)	3 tier-1 operators, two cross-US routes (NUWiNS, PAM 2025)
Event clock	RRC signalling: command and complete	serving-cell identifier changes only	XCAL handover event column
Row semantics	aggregate over the second, written at its end	instantaneous sample	set by us: last, or first, sample of the second
Handovers audited	938	1,175 cell changes	4,712 in 12 files
Row t shows the target cell	75.9%	29.4%	93.2%
Isolated handovers only	74.6% (n = 291)	not separable (no signalling clock)	92.8% (n = 3,160)
Row $t-1$ shows the target cell	15.6%	no step across the integer boundary	0.1%
Same rows, start-of-second rule	not available (the tool aggregates)	not applicable (rows are samples)	4.1%
Update delay observed	p90 99 ms from RRC complete messages	not observable on a 1 Hz grid	50-75 ms, recovered from the contamination cliff (98% at 75-100 ms left, 44% at 50-75 ms, 0% below 50 ms)
Verdict	defect present, attenuated by a one-row KPI refresh lag	no defect: sampled rows are not aggregates	defect present under end-of-second, absent under start-of-second

Table A.4: Per-campaign calibration at the one-second horizon (coherence arms, corrected parser).

Arm	**1st Campaign (urban arterial) ECE / Brier**	**2nd Campaign (urban loop) ECE / Brier**	**3rd Campaign (dense urban) ECE / Brier**	**4th Campaign (highway) ECE / Brier**
independent	0.057 / 0.078	0.047 / 0.071	0.050 / 0.061	0.033 / 0.049
independent + cumulative max	0.059 / 0.078	0.047 / 0.071	0.049 / 0.061	0.033 / 0.049
independent + PAV	0.056 / 0.078	0.048 / 0.071	0.050 / 0.061	0.032 / 0.049
independent + isotonic	0.024 / 0.073	0.020 / 0.068	0.014 / 0.058	0.047 / 0.052
independent + isotonic + PAV	0.021 / 0.073	0.016 / 0.068	0.015 / 0.058	0.046 / 0.051
hazard	0.048 / 0.076	0.032 / 0.068	0.043 / 0.060	0.028 / 0.051

Table A.5: Sensitivity of every window chosen after the data were collected. Each is varied across a factor of two. These runs use a single seed and default LightGBM hyper-parameters without nested tuning, so absolute values differ from the tuned three-seed results of Chapter 5 (e.g. the 10 s cutoff gives 2.53× vs 1.59× here against 3.0× vs 1.8× in Table 5.13). Corrected-parser run; only the ordering across values is interpreted. This is local sensitivity around the chosen values, not a test of whether the choices themselves were unbiased.

Window varied	Value	Quantity it governs	Result
Random-block length	60 s	1 s AUPRC vs LOCO	+25.4%
Random-block length	120 s	1 s AUPRC vs LOCO	+15.8%
Random-block length	180 s	1 s AUPRC vs LOCO	+14.9%
Random-block length	240 s	1 s AUPRC vs LOCO	+8.2%
Random-block length	300 s	1 s AUPRC vs LOCO	+10.0%
Purge between blocked train and test	0 s	1 s AUPRC vs LOCO	+10.7%
Purge between blocked train and test	30 s	1 s AUPRC vs LOCO	+8.2%
Purge between blocked train and test	60 s	1 s AUPRC vs LOCO	+7.9%
Purge between blocked train and test	90 s	1 s AUPRC vs LOCO	+4.0%
Purge between blocked train and test	120 s	1 s AUPRC vs LOCO	+3.6%
Purge between blocked train and test	180 s	1 s AUPRC vs LOCO	+7.6%
Quiet / in-burst cutoff	5 s	1 s lift by stratum	2.43x vs 1.37x
Quiet / in-burst cutoff	10 s	1 s lift by stratum	2.53x vs 1.59x
Quiet / in-burst cutoff	15 s	1 s lift by stratum	2.76x vs 1.70x
Quiet / in-burst cutoff	20 s	1 s lift by stratum	3.01x vs 1.73x
Post-handover blank	0 s	1 s lift on the row set	2.01x on 10,236 rows
Post-handover blank	1 s	1 s lift on the row set	2.23x on 9,333 rows
Post-handover blank	2 s	1 s lift on the row set	2.17x on 8,586 rows
Post-handover blank	3 s	1 s lift on the row set	2.31x on 8,004 rows
Post-handover blank	5 s	1 s lift on the row set	2.53x on 7,074 rows

Table A.6: Per-handover evidence for the delayed-refresh explanation.

Quantity	**Value**
Handovers with a positive margin (row t predicted contaminated)	923
of which row t does carry the target	712
Residual: row t carries the source instead	211
of the residual, row $t+1$ carries the target	178 (84%)
Median margin, residual handovers	0.477 s
Median margin, contaminated handovers	0.522 s
Ten residual handovers, smallest margin first (unfiltered)	
1st Campaign, $\delta = 0.999$ s, margin 0.000 s	row t shows PCI 489, row $t+1$ shows PCI 380, target is PCI 380
1st Campaign, $\delta = 0.998$ s, margin 0.001 s	row t shows PCI 27, row $t+1$ shows PCI 91, target is PCI 91
2nd Campaign, $\delta = 0.998$ s, margin 0.002 s	row t shows PCI 346, row $t+1$ shows PCI 38, target is PCI 38
1st Campaign, $\delta = 0.995$ s, margin 0.004 s	row t shows PCI 13, row $t+1$ shows PCI 256, target is PCI 147
3rd Campaign, $\delta = 0.995$ s, margin 0.005 s	row t shows PCI 435, row $t+1$ shows PCI 303, target is PCI 303
3rd Campaign, $\delta = 0.984$ s, margin 0.015 s	row t shows PCI 30, row $t+1$ shows PCI 348, target is PCI 348
1st Campaign, $\delta = 0.980$ s, margin 0.019 s	row t shows PCI 270, row $t+1$ shows PCI 280, target is PCI 280
4th Campaign, $\delta = 0.978$ s, margin 0.022 s	row t shows PCI 39, row $t+1$ shows PCI 88, target is PCI 88
4th Campaign, $\delta = 0.975$ s, margin 0.024 s	row t shows PCI 193, row $t+1$ shows PCI 292, target is PCI 292
1st Campaign, $\delta = 0.876$ s, margin 0.026 s	row t shows PCI 280, row $t+1$ shows PCI 33, target is PCI 33

APPENDIX B

FEATURE INVENTORY AND RECONCILIATION

This appendix provides the exhaustive inventory of features, reconciles column counts across all manuscript tables, and specifies the zero-order hold and expiration rules applied to event-triggered neighbour measurements.

B.1 Exhaustive Inventory of the 112 Main Features

The primary predictive model uses 112 features, partitioned into three blocks: 89 radio features, 17 mobility features, and 6 session-level history features. All rolling statistics are computed strictly backward-looking per drive/campaign, closing at the lagged prediction instant $t-1$.

B.1.1 Radio Block (89 Features)

Because the vendor diagnostic CSV export omits periodic neighbour-cell logging (all candidate `Best_N` columns in the raw export are unpopulated), radio features combine synchronous 1 Hz serving RF measurements with causally projected neighbour measurements decoded from RRC `MeasurementReport` messages.

1. **Serving RF Features (55 features):** For each of the five serving-cell radio scalars—RSRP (`serving_rsrp`), RSRQ (`serving_rsrq`), SINR (`serving_sinr`), RSSI (`serving_rssi`), and CQI (`serving_cqi`)—the model computes 11 causal transforms:

    - Instantaneous scalar value (5 features);
    - First difference ($d_1 = x_t - x_{t-1}$) and second difference ($d_2 = d_{1,t} - d_{1,t-1}$) (10 features);
    - Causal rolling mean over backward windows of 3 s, 5 s, and 10 s (15 features);
    - Causal rolling standard deviation over backward windows of 3 s, 5 s, and 10 s (15 features);
    - 10-second least-squares causal slope (5 features);
    - 3-second fading range ($\max_{3s} - \min_{3s}$) (5 features).

2. **Per-Candidate Neighbour Features (18 features):** For the top-3 candidate neighbours ranked by RSRP in the most recent valid measurement report:

    - Projected neighbour RSRP (`nbr1_rsrp`, `nbr2_rsrp`, `nbr3_rsrp`) (3 features);

- Serving-to-neighbour RSRP gap ($\text{RSRP}_{\text{serving}} - \text{RSRP}_{\text{nbr},i}$) (3 features);
- First difference of neighbour RSRP (d_1) (3 features), tracking the *rank position* (the difference between the i-th ranked candidate slot across successive time steps, rather than tracking a fixed physical cell identity). Slot identity is not stable: on the four campaigns, the cell in slot 1 changes between consecutive seconds in 34% of second pairs where both carry a neighbour (29–38% by campaign), so differences, rolling means and the lead streak mix cells about a third of the time. Rank slots were chosen because the quantity of interest is the strongest competitor, whichever cell it is; PCI-keyed tracking would describe a fixed cell that may no longer be the competitor;
- First difference of serving-to-neighbour gap (d_1) (3 features), likewise tracking the rank position;
- Rolling mean gap over 3 s and 5 s windows for each candidate (6 features).

3. **Candidate Summary Features (10 features):** Summary statistics evaluated across all active candidates:
   - Best candidate RSRP (`nbr_best_rsrp`) and mean candidate RSRP (`nbr_mean_rsrp`) (2 features);
   - Candidate spread ($\text{RSRP}_{\text{best}} - \text{RSRP}_{\text{worst}}$) (`nbr_spread_rsrp`) (1 feature);
   - Serving-to-best neighbour gap ($\text{RSRP}_{\text{serving}} - \text{RSRP}_{\text{best}}$) (`gap_serving_best_nbr`) (1 feature);
   - First difference of serving-to-best gap (d_1) (1 feature);
   - Rolling mean of serving-to-best gap over 3 s, 5 s, and 10 s windows (3 features);
   - Duration of the A3-like lead streak, measuring how many consecutive seconds the best candidate's RSRP has strictly exceeded the serving RSRP (`nbr_better_streak_s`); the test is the raw inequality, with no offset or hysteresis, and a missing neighbour counts as not better (1 feature);
   - Serving cell rank among candidate cells (`serving_rank_among_cands`) (1 feature).
4. **Missingness Indicator Masks (6 features):**
   - Binary missingness flags for serving RSRP, RSRQ, and SINR (3 features, constant zero on this corpus but retained so feature construction is independent of dataset completeness);
   - Binary missingness flags for candidate 1, candidate 2, and candidate 3 RSRP (3 features, active when no valid neighbour report exists within the expiration window).

Total for Radio block: $55 + 18 + 10 + 6 = 89$ features.

Diagnostic Staleness Tracking vs. Model Inputs. During dataset ingestion and signalling merging, the pipeline derives two continuous staleness features: `mr_age_s` (time elapsed since the most recent `MeasurementReport` of any type) and `nbr_age_s` (time elapsed since the most recent report carrying neighbour measurements). These columns are used exclusively for quality control, audit characterisation (e.g., verifying that 56% of sample rows carry neighbour reports within 3.0 s, with median age 0.27 s), and case-study visualisation. They are omitted from the primary 89-dimensional radio feature vector, which relies instead on the explicit missingness masks and causal rolling gaps. The binary missingness mask operates as a coarse expiration gate; continuous decay weighting remains an area for future investigation.

B.1.2 Mobility Block (17 Features)

Constructed from the 1 Hz GPS kinematic stream, lagged one row:

1. Instantaneous speed in km/h (`speed_kmh`) and longitudinal acceleration (`accel_kmh_s`) (2 features);
2. Causal rolling mean and standard deviation of speed over 3 s, 5 s, and 10 s windows (6 features);
3. Sine and cosine of vehicle bearing angle (`bearing_sin`, `bearing_cos`) (2 features);
4. Heading angular difference (`bearing_change_deg`) and rolling mean of absolute heading change over 10 s (`bearing_change_abs_mean`) (2 features);
5. Cumulative distance travelled in drive (`dist_travelled_m`) (1 feature);
6. Stationarity indicator (speed < 2 km/h) (`is_stopped`) and stationary duration streak (`stop_duration_s`) (2 features);
7. Mobility speed regime indicators: mid-speed ($15 < v \leq 40$ km/h) and fast ($v > 40$ km/h) (2 features).

Total for Mobility block: 17 features.

B.1.3 History Block (6 Features)

Constructed from millisecond-resolution decoded signalling over the full continuous session:

1. Time elapsed since the most recent completed handover command (`t_since_prev_ho_s`) (1 feature);
2. Prior handover indicator (`has_prev_ho`), set to 1 if a prior command exists in the session, 0 otherwise (1 feature);

3. Time elapsed since the second-most-recent completed handover command (`t_since_prev2_ho_s`) (1 feature);

4. Handover command count within the preceding 30 s (`ho_count_30s`) and 60 s (`ho_count_60s`) (2 features);

5. Return-to-source indicator (`last_ho_was_return`), indicating whether the current serving cell matches the cell occupied two handovers prior (1 feature).

Total for History block: 6 features.

B.2 Neighbour Hold, Staleness, and Expiration Rules

Because neighbour cell readings are event-triggered rather than periodic, they are held across the 1 Hz grid according to the following deterministic rules:

- **Zero-Order Hold Window:** Each 1 Hz sample grid point t takes the candidate measurements from the most recent preceding `MeasurementReport` carrying neighbour cells, provided the report timestamp $\tau_{\text{MR}} \leq t$ falls within a backward tolerance window of tolerance_s $= 3.0$ s. Reports are not filtered by measurement object: the neighbours of intra-frequency and inter-frequency reports (and the best secondary-carrier neighbour carried by carrier-aggregation reports) are pooled and ranked by RSRP, so the gap to the "best neighbour" can be a gap across carriers. This is the same quantity a handset sees in its reports, but it is not the intra-frequency A3 margin of Equation (2.1), and it is one candidate reason why the neighbour block transfers poorly to the second operator, where most handovers change carrier (Section 5.12).

- **Expiration Policy:** When no report carrying candidate neighbour cells has occurred within the preceding 3.0 s, the held neighbour readings expire. All neighbour RSRP and gap features revert to NaN, and the corresponding neighbour missingness indicators (`mask_nbr*_rsrp`) are set to 1. Differences and rolling statistics are computed on these series with the gaps left missing (a rolling mean averages the values present in its window; the first difference of the gap is set to 0 where it is undefined). No constant is written into `features.parquet`. LightGBM receives the NaN values and routes them natively; for the other learners each column is imputed inside every fold with its training-partition median (0 if the column is entirely missing there), then standardised and clipped to ± 8 standard deviations (Section 4.6).

- **Serving Cell Handover Reset:** When an `RRCConnectionReconfiguration` with `mobilityControlInfo` completes, the serving cell changes. All serving-to-neighbour gaps are evaluated against `serving_rsrp` provided by the 1 Hz synchronous measurement stream. When a handover completes, the newly committed serving cell supplies subsequent serving measurements; if the new serving cell was previously

ranked as a neighbour candidate, candidate rankings are updated to prevent circular self-referencing.

- **Pipeline Alignment Order:** Zero-order holding operates on the synchronous 1 Hz grid *prior* to the one-row shift. When the one-row lag is applied to remove end-of-second aggregation leakage, the entire held vector is shifted by $+1$ sample interval ($t-1$), ensuring that all held neighbour values, differences, and rolling statistics close strictly at or before $t-1$. Serving and neighbour columns are shifted together, so a serving-to-neighbour gap used at t compares the serving RSRP of row $t-1$ with the neighbour readings held at grid instant $t-1$; reports decoded in $(t-1, t]$ are not used at t. The 0.27 s median report age and the 3.0 s expiry are measured at the grid instant where the hold is taken, that is, relative to $t-1$ in the lagged frame.

B.3 Column Reconciliation Across Tables

To prevent ambiguity between the raw CSV export, the lagged feature subsets, and the final design matrix, Table B.1 presents the exact column mapping across all chapters.

We emphasize two essential provenance distinctions: First, the raw vendor diagnostic CSV exports contain 113 columns of physical RF and GPS telemetry, without A3 offsets, hysteresis, or time-to-trigger fields. The three configuration parameters (`a3_offset_db`, `hysteresis_db`, `time_to_trigger_ms`) were decoded from Layer 3 RRC `measConfig` signalling messages and merged during preprocessing into intermediate feature matrices (`features.parquet`). Second, in the 109-column lagging experiment (Table 5.3), all 109 columns present in the intermediate table (89 radio + 17 mobility + 3 decoded configuration parameters) were shifted by one row to evaluate the spatial and semantic scope of the aggregation leak across feature subsets. In contrast, the primary prediction model (Table 5.4 and Sections 5.2–5.7) is trained strictly on the 112-feature vector spanning `rf` (89), `mobility` (17), and session `history` (6). The three configuration parameters were shifted in intermediate storage during the lagging experiment but excluded from the primary predictor's input set; they enter only as part of the 10-feature control-plane signalling block in the ablation of Section 5.8 ($112 + 10 = 122$ total columns).

Table B.1: Reconciliation of feature counts across manuscript tables. The 56-column and 31-column subsets share one column (`serving_rank_among_cands`), and the six missingness masks belong to neither, so the serving-relative subset is $56 + 31 - 1 + 6 = 92$ columns.

Designation	Count	Exact Composition and Table Reference
Lagged Export Columns	109	Total columns lagged in Table 5.3: 89 radio + 17 mobility + 3 configuration parameters decoded from RRC `measConfig` and merged into the intermediate feature matrix (`a3_offset_db`, `hysteresis_db`, `time_to_trigger_ms`; absent from raw 113-column vendor CSVs).
Serving-Relative Radio Columns	56	The 55 serving RF features plus `serving_rank_among_cands` lagged in Table 5.3.
Neighbour and Gap Columns	31	The 18 per-candidate neighbour features + 9 candidate summary features + 1 `serving_rank_among_cands` + 3 decoded configuration parameters lagged in Table 5.3.
Serving-Relative Subset	92	89 radio features + 3 decoded configuration parameters lagged in Table 5.3 (mobility left unlagged).
Main Feature Set	112	Primary input vector for all models in Sections 5.2–5.7: 89 radio + 17 mobility + 6 session history. (The 3 intermediate configuration parameters are shifted during the lagging experiment but excluded from this primary vector).
Full Design Matrix on Disk	122	112 main features + 10 control-plane signalling/configuration features (the 3 configuration parameters re-enter here alongside 7 signalling history counters in Section 5.8).

APPENDIX C

REPRODUCTION ARCHIVE AND PIPELINE SPECIFICATION

To ensure full reproducibility and enable independent verification of all empirical findings reported in Chapters 3, 4, and 5, the complete computational framework is structured as a deterministic, modular pipeline. This appendix documents the pipeline architecture, the execution directed acyclic graph (DAG), the repository layout, the configuration freeze manifest, and cryptographic SHA-256 verification checksums for all derived data frames and experimental artifact tables.

C.1 Pipeline Architecture and Execution Directed Acyclic Graph

The analysis is executed as a strictly ordered sequence of stages under Python 3.11 or later with the package versions of `requirements.txt` (the 17 October 2026 re-run used Python 3.13, pandas 2.3.3, scikit-learn 1.8.0, LightGBM 4.7.0 and PyTorch 2.14). Dependency directionality is strictly enforced: no stage may read artifacts from a downstream stage, ensuring an acyclic dependency graph that precludes reverse information leakage. Furthermore, every empirical table and figure presented in the manuscript is generated programmatically from primary stage outputs without manual intervention.

Table C.1 provides the inventory of core pipeline stages, specifying their primary source scripts, execution purpose, input dependencies, and output artifacts.

C.2 Repository Layout and Manifest Specifications

The reproduction directory structure is organized hierarchically into code, configuration, derived data, and reproducible reporting outputs:

```
pipeline/
  sept10/ sept12/ sept13/ sept15/ oct09/
    data/raw/                    # Raw XCAL CSV export + signalling text, per capt
  artifacts/
    freeze_manifest.json         # Configuration freeze (not registered)
  configs/
    models/                      # Architecture and hyperparameter grids
    adapters/                    # Hardware and logging schema mappings
  data/
    processed_xcal/              # Derived, quality-controlled row/event frames
    rev/                         # Decoded-signalling pickles (stage 12b)
```

Table C.1: Inventory of core pipeline execution stages in `hoproj.pipeline`.

Stage	Script	Scope and Primary Operation	Key Outputs
12	`stage12_xcal_prepare.py`	Ingests 4 campaigns; decodes ASN.1 RRC timeline; applies 1-row lag correction; enforces QC boundary cuts (957 → 938 HOs).	`df_usable_rows.csv`, `df_events_qc.csv`
12b	`stage12b_rev_pickles.py`	Builds the decoded-signalling pickles that stages 22–31 read (same builder as stage 32).	`data/rev/*.pkl`
13–18	`stage13–stage18`	Pre-revision benchmark, hazard, point-process, ping-pong, figure and tuning stages, retained for provenance only. They used settings the thesis no longer reports — six model families including TabNet (stage 13), ping-pong windows of 30/60/120 s (stage 16), a 50-trial Optuna search (stage 18) — and *no table or figure in Chapter 5 or Appendix A is generated by them*; the learners, windows and tuning reported are those of stage 22.	`reports_xcal/` (superseded)
19	`stage19_transfer_adaptation.py`	Unsupervised domain adaptation (per-drive robust standardisation and CORAL) on the configuration-regime transfer and on the public dataset of Shafi et al. [38]; the null of Table 5.17.	`transfer_adaptation.csv`
20	`stage20_departmental_baseline.py`	Re-implementation and head-to-head benchmarking of the contextual-bandit reinforcement-learning method of Shafi et al. [39], implemented as published (tabular Q-learning over a five-feature state).	`departmental_baseline.csv`
21	`stage21_capture_transfer.py`	Evaluates out-of-sample highway corridor transfer (Campaign 4 holdout).	`capture_transfer.csv`
22	`stage22_revision.py`	Source of the alignment, prediction, protocol, coherence, risk-control, ablation and event-level tables of Chapter 5 and Appendix A (Tables A.3, A.5 and A.6 come from stages 28–30): alignment audit; four-protocol ladder (fixed hyper-parameters); LOCO with nested tuning of **20 random-search trials per outer fold** over the seven learners of Section 4.7 (A3 proxy, logistic regression, LightGBM, MLP, TCN, Transformer, GRU), seeds 0, 1, 2; coherence arms; conformal risk-control rotations; ablations; event-level metrics.	`reports_rev/tables/c*.csv`
23–31	`stage23–stage31`	Signalling and external revisions, NUWiNS audit (stage 28), alignment mechanism (stage 29), single-seed sensitivity analyses (Table A.5), cell contamination audits, burst/isolated stratifications, and case-study extractions.	`c9_row_alignment_audit.csv`, `c19_sens_*.csv`
22 (re-run)	`stage22_revision.py` with the corrected parser	Stage 22 re-run on the four frozen campaigns after the optional-RSRQ fix in `hoproj/data/signalling.py` (Section 3.3). All experiments (`--exp all`, including `selective_lag`) and stage 30 were re-run; this is the source of every model table in Chapter 5 and Appendix A. The dwell, history and Hawkes arms reproduce the original tables exactly.	`reports_rev_parserfix/tables/c*.csv`
32	`stage32_second_operator.py`	Builds the row-level pickles (`build_pickles`) for a separate five-campaign dataset; runs the blind second-operator fold under the frozen stage-22 protocol; alignment audit, lag ablation, segment bootstrap, cross-operator risk control and per-campaign ping-pong (Section 5.12). Does not add the capture to the stage-12 corpus.	`second_operator/reports/tables/c32_*.csv`

```
    external/                    # Resampled public corpora (NUWiNS, Astana)
  reports_rev/
    tables/                      # Original-parser tables (superseded)
    figures/                     # Publication-ready figures
  reports_rev_parserfix/
    tables/                      # Corrected-parser re-run of stage 22
  second_operator/
    reports/tables/              # Stage 32: blind second-operator test
  src/
    hoproj/
      core/                      # Signalling decoder & discrete hazard math
      features/                  # 112 causal rolling and lag transforms
      models/                    # learners of the pre-revision stages 13-18
      pipeline/                  # Numbered stage execution scripts
```

C.3 Configuration Freeze Manifest and Cryptographic Checksums

Before Campaign 4 was driven, a configuration freeze was written to `freeze_manifest.json` (configuration fingerprint `ead9c9d0bfad`; file last modified 14 September 2026). The checksum in Table C.2 is that of this original, pre-audit file, which has not been edited since. It records the horizons (1, 2, 3, 5 s), a 10 s window, four feature blocks (radio, mobility, history, QoE) and a frozen model family (GRU); it does not record search spaces or the evaluation protocol. As Section 3.2 sets out, the final protocol departs from it (LightGBM carried as the primary learner, a 0.5 s horizon added, the QoE block dropped, the one-row lag introduced), and the `artifacts/` directory is not under version control, so the file's date is not corroborated by commit history.

To guarantee artifact integrity and permit exact verification against pipeline bit rot, Table C.2 documents the SHA-256 cryptographic hashes for the configuration manifest, derived intermediate data frames, and core empirical results tables.

C.4 Reproduction Commands and Environment Setup

The full execution environment is reproducible via Python standard packaging. With dependencies installed from `requirements.txt`, the stages that produce every reported table are run in the order below:

```
# Set Python path to include source modules
export PYTHONPATH=$PWD/src           # Unix / macOS
$env:PYTHONPATH="$PWD/src"           # Windows PowerShell

# The stages that produce the reported tables, in order
```

Table C.2: Cryptographic SHA-256 verification hashes for pipeline configuration and empirical tables.

File Name	SHA-256 Checksum	Description / Manuscript Mapping
freeze_manifest.json	98B9887F365BBA14BBA267D70DDBE174	Pre-Campaign 4 protocol freeze manifest
	BAF9117566A03915645BBAF87E38F80B	(config fingerprint ead9c9d0bfad).
c1_dataset_sessions.csv	D3DBA7B61537BE675D193AE6AF786EA4	Campaign drive-test summary metadata
	8AB7702015F7EFA2289CE8C761D84859	(Table 3.3).
c2_protocol_ladder.csv†	C82B45D7E84614CE15E9CA51E9BD66A3	Splitting protocol inflation ladder
	99041AEE57124B914E9F304FBBB235B3	(Table 5.6).
c3_main_loco.csv†	14AE1A0FB10FB71370BF6A252C3A902F	Primary multi-model LOCO benchmark
	675D93958D5DFF6DF3BC3BD13B7A9E1F	(Tables 5.4–5.5).
c4_hazard_prevalence.csv	B3A0D1797711D43427A06577DC3C7B70	Reconciled hazard expansion prevalence
	7E8DBEE8B5522945A7D5FE3AAD2FA799	(Section 4.3).
c9_row_alignment_audit.csv	F3D30CAF975F948749466CCAD86588F9	Alignment audit and cell contamination rates
	3AC59EC93B9EE1A13F9395530C8DE613	(Table 5.1).
c10_crc_frontier.csv†	7B6076664D82BCB3C4D6340D67CB31B4	Conformal risk control empirical loss frontier
	533B08FDDE6F7D5ED2367FA3273018A8	(Table 5.9).
c11_coherence_metrics.csv†	A67813CC27D8E4C7326EDC6B88733F92	Horizon coherence and calibration metrics
	D365E6057420BD60C217E5BC61A96AD4	(Table 5.7).
c13_pingpong_definitions.csv	32BA9D458F5F7B4DD2060CEE732AF65D	Ping-pong definitional sensitivity grid
	E087A59F4CA8043F21911094A897E553	(Table 5.15, Figure 5.15).
c18_ablation.csv†	6012AA7D8FC96B1FEC7158A36567823D	Feature group ablation comparison
	2FB0DDDABA5DC09084BD29AECACC86A4	(Tables 5.2, 5.17).
c32_blind_second_	25C4E1498BD539ACD2E9FE00B1CCDEAF	Blind second-operator test
operator.csv‡	BE1610FF76D77CDC74B3CDE61FD58B17	(Table 5.21).
c32_alignment_audit_	A9107B45A4E4D26F91C4C399801D6230	Per-campaign alignment audit
per_campaign.csv‡	B0A0379C94BB9630E893DECDDB30D4E9	(Table 5.20).
c32_lag_ablation_oct9.csv‡	EABA6324E00962CCF1D172866DB7290C	Lag ablation on the blind capture
	7B8BFBB5B1200967AF74F1075C81922A	(Table 5.20).
c19_sens_blank.csv†	FF36F1B118B3339FECF825785D24116E	Post-handover blanking window sensitivity
	8DB670D297B25FAE2015AE6B71D6C863	(Table A.5).

† `reports_rev_parserfix/tables/` (corrected parser). ‡ `second_operator/reports/tables/`. `freeze_manifest.json`: `pipeline/artifacts/`. All other files: `reports_rev/tables/`.

```
# (run_all is the older runner; it covers the pre-revision stages 12-21)
python -m hoproj.pipeline.stage12_xcal_prepare   # ingestion, QC, alignment
python -m hoproj.pipeline.stage12b_rev_pickles    # data/rev signalling pickles
python -m hoproj.pipeline.stage22_revision --exp all   # Chapter 5 model tables
python -m hoproj.pipeline.stage23_signalling_revision  # A3, profiles, Table 3.3
python -m hoproj.pipeline.stage26_rev_extras      # ping-pong, Hawkes, KS tests
python -m hoproj.pipeline.stage28_nuwins_audit --fetch all   # Table A.3 (NUWiNS)
python -m hoproj.pipeline.stage29_alignment_mechanism --exp all  # Tables A.3, A.6
python -m hoproj.pipeline.stage30_sensitivity     # Table A.5
# Blind second-operator test (raw files in oct09/data/raw/)
python -m hoproj.pipeline.stage32_second_operator
```

Execution is deterministic for a fixed thread count: random seeds are fixed across Python `random`, `numpy`, and `torch` (a re-run on 17 October 2026 reproduced the stage-22 tables to four decimals; with a different number of threads LightGBM varies by about ± 0.005 AUPRC). The reported tables of Chapter 5 use seeds 0, 1 and 2 with predictions averaged; the sensitivity runs of Table A.5 use the single seed 0 with default LightGBM hyper-parameters. Every stage writes its tables to `reports_rev/tables/`; there is no output-directory flag. The corrected-parser tables marked † in Table C.2 were produced by running the commands above with the corrected `signalling.py` and then copied to `reports_rev_parserfix/tables/`, so that the original-parser tables remain alongside them; stage 32 writes to `second_operator/reports/tables/`. A fresh run of the commands above reproduces the † files in `reports_rev/tables/`, where their checksums can be compared with Table C.2.

APPENDIX D

THE 22 AUDITED MODELS

only when a precision–recall curve or average precision is reported; precision and recall at a single threshold are recorded but not counted. Per-paper extraction: `pipeline/docs/literature/protocol_audit.csv`.

#	Model	Class	Year	Venue	Data	Split protocol	Prevalence-aware metric	Calibration	Imb. acc.	Code	Data
1	Chou et al., Learning-based RLF Prediction (Railway)	Direct	2026	arXiv:2511.08851 [57]	REAL-MEASURED	NOT STATED	no	no	yes	yes	no
2	Kabeer et al., Sequence-Based DL for Handover Optimization	Direct	2025	arXiv:2510.02958 [45]	REAL-MEASURED	NOT STATED	no	no	no	no	yes
3	Dzaferagic et al., ML-Based HO Prediction over Real O-RAN	Direct	2024/2025	IEEE TNSM 22(1) [54]	REAL TESTBED	60/20/20 stated; unit and order not specified	precision/recall	no	no	no	no
4	Panitsas et al., Predictive Handover Strategy in 6G	Direct	2024	arXiv:2404.08113 [58]	SIMULATED	80/20 with 10-fold CV (title changed between versions)	precision/recall	no	yes	no	no
5	Boutiba et al., Radio Link Failure Prediction in 5G	Direct	2021	IEEE GLOBECOM [59]	REAL TESTBED	two separate sets, unspecified	precision/recall	no	yes	no	no
6	Mandapati, Meta-Continual Mobility Forecasting	Direct	2025	arXiv:2512.11841 [46]	HYBRID (GeoLife + ray trace)	spatial zone hold-out	no	no	no	no	no
7	Chien et al., Privacy-Preserving HO Optimization (FL+LSTM)	Direct	2024	Sensors 24(20):6685 [60]	UNVERIFIED "real-world"	NOT STATED	no	no	yes	no	no
8	Wadud et al., AI-Driven Multi-Modal Adaptive HO Control	Direct	2026	arXiv:2603.17158 [61]	SIMULATED	NOT STATED for the ML components	precision/recall	CI over 30 runs	yes	no	no
9	Amirova et al., Data-Driven ML Analysis of HO Behavior	Direct	2026	Future Internet 18(6):290 [15]	REAL-MEASURED	device-level hold-out	YES (PR curve + AUC CI)	bootstrap CI on AUC only	no	no	no
10	Bartsioka et al., Intelligent Dynamic Handover via AI Signal Quality	Adj.: regression	2025	arXiv:2510.14832 [62]	SIMULATED	random 80/20	n/a (regression)	n/a	no	no	no
11	Sun et al., Proactive HO Type Prediction and Parameter Optimization	Direct	2025	IEEE TWC 24(4) [63]	SIMULATED	NOT VERIFIED	not verified	not verified	yes	no	no
12	Gonzalez Bermudez et al., GNNs for O-RAN Mobility Management	Direct	2025	arXiv:2502.02170 [64]	REAL-OPERATOR (Huawei)	random 80/10/10 edge split, structure-preserving	yes (AP alongside AUC)	threshold tuning for imbalance only	no	no	no
13	Hasan, Trappenberg & Haque, Generalized Transformer RLF Prediction	Direct	2024	arXiv:2407.05197 (journal version IEEE Xplore 11018489) [65]	REAL-OPERATOR (Turkcell)	rolling-origin temporal, 5 folds	no (F1 only)	no	no	yes	no
14	Santana & Dias, Open RAN-Enabled DL-Assisted Mobility Management	Direct	2024	arXiv:2412.21161 [66]	SIMULATED (OMNeT++/Simu5G) + real traces for seeding	NOT STATED	no	no	no	no	no
15	Narayana et al., AI-Native Mobility in 6G (dataset paper)	Adj.: dataset	2026	arXiv:2605.12453 [67]	REAL-MEASURED (commercial 5G SA, n78)	n/a - no models trained	n/a	n/a	n/a	no	no
16	Wu et al., DL for Moving Blockage Prediction (real mmWave)	Direct	2022	arXiv:2101.06886 / IEEE ICC [68]	REAL-MEASURED (mmWave testbed)	NOT STATED	no	no	yes	no	no
17	Lee et al., Prediction-Based Conditional Handover (5G mm-Wave)	Direct	2020	IEEE Veh. Technol. Magazine 15(1):54-62 [69]	SIMULATED	NOT STATED	no	no	no	no	no
18	Mehregan & De Grande, TH-GCN Throughput-Oriented Handover (GCN)	Direct	2025	arXiv:2505.04894 (IEEE DCOSS-IoT 2025) [12]	SIMULATED	NOT STATED	no	no	no	no	no
19	Dinh, Fazio & Voznak, Mobility and Resource Prediction for 5G Small Cells	Adj.: regression	2026	PLOS ONE 21(8):e0355372 [10]	SIMULATED (SUMO over a real Singapore map)	hold-out by travel day	n/a (regression)	n/a	no	no	no
20	Safiullin et al., RL-Based Handover Algorithm for 5G/6G	Adj.: RL policy	2026	Inventions 11(1):8 [70]	SIMULATED (ns-3)	n/a (RL)	n/a	n/a	n/a	no	no